

Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Abstract

This paper uses graduate admissions to study dynamic selection under costly evaluation. With a synthetic data protocol to protect applicants' privacy, I relate acceptance to measures of success, such as job placements. I then estimate structural models of an admissions committee's decision problem that capture first-round filtering and matriculation, and I develop statistical tests of whether observed decision rules are optimal given time and effort constraints. To determine which if any applicant characteristics are incorrectly valued, I quantify deviation gains from adopting counterfactual decision rules. This framework is applicable to multiple settings where decision-makers face uncertain outcomes and evaluation costs.

Keywords: Costly Evaluation, Dynamic Optimization, Education Economics, Graduate Admissions, Optimal Selection, Structural Estimation, Synthetic Data

JEL Codes: C61, C81, D81, I23

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1 Introduction

How do decision-makers select among a set of options in which each option's true quality is unobserved and costly to evaluate in terms of time and effort? In many such problems, decision-makers rely on a coarse, low-cost screening of all the options, followed by a precise, high-cost evaluation of the subset of options that survive the screening. Moreover, if the selected options have agency over whether to match with the decision-maker, final selections are often staggered over time through a deferred process to ensure the number of matches meets a target number.

Admissions and hiring are two prominent examples of such selection problems. There, committees first screen all the applications to filter out unqualified applicants, then read a subset of promising applications in full detail, then potentially interview the most promising applicants, and finally make early and late offers, depending on how many early offers are accepted. Other examples of selection with costly evaluation, and in some cases deferred offers, include conducting multi-stage clinical trials for medicines, messaging virtually on dating apps before meeting in real life, and simulating product prototypes on a computer before physically building them.

A fundamental question in such settings is how decision-makers should interpret the available information at each stage of evaluation to select either the best subset of options or all options above a quality threshold. A related question is whether observed decision rules reflect optimal dynamic selection given evaluation costs and other constraints, or whether those rules systematically deviate from optimal behavior. One could imagine a hiring committee undervaluing a skill or discriminating against a demographic. Moreover, if there are deviations, which decision rules should decision-makers implement instead? Answering these questions requires moving beyond identification of the determinants of success and explicitly modeling the decision problem.

Graduate admissions is a meaningful setting to study these questions, as evidenced by economics papers from Krueger and Wu (2000) to Bai, Esche, MacLeod, and Shi (2022). Graduate programs, especially PhD programs, invest at least a few years of time, along with tens of thousands of dollars of funding, into students' development. In addition, high-ranking academic placements boost a program's reputation, and so increase its ranking from sources such as U.S. News. Better students also lead to better coauthoring opportunities for faculty, and since people with graduate degrees often find themselves in influential positions, such as professorships, studying how best to admit them is important for programs and society. In particular, if schools are undervaluing applicants with certain characteristics (e.g. from a particular demographic), future applicants with such characteristics may receive opportunities they would not have otherwise. For example, while Chetty, Deming, and Friedman (2026) focus on undergraduate admissions, they find that applicants from high-income families have higher admissions rates at Ivy-Plus colleges even if their standardized test scores are no higher than those of less fortunate applicants.

Moreover, graduate admissions is a setting with costly evaluation. Admissions committees observe for all applicants a limited set of objective characteristics, such as standardized test scores, but typically lack the time to read every file in detail. As such, applicants are initially

filtered based on coarse signals, with more intensive evaluation reserved for the subset of applicants with promising objective characteristics. Final decisions may also involve waitlists or other forms of deferred acceptance, as uncertainty about applicant quality and matriculation status resolves over the period between the application and admissions deadlines. These features make graduate admissions well-suited for modeling dynamic selection under costly evaluation.

This paper addresses three specific research questions. The first question asks which *ex-ante* factors in application files determine admission and success in graduate school. Following the literature, success may be defined in multiple ways, including graduate course grades, passing comprehensive examinations, program completion, and job placement outcomes. I study this question as comprehensively as possible by extracting all available data from application files, including recommendation letters, application essays, and CVs. My data consist of all applicants to a research university’s economics PhD program from 2015 to 2023, as well as all applicants to that university’s government, history, linguistics, and sociology PhD programs from 2020 to 2023. Because these data are confidential, I implement a privacy-protecting research protocol that uses synthetic data for model development and validation (see Section 2). This protocol also serves as a template for researchers who wish to work with sensitive data in general.

The second question asks whether admissions committees optimize or make mistakes when admitting applicants. Specifically, if a factor predicts admission but not success, the committee may be overvaluing it, whereas if a factor predicts success only, the committee may be undervaluing it. For instance, the committee may be putting too much or too little weight on an elite undergraduate institution. To determine if the committee is optimizing, it is necessary to model the committee’s optimization problem, estimate the parameters, and formally test the hypothesis of optimization, which to date, no other paper has done. To make such models tractable, per Bai et al. (2022), I reduce the committee’s portfolio choice problem (i.e. from N applicants, choose the optimal subset N_A) to a cutoff rule problem (rank the N applicants by their expected success probabilities and select the N_A above the cutoff). I estimate multiple structural models in increasing order of complexity, modeling such features as first-round filtering and matriculation. Predictors that are strongly associated with admission but weakly associated with success may be overvalued, while predictors of success that play little role in admissions decisions may be undervalued. For example, committees may place too much or too little weight on elite undergraduate institutions, standardized test scores, or technical skills. Determining whether such patterns are mistakes requires modeling the committee’s optimization problem directly. That said, the quality of the PhD program also affects success (Angrist, Diederichs, & Ellison 2026), so it is important to account for that lest the model encourage the committee to put more weight on a variable that only predicts success by way of predicting admission into a highly-ranked program.

Perhaps as Simon (1956) suggests, if committees are not optimizing, they may be satisficing (i.e. aiming for a satisfactory result), namely because reading every application file would

require too much effort.¹ This idea lends itself to the third question of whether committees can make better decisions without having to expend additional effort. To answer this question, I use the structural models to calculate counterfactual deviation gains from adopting the optimal admissions strategy, while respecting the committee’s time constraint of not being able to read every application file in detail. For robustness, I perturb the success parameter estimates about their asymptotic distribution when comparing the expected success probability for the class admitted by the model’s decision rule vs. the committee’s. Perhaps committees have other priorities than maximizing success probabilities, such as admitting a diverse class, but I show how the models can be extended to factor in those priorities. Using Monte Carlo simulated data whose data-generating parameters I calibrate with reduced-form results from actual data, I show that in a dataset where the committee behaves optimally, a Kolmogorov-Smirnov test fails to reject the null hypothesis of no deviation gains. But in a dataset where the committee makes mistakes, such as discriminating against a demographic, the test rejects the null with $p < .0001$.

In general, when studying admissions data, one may find evidence of discrimination against applicants with certain features (e.g. a low-income background, as shown in Chetty et al. 2026), or of academic network effects. For example, like Bai et al. (2022), I find that having a recommender who also wrote letters for other applicants is predictive of acceptance, which implies that it helps to have well-connected recommenders. Given that the university in this study is not necessarily representative of all graduate programs, some of this paper’s results might not generalize to other settings. However, the methods are generalizable to admissions, hiring, and any other setting where decision-makers must decide which options to select based on *ex-ante* information.

1.1 Related Literature

While there have been earlier papers on graduate education, Ehrenberg and Mavros (1995) were the first to include predictors of success found in application files. They analyze Cornell PhD students, including economics, and find that verbal but not quantitative GRE scores are predictive of program completion. Subsequent research implies the reason is likely that their sample is restricted to only the applicants who attended Cornell, nearly all of whose quantitative GRE scores were concentrated in the top percentiles. Two years later, Attiyeh and Attiyeh (1997) use a sample across 48 universities and multiple programs from 1990 to 1991 to study determinants of admission. They find that all else equal, committees prioritized underrepresented minorities.

Krueger and Wu (2000) and Grove and Wu (2007) then study admission and success at a single top economics department in 1989, including all applicants in their success regressions and tracking placement outcomes for non-matriculants. They find that admissions committee scores, GRE scores, and the prominence of recommendation letter writers predict good placements, but

¹Satisficing is distinct from rational inattention (see Section 1.1), which assumes that decision-makers are optimizing and so embeds the choice of how much effort to allocate into an optimization framework.

there is a lot of uncertainty. Finally, Athey, Katz, Krueger, Levitt, and Poterba (2007) analyze the determinants of success for Chicago, Harvard, MIT, Princeton, and Stanford economics PhD students from 1990 to 1999, adding graduate course grades into their models. While GRE scores and other *ex-ante* factors predict good placements, they do not when controlling for grades.

Stock, Finegan, and Siegfried (2006) begin a series of papers by compiling a 586 student dataset of Fall 2002 economics PhD matriculants. They start by looking at attrition rates in the first two years and find that low GRE scores predict attrition, but RAships and shared department office space help programs retain students. Siegfried and Stock (2007) compile another, larger dataset of everyone who earned a U.S. economics PhD from 1966 to 2003 and observe that top liberal arts colleges feed economics PhD programs at disproportionately high rates.

Stock, Finegan, and Siegfried (2009a) and Stock, Finegan, and Siegfried (2009b) follow up on the progress of the students in Stock et al. (2006) and find similar results but emphasize that there is a lot of uncertainty in predicting completion after five years. Stock, Siegfried, and Finegan (2011) and Stock and Siegfried (2014) track the students after eight years and summarize their multi-paper study, concluding that while some factors like high GRE scores, a liberal arts undergraduate institution, and financial support predict completion, uncertainty remains very high. Finally, Stock and Siegfried (2015) add more recent years to Siegfried and Stock (2007)'s larger dataset, noting that foreign undergraduate institutions have become more common over time.

Meanwhile, Grove, Dutkowsky, and Grodner (2007) study Syracuse economics PhD students, who have different characteristics than students attending top-ranked programs. They also incorporate factor analysis into their regressions, finding that different attributes predict success at different stages. For example, GRE scores predict passing comprehensive exams, whereas research motivation (a factor from their factor analysis) predicts program completion. Schlauch and Startz (2018) study economics PhD candidates from top 50 programs posting their CVs for the 2016–2017 job market and find that while RA experience predicts attending a highly-ranked PhD program, a masters degree does not. This result tracks with the increasing prevalence of predocs in economics. Meanwhile, Jones, Schuhmann, Soques, and Witman (2020) do not run regressions but rather survey individuals involved in admissions at 132 economics PhD programs. They find that higher-ranked programs care about different factors than lower-ranked programs. For example, they care more about undergraduate rankings. Also, while recommendation letters matter for admission, programming skills usually do not, though they may matter for success.

Bai et al. (2022) represent the frontier of economics graduate admissions research. They study economics PhD applicants at a single top program from 2013 to 2019 and like Krueger and Wu (2000) collect data on admissions and placement outcomes for all applicants. They start by proving that a committee's portfolio choice problem simplifies to a cutoff rule problem, which is more tractable. I build on this result by directly modeling the cutoff rule problem and estimating the parameters, which is necessary to make claims as to whether committees are optimizing. In addition, they perform textual analysis of recommendation letters and find that subjective factors, such as letters, matter more for both admission and success more than objective factors, such as

GREs. As such, the models in this paper incorporate subjective as well as objective factors.

This paper also relates to a broader methodological literature on optimal dynamic selection with costly evaluation, as well as a literature on synthetic data and privacy protection in empirical economics. Within the former literature, earlier papers develop classic models of sequential search where agents face uncertainty about option quality and must decide when to stop searching given search costs. McCall (1970) and Mortensen (1970) model job search as a dynamic stopping problem in which workers optimally accept offers that exceed a reservation wage. Weitzman (1979) extends this logic to settings in which decision-makers can acquire information about multiple alternatives at a cost, which he denotes as “Pandora’s Problem.” Together, these papers establish that when information is costly, optimal decision rules are often threshold-based.

A more recent group of papers studies costly evaluation outside of search problems. In models of rational inattention, agents are fully optimizing but face constraints on their ability to process information. Sims (2003) introduces rational inattention as a framework in which information acquisition is costly, leading agents to optimally trade off decision accuracy with information costs. Matějka and McKay (2015) show that rational inattention can generate probabilistic discrete choice rules in line with multinomial logit models, while Caplin and Dean (2015) provide a revealed-preference characterization of optimal information acquisition that separates rational inattention from actual mistaken decisions. These models differ from Simon (1956)’s concept of satisficing, which proposes heuristic, “good enough” decision rules. In contrast, rational inattention models retain the mathematical framework of rationality but endogenize information acquisition. Notably, unlike this paper, none of these papers estimate their models on data.

This paper differs from both rational inattention and satisficing. I take the information environment faced by decision-makers as given, since admissions committees, at least in this framework, do not choose how much information is available at each stage of evaluation. Instead, they interpret the information they observe at each stage and decide how to act on it. The contribution of this paper is to model this dynamic selection problem explicitly, estimate parameters governing the resulting decision rules, and determine whether observed behavior is optimal conditional on respecting the committee’s constraints. By doing so, this paper makes optimality testable and allows for the computation of counterfactual decision rules and deviation gains.

The second methodological literature addresses synthetic data and privacy protection. Rubin (1993) introduces synthetic data as a tool for protecting confidentiality by replacing sensitive microdata with simulated observations drawn from an estimated data-generating process. Before synthetic data, privacy protection often involved masking, or adding random noise to the data. Masking, however, induces a trade-off between privacy and information loss (Little 1993), and the rows in a masked dataset still correspond to real subjects. More recent work, such as Abowd and Schmutte (2015), emphasizes the challenges posed by restricted access to data. Meanwhile, Koenecke and Varian (2020) propose that researchers publicly release synthetic data, such as those generated by the Synthetic Data Vault (Patki, Wedge, & Veeramachaneni 2016), when they cannot release the actual data. Their proposed protocol differs from this paper’s protocol, as this

protocol is designed for a setting where the *researcher* cannot access the actual data.

The rest of the paper proceeds as follows. Section 2 describes the data and synthetic data protocol. Section 3 presents reduced-form evidence that economics admissions is more data-driven than the university’s other social science programs, along with an omnibus Wald test that shows that the determinants of acceptance differ significantly across programs. Section 4 develops and estimates structural models of admissions that incorporate first-stage filtering, matriculation probabilities, and the waitlist. Section 5 implements counterfactual deviation gains tests that reject optimality only when the committee misvalues applicant characteristics. Section 6 concludes.

2 Data

This paper uses two datasets: one consisting of all applicants to a research university’s economics PhD program from 2015 to 2019, and another of all applicants to that university’s economics, government, history, linguistics, and sociology programs from 2020 to 2023. Unlike the latter dataset, the former dataset contains subjective variables (i.e. those derived from textual documents such as CVs and recommendation letters) and program outcomes data (e.g. post-program job placements). Therefore, I estimate this paper’s structural models on the former dataset only. However, the latter dataset, which still contains objective variables and each applicant’s admission status, is necessary for making reduced-form comparisons across programs.

Also, the implementation of this paper’s privacy-protecting synthetic data protocol is in progress. Therefore, while Section 3 contains reduced-form results from the latter dataset, the structural results in Section 4 are from Monte Carlo simulated data whose data-generating parameters were calibrated with the reduced-form results. The simulated data estimates verify that the structural models’ parameters are identifiable and that the hypothesis tests based on those models reject optimality only when the simulated admissions committee misvalues applicant characteristics. I will replace the simulated results with actual results and move the simulated results to an appendix in a future version of this paper. In this section, I describe the variables and synthetic data protocol, the latter of which serves as a template for conducting research on confidential data.

2.1 Variables

There are three types of variables: dependent variables, objective independent variables, and subjective independent variables. Table 9 of Appendix A contains a full list of the variables. The dependent variables consist of measures of both admission and success. For admission, such measures include binary variables of whether the applicant makes it past the first-round filtering stage, whether they are accepted into the program, whether they are waitlisted, whether they attend the university’s open house, whether they matriculate at the university, and whether they attend (and were therefore accepted at) a different university’s PhD program. I also have the U.S. News American and global rankings of the program they attended. If they attended a

U.S. program, I use their American ranking, whereas if they attended a non-U.S. program, I use their global ranking times 3/8. The reason is that the maximum American ranking is 147, the maximum global ranking is 400, and I assign 150 to anyone who did not attend a PhD program. This procedure ensures that all U.S. and non-U.S. programs are ranked on the same scale.

For success in economics, I have a binary variable of whether they complete the program, along with variables for the type and ranking of their eventual job placement. Placement types include academic, government, consulting, and technology. For academic placements, I use the department’s IDEAS/RePEc American and global rankings with a plus five penalty for business schools (Krueger & Wu 2000). For postdocs, I use a 2/3 to 1/3 weighted average of the university’s ranking and the maximum ranking, which is 138 for American and 340 for global. For non-tenure-track positions, I use a 1/3 to 2/3 weighted average. For unranked tenure-track positions, I assign the maximum ranking plus one; for unranked postdocs and non-tenure-track positions, I assign 150 for American and 350 for global. Ultimately, for U.S. placements, I use the American ranking, whereas for non-U.S. placements, I use the global ranking times 3/7.

Meanwhile, for non-academic placements, I construct academia-equivalent rankings shown in Appendix A. Following Krueger and Wu (2000), the most prestigious non-academic placements (World Bank, IMF, Federal Reserve Board of Governors, European Central Bank, Amazon Science, Google Research, and Microsoft Research) are given a ranking of 40, with lower rankings down to 135 for less prestigious placements.² Per Eberhardt, Facchini, and Rueda (2026), economics PhD graduates, even those with Nobel laureate advisors, are increasingly taking non-academic jobs, especially in technology. As in Krueger and Wu (2000) and Bai et al. (2022), to avoid selection bias, I use publicly available LinkedIn profiles, graduate program websites, personal websites, PandaInUniv (2026), and García Guzmán (2026)’s *The Econ PhD placements dataset* to obtain outcomes data for all applicants, not just matriculants. The minority of applicants whose outcomes data are not available receive the lowest possible ranking of 150.

Regarding independent variables from the application files, I distinguish objective variables from subjective variables. Objective variables are those variables that the application software compiles into a spreadsheet for the committee’s convenience, whereas subjective variables are those that the applicant uploads as pdf files that cannot be easily converted into a spreadsheet.³ Because the economics committee does not have time to read everyone’s file, it uses the objective variables to filter out a predetermined percentage of the applicants before reading any files. For example, they filter out applicants with low quantitative GRE scores. For this reason, the structural models in this paper treat objective variables separately from subjective ones.

Examples of objective variables include demographics, such as application year, age, gen-

²I use an average of the rankings provided by ChatGPT 5.5, Claude Opus 4.7, and Gemini 3 in temporary chat mode. The prompt, along with screenshots of the outputs, are available on request.

³Bai et al. (2022) define objective variables as “verifiable performance measures” of applicant quality and subjective variables as committee member assessments. For most variables, our definitions coincide. While they do not categorize demographics, such as gender and nationality, I categorize demographics as objective variables.

der, and an ethnicity/region of the world categorical variable, in which the categories are U.S. African/Hispanic/Native American, U.S. Asian, U.S. Other, International Asian, and International Other. There are also performance variables; such as verbal, quantitative, and analytical GRE percentiles; undergraduate GPA; TOEFL/IELTS scores for non-native English speakers; two binary variables for having a graduate degree and work experience respectively; and a categorical variable for the type of undergraduate institution (elite U.S. university, elite U.S. liberal arts college, other U.S. college/university, elite foreign university, and other foreign university), where elite is always defined as top 50. Moreover, recommender variables, namely the number of professor (Krueger & Wu 2000) and prolific recommenders the applicant has, measure how well-connected the applicant is. A prolific recommender is one who writes a letter for at least one other applicant in the sample, which Bai et al. (2022) find predictive of admission and success. For non-economics applicants, there is also a categorical variable for concentration, such as U.S. vs. world history. Finally, for economics applicants whose files the committee read, I have the subjective score that their assigned reader gave them relative to the scores that reader gave other applicants. This score is a proxy for information that the committee can observe but I cannot.

Meanwhile, subjective variables in economics come from recommendation letters, application essays, CVs, and supplemental forms. I use textual processing algorithms on the letters to measure the overall word count and rates of positive, negative, standout, ability, research, grindstone, teaching, communal, and agentic words (Bai et al. 2022). From application essays, the algorithms extract the word count; rates of positive, negative, economics, and research words; and whether the applicant mentioned a professor’s name from this university. The last three are my analogues of Grove et al. (2007)’s *Specific Topic*, *Mention Paper*, and *Specific Member*. From CVs, there are the word count, a binary variable for whether the applicant has a publication or working paper, and the number of programming languages the applicant listed. Jones et al. (2020) find that economics committee members, especially at high-ranking programs, do not put much weight on coding skills. But given the increasing prevalence of empirical research that uses techniques like webscraping and machine learning, along with the increasing use of AI tools in economics, such skills have become more relevant. Economics also has a supplemental form, in which the applicant lists the number of math courses they took, as well as the names of advanced math courses such as real analysis. I find that more math is predictive of getting accepted into the economics program, and I will determine whether it is also predictive of success. I also have a categorical variable of the applicant’s indicated field of interest (e.g. econometrics, macroeconomics, etc.). Compared with concentrations in other programs, fields of interest in economics are less binding because applicants can costlessly change their field after entering the program.

The economics dataset contains all of the above variables, albeit the committee score is available for only the applicants whose files were read, and the other subjective variables from the textual documents are available for only a random subset of applicants. In contrast, the dataset of all programs’ applicants is more limited because it does not contain success data or subjective variables. But the latter dataset is still necessary for reduced-form analyses and statistical tests

comparing the determinants of admission across programs. I present summary statistics from this dataset in Table 11 of Appendix B, as well as reduced-form results in Section 3. I find that for better or worse, economics admissions is more data-driven than the other programs.

2.2 Synthetic Data Protocol

In this section, I describe the synthetic data protocol used to protect the applicants’ privacy, which I employ in conjunction with a university administrative office that is responsible for stewarding the data. While a handful of recent papers, such as Stanley and Totty (2024) and Carr, Wiemers, and Moffitt (2025) use synthetic data, the synthetic data in those papers had already been made available to researchers by the U.S. Census Bureau. In contrast, this paper’s protocol covers all the steps from creation of the synthetic data to obtaining the final results on the actual data.

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	.	150
0	0	0	0	0	80	1	1	0	1	.	150
0	0	0	0	1	99	0	1	0	0	.	29
0	0	0	0	0	81	0	0	1	1	.	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	.	48
1	1	0	0	0	87	0	1	0	1	5	34
⇓											
Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	.	30
0	0	0	0	1	89	0	0	0	0	.	25
0	0	0	0	0	85	1	0	0	0	.	150
0	0	0	0	0	76	0	1	0	1	.	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	.	35
0	0	0	0	0	70	1	0	0	0	.	150
1	1	0	0	1	93	0	1	0	0	5	57

Figure 1: Synthetic Data Example

The steps of the protocol are as follows. First, the administrative office receives the data from the graduate admissions office and individual departments, such as economics. They remove all direct identifiers, such as names and email addresses, and run algorithms I provide to extract variables from textual documents, such as recommendation letters. Second, they use Patki et al. (2016)’s publicly-available Synthetic Data Vault (SDV) to generate the synthetic data. The SDV estimates a Gaussian copula to capture correlations between variables and then simulates a new, synthetic dataset from the copula using a random number generator. Crucially, the rows in the synthetic dataset do not correspond to real subjects, which prevents identity disclosure (Chapelle & Falissard 2023). The SDV works best when the final versions of the variables are already created (e.g. turning a list of country names into a categorical variable), so without seeing any actual data, I work with the office to create the variables before implementing the SDV.

However, there remains the possibility of attribute disclosure. For example, if an applicant from an unusual country has an unusually low GRE percentile, information about that applicant may leak into the synthetic dataset. As such, if any cell in the synthetic data has fewer than five other cells from that variable with the same value, the variable’s observations are binned and in each bin, replaced with the bin’s sample mean. Only at this point do I receive the synthetic data. I use the data to finalize my estimation code, which I then give to the administrative department. The department takes that code, runs it on the actual data, and returns the parameter estimates.

As a demonstration of synthetic data, I use a Gaussian copula to simulate 1,000 observations of the following independent variables: *GRE Quantitative Percentile* (continuous), *Gender* (binary), *Demographic* (categorical with three demographics), *Advanced Math* (binary), and *Committee Score* (continuous). Then I use the standard logistic function to simulate binary dependent variables for transitioning past the first-round filter, getting waitlisted, getting accepted into the program, matriculating at the program if accepted, and obtaining a successful placement above a threshold ranking. I also generate the (continuous) rank of the PhD program each applicant attends, which is constant for the applicants with *Matriculate* = 1 and equals 150 for applicants who do not attend a PhD program (for these applicants, *Success* = 0). The top frame of Figure 1 contains the first ten observations of this simulated dataset. *Committee Score* is missing for all applicants who did not survive the first-round filter, which is a feature of the actual data.

I then pass this dataset through a Gaussian copula and use the copula to generate a synthetic dataset, the first ten observations of which are in the bottom frame of the figure. The means, variances, and correlations of the variables are similar in both datasets, with only the correlations in the synthetic dataset slightly attenuated. However, none of the rows of the simulated dataset are present in the synthetic dataset. In addition, important statistical properties; such as *Accept* equaling 1 only when *Transition* equals 1, *Demographic 1* and *Demographic 2* never both equaling 1, and *Committee Score* missing when *Transition* equals 0; are preserved.

3 Reduced-Form Analysis

In this section, I perform two reduced-form analyses. First, using the dataset of all 6,222 applicants to the university’s economics, government, history, linguistics, and psychology PhD programs from 2020 to 2023, I regress a binary variable for acceptance on a set of objective variables, such as GRE percentiles and undergraduate GPA. Summary statistics of the data are in Table 11 of Appendix B, which shows that the economics program has the highest acceptance rate at 11.21%, and the government program has the lowest acceptance rate at 4.30%. I also run lasso and elastic net regressions to determine which variables are most predictive of acceptance, along with post-lasso logistic regressions, in which I keep only the variables that survive the lasso. In a future version of this paper, I will regress success measures on objective and subjective variables for the economics program. The success measures will include binary variables for

program completion and for obtaining a placement above a threshold ranking.

Second, I implement an omnibus Wald test of whether the determinants of acceptance are the same for the economics program as for the other programs in aggregate. Because economics places a high weight on the quantitative GRE, along with GPA and a strong math background, I reject the null of equal determinants with $p < 0.0001$. This test can also be used to determine whether the determinants of success are equal across programs, but crucially, it cannot be used as a test of optimality by comparing the determinants of acceptance vs. success within a program. To test optimality, a structural model is necessary to relate acceptance effects to success effects.

3.1 Regressions and Lasso

In Table 1, I present logistic regressions of acceptance on objective independent variables for all five programs.⁴ The left subtable contains the regression coefficients, whereas the right subtable contains the average marginal effects (AMEs). Most notably, relative to the other programs, economics places much larger weight on undergraduate GPA and the quantitative GRE. A one point increase in an applicant's quantitative GRE percentile increases their acceptance probability by over one percentage point. As Bai et al. (2022) find, having a prolific recommender, defined as one who also recommended other applicants in the sample, matters as well. Going from zero to three prolific recommenders increases an applicant's acceptance probability by over six percentage points. This result provides evidence of academic network effects in the social sciences.

There are a few more results worth describing. First, across programs, acceptance rates were lowest in Spring 2021, which was the year that Covid reduced department funding. Second, all else equal, economics favors younger applicants, and economics, government, and history favor women. Third, government and history favor underrepresented minorities, whereas economics disfavors applicants from East and South Asia. Fourth, relative to a non-elite U.S. undergraduate institution, an elite institution matters more than a graduate degree or work experience. Fifth, taking a high number of math courses matters for economics. Note, however, that these results are descriptive and by themselves do not support any normative claims about admissions.

In addition, both in-sample and 5-fold-cross-validated (CV) Pseudo- R^2 values never exceed 0.3 for any program. However, they are highest for economics, meaning that economics admissions is more data-driven than the other programs, at least with respect to the observable variables. For economics, the CV Pseudo- R^2 of 0.2102 is within a few percentage points of the in-sample Pseudo- R^2 of 0.2677, but that is not the case for the other programs. In fact, for linguistics and psychology, the CV Pseudo- R^2 s are negative. Some of the other programs have additional admissions criteria, such as psychology, which interviews applicants. Although I cannot observe interview results, the interviews may explain some of the variation in acceptances.

⁴In Table 10 of Appendix B, I present the regression results on synthetic data. The synthetic estimates are more often than not similar to the actual ones, but the synthetic ones are generally less statistically significant.

Because of the low to negative CV Pseudo- R^2 s, I run lasso and elastic net, with k-fold cross-validation for variable selection. In Table 2, I present the lasso estimates. All the CV Pseudo- R^2 s are positive, and lasso selects more variables for economics than for any other program. The last three variables in this table are missing variable dummies. Any time an applicant's GRE percentiles, undergraduate GPA, or TOEFL/IELTS scores are missing, I impute the variable's sample mean for that observation and encode the missing variable dummy equal to 1 (Krueger & Wu 2000). I could implement group lasso (Yuan & Lin 2006) to force lasso to treat a variable and its corresponding missing dummy as a package, but allowing lasso to select one but not the other (see Bai et al. 2022) has a meaningful interpretation. For example, when lasso selects undergraduate GPA but not its missing dummy for economics, it means that undergraduate GPAs, as opposed to TOEFL/IELTS scores where lasso selects the missing dummy, are missing at random.

Table 1: Logistic Regression Results

(a) Coefficients						(b) AMEs						
Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.3966** (0.2010)	-0.4332* (0.2612)	-0.5752* (0.3479)	-0.5501 (0.4129)	-0.1766 (0.5545)	Covid Year	-0.0303*** (0.0153)	-0.0164 (0.0100)	-0.0431* (0.0261)	-0.0298 (0.0224)	-0.0075 (0.0238)	-0.0218*** (0.0084)
Winsorized Age	-0.0995*** (0.0358)	0.0454* (0.0271)	0.0380 (0.0327)	0.0222 (0.0532)	-0.0520 (0.0899)	Winsorized Age	-0.0076*** (0.0027)	0.0017* (0.0010)	0.0028 (0.0025)	0.0012 (0.0029)	-0.0022 (0.0038)	0.0016* (0.0009)
Female	0.4079** (0.1681)	0.6723*** (0.2299)	0.6539** (0.2971)	0.1311 (0.3452)	-0.5060 (0.5606)	Female	0.0312** (0.0129)	0.0254*** (0.0088)	0.0490** (0.0220)	0.0071 (0.0187)	-0.0216 (0.0239)	0.0207*** (0.0075)
U.S. African/Hispanic/Native	0.4221 (0.5258)	1.0081*** (0.3545)	1.1601*** (0.4426)	0.0120 (0.8411)	-0.8263 (1.1611)	U.S. African/Hispanic/Native	0.0323 (0.0402)	0.0381*** (0.0137)	0.0869*** (0.0332)	0.0007 (0.0455)	-0.0352 (0.0487)	0.0384*** (0.0119)
U.S. Asian	-0.5394 (0.4199)	-0.5669 (0.6504)	-0.3260 (1.2512)	0.8638 (0.7201)	1.1711 (0.7276)	U.S. Asian	-0.0413 (0.0322)	-0.0215 (0.0247)	-0.0244 (0.0936)	0.0468 (0.0391)	0.0499 (0.0313)	0.0062 (0.0185)
International Asian	-0.8445*** (0.2945)	-0.1853 (0.4146)	0.9405 (0.7233)	0.4890 (0.6729)	-0.7827 (0.9323)	International Asian	-0.0646*** (0.0224)	-0.0070 (0.0157)	0.0704 (0.0543)	0.0265 (0.0363)	-0.0334 (0.0393)	0.0069 (0.0132)
International Other	0.5034 (0.3172)	0.3781 (0.4131)	1.3523*** (0.5086)	-0.1061 (0.7483)	0.6239 (1.1070)	International Other	0.0385 (0.0243)	0.0143 (0.0157)	0.1013*** (0.0385)	-0.0057 (0.0406)	0.0266 (0.0471)	0.0238* (0.0132)
GRE Verbal Percentile	0.0155*** (0.0060)	0.0281* (0.0160)	0.0753** (0.0344)	-0.0092 (0.0154)	0.0607*** (0.0218)	GRE Verbal Percentile	0.0012*** (0.0005)	0.0011* (0.0006)	0.0056** (0.0026)	-0.0005 (0.0008)	0.0026*** (0.0010)	0.0015*** (0.0005)
GRE Quantitative Percentile	0.1547*** (0.0186)	0.0161* (0.0086)	0.0210* (0.0116)	0.0211 (0.0150)	-0.0138 (0.0159)	GRE Quantitative Percentile	0.0118*** (0.0014)	0.0006* (0.0003)	0.0016* (0.0009)	0.0011 (0.0008)	-0.0006 (0.0007)	0.0006** (0.0003)
GRE Analytic Percentile	0.0179*** (0.0042)	0.0199** (0.0094)	0.0030 (0.0134)	0.0175 (0.0127)	-0.0202 (0.0218)	GRE Analytic Percentile	0.0014*** (0.0003)	0.0008** (0.0004)	0.0002 (0.0010)	0.0009 (0.0007)	-0.0009 (0.0009)	0.0007*** (0.0003)
Undergraduate GPA	2.2212*** (0.7219)	0.6968 (0.5476)	1.3648** (0.5513)	1.2645 (0.9703)	1.7772* (0.9487)	Undergraduate GPA	0.1699*** (0.0551)	0.0264 (0.0208)	0.1022** (0.0423)	0.0684 (0.0522)	0.0757* (0.0418)	0.0488*** (0.0176)
TOEFL/IELTS	0.3878 (0.2883)	0.6456 (0.4708)	1.0963** (0.5526)	0.9655* (0.5456)	-0.4656 (0.6112)	TOEFL/IELTS	0.0297 (0.0219)	0.0244 (0.0179)	0.0821** (0.0417)	0.0523* (0.0301)	-0.0198 (0.0259)	0.0326*** (0.0123)
Elite U.S. University	0.8850*** (0.2960)	0.3943 (0.3135)	0.6541* (0.3802)	-0.6057 (0.4985)	1.4303** (0.6192)	Elite U.S. University	0.0677*** (0.0225)	0.0149 (0.0119)	0.0490* (0.0281)	-0.0328 (0.0264)	0.0610** (0.0271)	0.0233** (0.0098)
Elite U.S. Liberal Arts	1.1173*** (0.3813)	0.6276* (0.3754)	1.0698** (0.5399)	-0.6088 (1.4186)	2.3151** (1.0642)	Elite U.S. Liberal Arts College	0.0855*** (0.0288)	0.0238* (0.0143)	-0.0330 (0.0402)	0.0987** (0.0761)	0.0420*** (0.0480)	0.0420*** (0.0136)
Elite Foreign University	1.0061* (0.6060)	0.4759 (0.8224)	1.7715** (0.7938)	1.6195* (0.8587)	0.3889 (1.9889)	Elite Foreign University	0.0770* (0.0463)	0.0180 (0.0311)	0.1327*** (0.0591)	0.0877** (0.0481)	0.0166 (0.0850)	0.0479*** (0.0241)
Other Foreign University	0.5442 (0.5765)	0.5777 (0.7686)	1.3733** (0.6992)	0.4188 (0.7157)	-1.0737 (1.3016)	Other Foreign University	0.0416 (0.0441)	0.0219 (0.0291)	0.1028* (0.0527)	0.0227 (0.0391)	-0.0458 (0.0556)	0.0273 (0.0221)
Graduate Degree	0.0467 (0.2238)	0.0733 (0.2583)	0.3128 (0.3280)	0.4049 (0.4896)	0.4892 (0.6399)	Graduate Degree	0.0036 (0.0171)	0.0028 (0.0098)	0.0234 (0.0247)	0.0219 (0.0262)	0.0208 (0.0273)	0.0106 (0.0089)
Work Experience	0.3932** (0.1889)	0.4027 (0.2993)	0.4972 (0.3295)	0.1495 (0.3778)	1.8201** (0.8564)	Work Experience	0.0301** (0.0144)	0.0152 (0.0113)	0.0372 (0.0248)	0.0081 (0.0205)	0.0776** (0.0368)	0.0221** (0.0089)
#Professor Recommenders	0.0340 (0.1317)	-0.0179 (0.1339)	0.5511** (0.2554)	0.4781* (0.2662)	-0.0981 (0.2660)	#Professor Recommenders	0.0026 (0.0101)	-0.0007 (0.0051)	0.0413** (0.0190)	0.0259* (0.0142)	-0.0042 (0.0113)	0.0063 (0.0047)
#Prolific Recommenders	0.2635*** (0.0854)	0.3025** (0.1262)	0.1880 (0.1576)	0.3284** (0.1544)	0.5901* (0.3145)	#Prolific Recommenders	0.0202*** (0.0065)	0.0114** (0.0048)	0.0141 (0.0118)	0.0178** (0.0082)	0.0251* (0.0130)	0.0133*** (0.0038)
#Math Courses	0.1689*** (0.0596)					#Math Courses	0.0129*** (0.0045)					
Advanced Math	0.2319 (0.1790)					Advanced Math	0.0177 (0.0137)					
Missing GRE	0.1899 (0.7646)	-0.3259 (0.3100)	0.6282 (0.4275)	-0.0831 (0.3761)	-1.1514** (0.5702)	Missing GRE	0.0145 (0.0585)	-0.0123 (0.0118)	0.0470 (0.0319)	-0.0045 (0.0204)	-0.0491* (0.0252)	-0.0062 (0.0088)
Missing Undergraduate GPA	-0.4745 (0.4891)	-0.4917 (0.5440)	-1.5492** (0.6893)	-1.1097* (0.6481)	0.4690 (1.1831)	Missing Undergraduate GPA	-0.0363 (0.0375)	-0.0186 (0.0206)	-0.1160** (0.0524)	-0.0601* (0.0355)	0.0200 (0.0505)	-0.0291 (0.0179)
Missing TOEFL/IELTS	-0.3786 (0.2432)	0.6188 (0.5657)	0.6745 (0.5190)	-0.0440 (0.5099)	-1.7224* (0.9271)	Missing TOEFL/IELTS	-0.0290 (0.0188)	0.0234 (0.0215)	0.0505 (0.0388)	-0.0024 (0.0276)	-0.0734* (0.0417)	0.0108 (0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	Observations	2078	2231	690	737	486	4144
Pseudo- R^2	0.2677	0.1525	0.1982	0.1871	0.2475	Wald Statistic DF P-Value	115.8837	23	0.0000			
CV Pseudo- R^2	2.102	0.0720	0.0604	-0.0265	-0.2787	Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.						

Moreover, per Table 12 in Appendix B, several of the variables are correlated with one another. When variables are correlated, and lasso selects one but not the other, its decision may not be robust to sampling error. Therefore, in Table 13 of this appendix, I present elastic net results. Elastic net is a generalization of lasso, in which the estimator can select any combination of the L1 (lasso) or L2 (ridge) penalty. The closer α^* is to 0, the closer the penalty is to lasso, and the closer it is to 1, the closer the penalty is to ridge. Unlike lasso which removes some variables entirely, ridge only shrinks coefficients closer to 0. For all but economics, α^* is equal or close to 0. But for economics it equals 0.8, meaning that elastic net wants to keep most of the variables in the model. Finally, in Table 14 of the appendix, I present post-lasso regression results, in which I regress admission only on the predictors that lasso selects; these estimates are not shrunk.

Table 2: Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R ²	0.2297	0.0912	0.0592	0.0850	0.0537

Finally, in Tables 15–17 of Appendix B, I limit the sample to the 231 accepted economics applicants and regress a binary variable for matriculation on the same independent variables (the sample sizes for the other programs are too small for this analysis, so I stick to economics only). I find that variables that predict acceptance often indicate that the applicant is less likely to matriculate. Specifically, the effects of undergraduate GPA, work experience, and advanced math are significantly negative. In Section 4.2, I discuss in more detail the tradeoff the committee faces

between accepting applicants it deems likely to succeed vs. those it deems likely to matriculate.

Before proceeding to the Wald tests, I briefly address inference vs. prediction, as well as sample selection. Both are important when running regressions with success measures as dependent variables. Regarding inference vs. prediction, omitted independent variables make it challenging to obtain consistent estimates in a success regression. For example, I do not observe the effort that students put in after matriculating. Even effort, let alone any other variables about the applicants that I do not observe, can bias success parameter estimates upward if effort is correlated with the variables corresponding to those parameters and with success irrespective of those variables.

However, there are reasons for reassurance. First, in a later version of this paper, I will incorporate committee scores as an independent variable, which will proxy variables that the committee observes and I do not. I will also follow Bai et al. (2022) and include as a proxy of effort the number of “grindstone terms,” such as *depend** and *diligen**, in recommendation letters. In addition, in this paper, prediction matters as much as inference. Mullainathan and Spiess (2017) distinguish $\hat{\beta}$ (e.g. the effect of hiring an additional teacher) from $\hat{y}$ (e.g. which teacher to hire) problems. For graduate admissions, if GRE scores predict success, the committee should use them regardless of whether it knows the mechanism through which they affect success.

In addition, the determinants of success conditional on applying (see Krueger and Wu 2000, and Grove and Wu 2007) are different from those conditional on matriculating (see Athey et al. 2007). Bai et al. (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples, a theoretical finding consistent with the significance of the quantitative GRE being weaker in studies on matriculants only. This problem is akin to height not predicting success in a sample of professional basketball players. Like Bai et al. (2022), I collect placement outcomes for all applicants, not just those who attended the university. But there are other measures of success, such as course grades and passing comprehensive examinations, that are only available for matriculants, and that have been used in papers like Athey et al. (2007).

As such, for these measures, one can implement a Heckman (1979) correction. Notationally, let $y_n^* = x_n\beta + u_n$, $d_n = 1(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$. y is a measure of success, x are variables in applications, and d is matriculation. The first step is to run a probit regression of d on z and compute the inverse Mills ratio $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$. The second step is to run OLS or heckprobit of y on $(x, \hat{\lambda})$ to obtain $\hat{\beta}$. For a Heckman instrument, one needs a variable that affects d but not y . An example is a variable measuring the applicant’s enthusiasm about the program, such as whether they included a specific professor’s name in their application essay. If the correction works well for success measures for which I have data for all applicants and can therefore verify its effectiveness, then it may also work well for measures for which I do not. In Appendix C, I incorporate a Heckman correction into a structural model estimated on simulated data.

3.2 Wald Tests Across Programs

I now implement a Wald test of whether the determinants of acceptance are the same across programs. The test's setup is as follows. Because the samples across programs are different, I assume that $(Y^E \perp\!\!\!\perp Y^{NE})|X$; that is, conditional on the independent variables X , the dependent variable for economics Y^E is independent of its non-economics counterpart Y^{NE} . Also, suppose the dependent variable is binary; for example, accepted vs. rejected. As such, I parameterize the economics and non-economics outcome probabilities with the inverse logit functional form: $P(Y_n^E = 1|x_n^E; \theta^E) = \frac{\exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}{1 + \exp(x_{n,0}^E \theta_0^E + x_{n,1}^E \theta_1^E)}$ and $P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE}) = \frac{\exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}{1 + \exp(x_{n,0}^{NE} \theta_0^{NE} + x_{n,1}^{NE} \theta_1^{NE})}$.

Notably, I separate the intercept θ_0 from the slope θ_1 , since the goal is to test for equal slopes, not equal intercepts. That said, I can add slopes I do not test to θ_0 . Then the likelihood is:

$$\mathcal{L}_N^U(y|x; \theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E; \theta^E)^{Y_n^E} [1 - P(Y_n^E = 1|x_n^E; \theta^E)]^{1-Y_n^E} \prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})^{Y_n^{NE}} [1 - P(Y_n^{NE} = 1|x_n^{NE}; \theta^{NE})]^{1-Y_n^{NE}}. \quad (1)$$

Let $g(\hat{\theta}) = \frac{1}{N_E} \sum_{n=1}^{N_E} \hat{P}_n^E (1 - \hat{P}_n^E) \hat{\theta}_1^E - \frac{1}{N_{NE}} \sum_{n=1}^{N_{NE}} \hat{P}_n^{NE} (1 - \hat{P}_n^{NE}) \hat{\theta}_1^{NE}$. The Wald test statistic is therefore $W = g(\hat{\theta})' [\nabla_{\theta} g(\hat{\theta}) \hat{V}(\hat{\theta}) \nabla_{\theta} g(\hat{\theta})']^{-1} g(\hat{\theta}) \sim \chi_Q^2$, where Q is the number of restrictions; that is, the number of determinants I test at a time. Note that the null hypothesis is of equal AMEs, not equal logit coefficients, since only AMEs can be compared across samples (Mood 2010).

If the dependent variable were continuous (e.g. placement rank), I would parameterize its economics and non-economics distributions as: $f(y_n^E|x_n^E; \beta^E, \sigma_E^2) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2} (y_n^E - x_{n,0}^E \beta_0^E - x_{n,1}^E \beta_1^E)^2\right]$ and $f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2} (y_n^{NE} - x_{n,0}^{NE} \beta_0^{NE} - x_{n,1}^{NE} \beta_1^{NE})^2\right]$. Then:

$$\mathcal{L}_N^U(y|x; \beta, \sigma^2) = \prod_{n=1}^{N_E} f(y_n^E|x_n^E; \beta^E, \sigma_E^2) \prod_{n=1}^{N_{NE}} f(y_n^{NE}|x_n^{NE}; \beta^{NE}, \sigma_{NE}^2). \quad (2)$$

Since linear coefficients can be compared across samples, the test is simpler (and can also be done with a likelihood-ratio test). Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. Under homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$), the Wald statistic is $W = g(\hat{\beta}_1)' [\nabla_{\beta, \sigma^2} g(\hat{\beta}_1) \hat{V}(\hat{\beta}, \hat{\sigma}^2) \nabla_{\beta, \sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$.

Returning to Table 1b, the first column contains the economics acceptance AMEs, the sixth column contains the non-economics acceptance AMEs, and the Wald test resoundingly rejects the null hypothesis of equal AMEs with $p < 0.0001$. Therefore, the differences in AMEs for economics vs. non-economics are highly statistically significant. To ensure that this test does not over-reject equal AMEs in general, I simulate two datasets, which for simplicity are for only the economics program and two non-economics programs. In each dataset, I simulate acceptance and success dependent variables. In the first dataset, the means, standard deviations, and correlations (see Tables 18 and 19 in Appendix B) are similar to the actual data, meaning that the determinants

of acceptance and success are different across programs.⁵ In the second dataset, I modify the correlations, so the non-economics determinants are the same as those for economics.

In Table 20 of Appendix B, I present Wald test results on the simulated data. In the dataset where the data-generating parameters are such that the true AMEs of the determinants of acceptance and success are the same, the test does not reject the null; I obtain $p = 0.6843$ for the acceptance dependent variable and $p = 0.6993$ for the success dependent variable. However, in the dataset where the determinants are different, the test rejects with $p < 0.0001$. Therefore, on the simulated data, the Wald test works properly, rejecting resoundingly when it should and not rejecting when it should not. Because the simulated dataset has 5,000 observations, which is comparable to the number of observations in the actual data, the test is sufficiently powerful.

In addition to testing if the determinants of acceptance or success are the same across programs, could I test if the determinants of acceptance are the same as those of success within a program? For example, does the quantitative GRE have the same AME for admission as for success? Intuitively, provided the committee wants to select applicants based on a “success” model, predictors of admission only may be overvalued, and vice versa. Such a test would be fatally flawed, however. First, there may be unwanted rejections of the null. For example, the committee may value diversity if it does not hurt $P(\text{Success})$ too much, in which case it may favor demographics not strictly more likely to succeed. There may also be unwanted non-rejections. If there is discrimination in the job market, the committee may rationalize discriminating in the same way, which would make the determinants of admission equal to those of success.

But more importantly, this test would not account for the mechanics of the admissions process. As a simple example, suppose the following conditions hold: (1) Advanced math has a large positive effect on success, and coding skills have a smaller positive effect. (2) 25% of applicants have both skills and therefore the highest $P(\text{Success})$, 25% have math only (2nd-highest), 25% have coding only (3rd-highest), and 25% have neither (lowest). (3) This university accepts half the applicants, and the other half attend a lower-ranked school, which ensures that everyone’s outcomes are observable. (4) Unlike applicant quality, school quality does not affect outcomes.

Under these suppositions, an optimal committee will accept the applicants with math and ignore coding. As such, an acceptance regression will have a degenerate math AME of 1 and a coding AME of 0, while a success regression will have AMEs as follows: $1 > \text{math} > \text{coding} > 0$. Also, for the acceptance AMEs to equal the success AMEs, the committee would suboptimally have to accept applicants without math. Therefore, unlike with, for instance, wages equaling marginal revenue product in a competitive labor market which allows for cross-regression comparisons, the acceptance and success AMEs need not be equal for the committee to be optimal. In turn, to draw conclusions about optimality, the relationship between the AMEs must be interpreted through models of the committee’s selection problem, which are the basis for Section 4.

⁵For some programs, the initial job placement matters less career-wise than it does for economics, which is worth considering when interpreting Wald test results for success across programs.

That said, within the Wald test framework, I can answer additional questions, as long as those questions are not about optimality. For example, I can test if there are different determinants of different types of success. Specifically, applicants may need different skills at different stages of the program; Athey et al. (2007) find that the quantitative GRE matters for course grades, whereas research experience matters more for placements. Statistically testing this hypothesis may be useful for committees who must weigh the early vs. late stages when deciding which applicants to accept. I can also test if the determinants of admission or success are different for U.S. vs. international applicants (see Krueger and Wu 2000). International applicants may have different levels of preparation or research interests, and be more likely to complete the program for lack of outside options and return to their native country after graduation. Finally, I can test if the determinants have changed over time. Reasons for such changes may include Covid and other macro/geopolitical factors, and for economics in particular, a shift in the profession toward empirics, coding/AI-intensive research, and elite predocs and Federal Reserve RAships.

4 Structural Analysis

There are two main reasons to estimate structural models of the admissions committee’s decision problem. First, to make any claims about whether committees are optimizing, it is necessary to define what optimization is by explicitly modeling their selection problem. Second, structural models can uncover feasible counterfactual deviation decision rules to help committees admit stronger applicants, and the models can quantify any deviation gains from such rules.

The question then is how to set up the models. Bai et al. (2022) prove that if all the variables in an application file can be reduced to a single number (i.e. an expected success probability),⁶ then accepting the optimal subset of applicants is the same as using a cutoff rule. A cutoff rule problem is far more tractable than a portfolio choice problem because I do not have to consider all the subsets of the hundreds of the applicants in each year who can be accepted; I only have to rank the applicants and choose the ones above the cutoff. Because the committee may be more selective in some years than others, I allow the cutoff probability to vary across years.

After generalizing into a two-round dynamic framework Bai et al. (2022)’s proof that the committee’s portfolio choice problem reduces to a cutoff rule problem, I start by modeling the committee’s problem as follows. In Round 1, the committee observes only each applicant’s objective variables, such as GRE scores. Based on those variables, it can transition the applicant to Round 2, or it can issue an immediate rejection. If the applicant survives Round 1, the committee then takes the time to read the application file and access all variables, both objective and subjective (e.g. recommendation letters). Finally, it accepts or rejects the applicant.

Rather than setting up a two-round model, why do I not use regressions, lasso, or any other

⁶The assumption that committees select based on expected success may have become more realistic in light of the 2023 Supreme Court ruling against race-based affirmative action in *Students for Fair Admissions v. Harvard*.

econometric or machine learning method to rank the applicants based on their success probabilities conditional on all data and compare this one-round model’s ranking to the committee’s? The reason is that a one-round model is unfair to the committee because the committee does not have time to read everyone’s files (though in the future, perhaps committees can use textual processing algorithms or local LLMs to “read” files quickly and generate additional objective variables without too much effort). Moreover, two-round models are useful not just for graduate admissions but for any other setting with first-stage filtering, such as hiring, clinical trials for medicines, etc.

I start with two different two-round models: dynamic and static. In Round 2 of the dynamic model, the committee accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. In Round 1, it forms a conditional expectation about those who survive to Round 1, meaning that the Round 1 cutoff is the level of that applicant’s expected Round 2 value. As such, the Round 1 cutoff can take a value greater than one. In contrast, in the static model, the committee ranks the applicants based on their objective variables, reads the files of those whose expected success probabilities are above the first cutoff probability, re-ranks the surviving applicants based on all their data, and accepts those above the second cutoff probability. I refer to this model as static because each round is a separate static problem.

The dynamic model appears more sophisticated because in Round 1, it favors applicants whose objective variables predict good subjective variables. However, the static model also correctly favors those applicants. While the static Round 1 success probability parameter estimates are inconsistent because the subjective variables are omitted, the fitted values estimate the success probability averaged over the subjective variables given the objective ones. Therefore, the component of the subjective variables that is predictable from the objective variables is absorbed into the estimates. Only the myopic model, which is identical in setup to the dynamic model except that the parameters for the objective variables in the distribution of the subjective variables are restricted to zero, does not favor such applicants in Round 1. Later in this section, when I present maximum likelihood estimation results of these models, I include the Akaike Information Criterion $(AIC) = 2[K - \log(\hat{\mathcal{L}}^R)]$ to compare the models and determine that the extra complexity does not improve the dynamic model’s fit. K is the number of parameters, and $\hat{\mathcal{L}}^R$ is the restricted likelihood’s value; lower AIC values indicate a better fit. The static model also better describes what real-world committees do, as they do not solve conditional expectations in Round 1.

Still later in this section, I present dynamic and static models in which not every accepted applicant matriculates, and which are also derived from two-round portfolio choice models, albeit models in which the objective is to maximize the sum of the success probabilities of the matriculants rather than the acceptances. In Round 2, the committee still accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. No weight is placed on matriculation probabilities in Round 2 because while likely matriculants are more likely to produce a positive payoff, they are also more likely to take up cohort spots and so are more “expensive.” However, weight is placed on matriculation probabilities in Round 1 because if the committee chooses to read the files of only the applicants who are most likely to

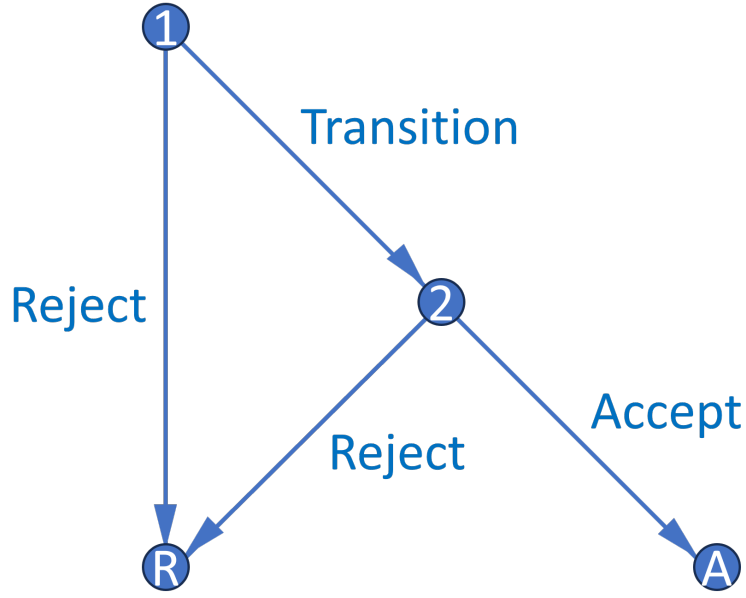


Figure 2: Decision Graph

succeed, it could find itself in a situation in Round 2 in which the only accepted applicants who matriculate are those with strong objective variables and weak subjective variables. This situation becomes increasingly likely the lower-ranked the committee's university is.

Finally, I present three-round models that include waitlists. The reason for these waitlist models is that the university in this study is highly-ranked enough that it does not consider matriculation probabilities in Round 1. However, unlike the very top universities, it still considers matriculation probabilities in terms of which applicants receive early vs. late acceptances. In Round 2 of these models, the committee does not accept or reject the surviving applicants. Rather, it early sends them an early acceptance, or it waitlists them. It then sees which of the early accepted applicants matriculate and which do not. Finally, in Round 3, it accepts or rejects the waitlisted applicants. The key dynamic is that the committee favors likely matriculants in Round 2, in addition to applicants with high expected success probabilities. It does so to avert a Round 3 situation in which it has barely filled up its class, applicants it could have accepted in Round 2 chose other schools, and now it has to accept weaker applicants. Finally, at the end of this section, I show how the models can be used to uncover counterfactual decision rules. I also show how much more effective these counterfactual rules are at admitting the best applicants.

4.1 Portfolio Choice to Cutoff Rule

Bai et al. (2022) prove that selecting the best N_A applicants out of N is equivalent to selecting

the N_A whose success probability exceeds a cutoff.⁷ Let x be a vector of objective variables (e.g. GRE scores), and z be a vector of subjective variables (e.g. variables from recommendation letters). Also, let d_n denote the accept or reject decision for the n th applicant, and y_n denote that applicant's outcome, where S = success and F = failure. The committee solves a constrained optimization problem, in which it maximizes the sum of the success probabilities of the accepted applicants under the constraint that it cannot accept more than N_A applicants. For simplicity, suppose there is one round in one year, and all accepted applicants matriculate:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A), \text{ such that } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad (3)$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(d_1, \dots, d_N; \lambda) &= \sum_{n=1}^N P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \left\{ N_A - \left[N - \sum_{n=1}^N \mathbb{1}(d_n = R) \right] \right\} \\ &= \sum_{n=1}^N [P(y_n = S | x_n, z_n) \mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)] + \lambda(N_A - N). \end{aligned} \quad (4)$$

Suppose without loss of generality that the applicants are sorted in decreasing order such that $P(y_1 = S | x_1, z_1) \geq \dots \geq P(y_N = S | x_N, z_N)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$. That is, in this setup, λ^* is set identified.⁸ Then the solution is to accept all applicants whose success probability exceeds λ^* and reject the rest, which means that λ^* is a cutoff probability:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda^*. \end{cases} \quad (5)$$

If $P(y_n = S | x_n, z_n) = \lambda^*$, then the committee is indifferent between acceptance and rejection, which means that in the event of a tie at the margin, the committee will randomly accept enough applicants to reach N_A acceptances. Also, while λ^* is set identified in this setup, I show in the proof of Theorem 2 in the next subsection how to point identify it with admissions data by modeling a committee that compares each applicant's success probability to a single-valued cutoff, and then by exploiting the observed committee's exact selectivity. Finally, given a large number of applicants, λ^* can be interpreted as the marginal accepted applicant's success probability. Therefore, I denote it going forward as $P(y^* = S)$, or the success probability of that applicant.

⁷To relax the assumption that the committee only cares about success, its objective can be generalized to maximizing the sum of a convex combination of success probabilities and other factors, such as diversity.

⁸If there were a continuum of applicants as in Bai et al. (2022)'s proof, the set's length would be 0, and so λ^* would be point identified. But all the models in this paper have N applicants, as there are in the data.

This version of the portfolio choice problem contains just one round. But the committee’s actual problem has two rounds: first-round filtering on x only, along with second-round final selections on x and z . In Appendix D.1, I generalize the one-round portfolio choice problem to a two-round portfolio choice problem, and show that it also reduces to a two-round cutoff rule problem. I do so for both the static and dynamic setups, and result is cutoff-rule problems that mirror the cutoff-rule problems in this section. The only difference is that in this section, I add type 1 extreme value preference shocks to the cutoff rule problems for estimation purposes.

That said, it is possible to add the preference shocks directly to the portfolio choice problem and carry them through the derivation, in which case the result is exactly the cutoff rule problems in this section. For notational simplicity, I do not do so throughout this paper, but I prove in Appendix G that for the one-round portfolio choice problem without matriculation (i.e. the portfolio choice problem I just presented), the preference shocks can be carried through the derivation:

Theorem 1. *For large N , the committee’s portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. See Appendix G.

4.1.1 Dynamic Model

Figure 2 portrays the dynamic model, whose notation is as follows. $r \in \{1, 2\}$ is the round, x is a vector of objective variables, z is a vector of subjective variables, t is the year, $\bar{k}$ is this university’s U.S. News program rank (k is the applicant’s observed future program rank), and ϵ_r is a type 1 extreme value shock with scale parameter σ . The shocks let the model fit the data, given that a real-world committee will not always accept the applicants with the highest success probabilities.

But what do the shocks represent? Mechanically, ϵ_r is the part of the committee’s decision that the observables $(x, z, t, \bar{k})$ do not explain. It is convenient to call ϵ_r a preference shock, but the label should not be read as pure idiosyncratic taste. The shock reflects my ignorance of what the committee conditioned on, not randomness in the committee’s behavior, and is therefore a residual defined relative to the observables. As such, anything the committee responded to that is absent from the observables is folded into ϵ_r , which means that what ϵ_r contains is an empirical matter. The more of the committee’s information I observe, the closer ϵ_r comes to genuine taste, and the more I do not observe, the more ϵ_r stands in for substantive variables that the committee acted on. This gap matters most when such variables also predict success, since the model then omits determinants of success. I discuss this situation and its implications in Section 5.

Starting in Round 2, the committee chooses $d_2 \in \{\text{Accept}, \text{Reject}\}$. If it accepts an applicant, it gets the applicant’s success probability $P(y = S|x, z, t, \bar{k})$; the probability of succeeding at this university is a function of applicant characteristics, the year, and this university’s quality. Whereas if the committee rejects the applicant, it gets the Round 2 cutoff probability $P(y_2^* = S|t)$, which is the success probability of t ’s marginal accepted applicant, whose success probability is conditional on $(x, z, t, \bar{k})$. There are various measures of success, from completing the program to

obtaining a job placement above a threshold ranking. The Round 2 value function is:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (6)$$

$$u_2(x, z, t, \bar{k}, d_2) = P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A) + P(y_2^* = S|t)\mathbb{1}(d_2 = R).$$

Meanwhile in Round 1, the committee chooses $d_1 \in \{\text{Transition}, \text{Reject}\}$. If it transitions an applicant to Round 2, it gets $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$, which is the expected value of Round 2, where z is integrated out conditional on (x, t) . However, if it rejects the applicant, it gets $L(y_1^* = S|t)$, or the expected Round 2 value of t 's marginal transitioned applicant. Because this value can exceed 1, it cannot be interpreted as a probability. The Round 1 value function is:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (7)$$

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x, t).$$

Each round's conditional choice probabilities (CCPs) then have the closed-form expression:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \tilde{d}_r)/\sigma)}. \quad (8)$$

To estimate this model, the first step is to set up the unrestricted likelihood function. I parameterize the unrestricted CCPs (P_{d_1}, P_{d_2}), success probability (P_y), and cutoffs ($L_{y_1^*}, P_{y_2^*}$) with the following flexible exponential and logistic specifications containing parameters θ :

$$P_{d_1} = P(d_1 = T|x, t; \theta_T) = \frac{\exp(f_T(x, t)\theta_T)}{1 + \exp(f_T(x, t)\theta_T)}, \quad P_{d_2} = P(d_2 = A|x, z, t; \theta_A) = \frac{\exp(f_A(x, z, t)\theta_A)}{1 + \exp(f_A(x, z, t)\theta_A)} \quad (9)$$

$$P_y = P(y = S|x, z, t, \bar{k}; \theta_S) = \frac{\exp(f_S(x, z, t, \bar{k})\theta_S)}{1 + \exp(f_S(x, z, t, \bar{k})\theta_S)} \quad (10)$$

$$L_{y_1^*} = P(y_1^* = S|t; \theta_{S_1^*}) = \exp(f_{S_1^*}(t)\theta_{S_1^*}), \quad P_{y_2^*} = P(y_2^* = S|t; \theta_{S_2^*}) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}. \quad (11)$$

Letting F_j be the conditional CDF for each z_j , I parameterize the integral in Equation (7) as:

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J|x, t). \quad (12)$$

Depending on if each z_j is continuous, binary, or multinomial, I use one of the following PDFs:

$$f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma_{F_{z|x,t,j}}^2) = \frac{1}{\sqrt{2\pi\sigma_{F_{z|x,t,j}}^2}} \exp \left[\frac{-1}{2\sigma_{F_{z|x,t,j}}^2} (z_j - g(x, t)\beta_{F_{z|x,t,j}})^2 \right], \quad P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}})}{1 + \exp(g(x, t)\theta_{F_{z|x,t,j}})}, \text{ or } P(z_j = (\ell \neq L)|x, t; \theta_{F_{z|x,t,j}}^\ell) = \frac{\exp(g(x, t)\theta_{F_{z|x,t,j}}^\ell)}{1 + \sum_{i=1}^{L-1} \exp(g(x, t)\theta_{F_{z|x,t,j}}^i)}, \text{ respectively.}$$

Let N_T be the number of N total applicants who survive to Round 2. Also, T = transition, A = accept, R = reject, S = success, and F = failure. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f(z_n|x_n, t_n; \theta_F) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N f(z_n|x_n, t_n; \theta_F) (1 - P_{d_{1,n}}(\theta_T)). \quad (13)$$

I estimate the restricted likelihood by substituting the restricted CCPs for the unrestricted ones. To save computation time, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R) . There is also a complication with the data, namely that *Committee Score* is unobservable for the $N - N_T$ of N applicants who were rejected in Round 1. The missing data pose a problem when estimating $f(z_n|x_n, t_n; \theta_F)$ in the likelihood because *Committee Score* is part of z . Because I cannot obtain *Committee Score* for everyone, I instead estimate the following Heckman-corrected likelihood:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \Phi(x_n, t_n; \theta_H) f(z_n|x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N (1 - \Phi(x_n, t_n; \theta_H)) f(z_n|x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F) (1 - P_{d_{1,n}}(\theta_T)). \quad (14)$$

The probit selection block is $\Phi(x_n, t_n; \theta_H)$, and the inverse Mills ratio is $\frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}$. Unfortunately, there are no available Heckman instruments that affect selection but not the distribution of z . Therefore, identification must come from the nonlinearity of the inverse Mills ratio.

After estimating the unrestricted likelihood, I can estimate the restricted likelihood $\mathcal{L}^R$ by substituting in the model's CCPs, which are functions of the success probability (θ_S) and structural $(\sigma, \theta_{S_1^*}, \theta_{S_2^*})$ parameters. Because I can observe whether the committee read each applicant's file and whether the committee accepted each applicant, I identify the cutoff parameters $(\theta_{S_1^*}, \theta_{S_2^*})$ for each year by how selective the committee is that year. Without loss of generality, every time $d_2 = A$, the term $\frac{\exp(P(y=S|x, z, t, \bar{k})/\sigma)}{\exp(P(y=S|x, z, t, \bar{k})/\sigma) + \exp(P(y_2^*=S|t)/\sigma)}$ enters $\mathcal{L}^R$. This term decreases in $P(y_2^*=S|t)$, so every time an applicant is accepted, it puts downward pressure on $P(y_2^*=S|t)$, considering the objective is to maximize the likelihood. Intuitively, the committee's selectivity identifies the cutoff probability. Also, if the committee engages in sorting, or selecting the applicants with the highest success probabilities $P(y = S|x, z, t, \bar{k})$, that improves the likelihood. Formally:

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A . Also, for any $\hat{P}_{y^*}$ and N_A , $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. See Appendix G.

When estimating this model, it is necessary to use public outcomes data to track success for all applicants, including those who did not matriculate at this university. Bai et al. (2022) prove that

success parameter estimates will be biased toward zero if the sample is restricted to matriculants only. The reason is that all matriculants must have been accepted, and accepted applicants tend to have high values of the determinants of success. For example, conditional on matriculating, the relationship between the quantitative GRE score and success is weakened because nearly all matriculants have very high scores (analogous to height not predicting success in the NBA).

In addition to tracking success for all applicants, it is also necessary to track which, if any, graduate program each applicant attended to avoid assuming that success depends only on applicant characteristics. For instance, a rejected applicant who succeeded at some other university might not have succeeded at this one, and so the committee may not have made a mistake by rejecting them. If a variable predicts success only through its effect on matriculation at a higher-ranked university, then this university should not use that variable in admissions. In estimation, I use each applicant’s observed future program rank k in the success probability block and this university’s fixed program rank in the restricted conditional choice probabilities.

I must also handle the problem of characteristics that the committee observes that I do not. For example, the committee may personally know a recommendation letter writer, or the letter may contain information that algorithms cannot detect, such as exactly how rigorous the applicant’s masters program was. Failing to address this problem may bias the success parameter estimates and make the model’s predictions worse than the committee’s. Therefore, including the committee score of each applicant in $\mathbf{z}$ to proxy such characteristics serves as an additional layer of robustness. Optimization may ultimately require merging human and data-driven approaches. While humans have access to more information and can use their intuition to evaluate less quantifiable characteristics, they may also fall prey to human biases that models do not.

Once I have estimated the unrestricted and restricted likelihoods, I perform the likelihood-ratio test $LR = 2[\log(\hat{\mathcal{L}}^U) - \log(\hat{\mathcal{L}}^R)] \sim \chi_Q^2$, where $Q = \dim(\theta^U) - \dim(\theta^R)$. The purpose of this test is to determine whether the model fits the data well. If it does, then $\hat{\mathcal{L}}^R$ will be close to $\hat{\mathcal{L}}^U$, and the test will not reject.⁹ With this test, I can determine whether the committee is optimizing in the same way the model says it should. If the test rejects, then the committee may be behaving suboptimally. However, the test can still reject even if the committee’s deviations from the model do not translate into a meaningfully weaker set of admitted applicants. Therefore, the more practical optimality test is the deviation gains test, which I present in Section 5.

In Appendix C, I present estimation results on simulated data for the dynamic model. First, in Tables 21 and 22 I provide summary statistics for two simulated datasets: one in which the committee is optimizing and one in which it is not. The data-generating parameters for the means, variances, and correlations across the independent variables, as well as the parameters governing the effect of the independent variables on acceptance on committee’s decisions, are calibrated

⁹Due to the σ and cutoff parameters, the restricted likelihood is not nested within the unrestricted one. While I could use Vuong (1989)’s non-nested specification test, I stick with the likelihood-ratio test, as Anderson, Rosen, Rust, and Wong (2025) do. The likelihood-ratio test is simpler, and there are relatively few non-nested parameters.

from the reduced-form results in Section 3. In the optimal dataset, the parameters governing the effect of the independent variables on success were chosen to be similar to the acceptance parameters. In contrast, the suboptimal committee discriminates against *Demographic 1*, undervalues *Advanced Math* (i.e. whether the applicant took advanced math, such as real analysis), and overvalues *Committee Score* (i.e. the committee reader’s score of the applicant).

In Tables 23 and 24, I present the unrestricted and restricted estimates, respectively. Because I have two z variables, namely *Advanced Math* and *Committee Score*, I have two PDFs to estimate. Committee score is a continuous z , and since I parameterize continuous z ’s with the normal distribution, I have an additional σ_F parameter that equals the square root of the distribution’s mean squared error. I also let *Committee Score* depend on *Advanced Math*, so I can estimate the joint distribution of the two z ’s without assuming the z ’s are independent. In general, I do not assume independent z ’s. I find that the likelihood-ratio test rejects the dynamic model even for the optimal committee. The reason for this rejection is that when I generated the simulated data, I had the committee optimize in a static manner. That is, like a typical real-world committee, I did not explicitly form an expectation of Round 2 by considering the distribution of z conditional on x and t . However, even though the data in the optimal dataset reject the dynamic model, I show in Section 5 that by following the dynamic model, the committee would admit a class nearly indistinguishable from the static model, which the data in the optimal dataset do not reject.

Finally, in Tables 25–26 of Appendix C, I present estimates from a myopic model. The myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t , dependence parameters among the components of z , and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. Because the myopic model is nested in the dynamic model, the dynamic model necessarily fits the data at least as well. This does not, however, imply that the dynamic committee admits a more successful cohort. The deviation gains metric I report, the mean predicted success of the acceptance or matriculation set, is a static quantity that the dynamic committee does not directly maximize, since its Round 1 rule also values the option of observing z before Round 2. Dynamic dominance on this metric is therefore not guaranteed, and the myopic model can slightly outperform the dynamic model.

4.1.2 Static Model

I now present the static model. Relative to the dynamic model, this model omits z in the Round 1 success probability specification, meaning that z ’s parameter estimates are inconsistent (though crucially, the fitted values are still correct). In addition, for any multiyear model, such as those in Appendix F, it is necessary for the Round 1 value function to be dynamically linked to the Round 2 value function, meaning that the static framework cannot work across years. However, this model has two advantages. First, it more accurately describes how real-world committees

behave. Second, the Round 1 cutoff can be interpreted as a probability and compared to the Round 2 cutoff probability. The static model's Round 2 value function is the same as the dynamic's:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (15)$$

$$u_2(x, z, t, \bar{k}, d_2) = P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A) + P(y_2^* = S|t)\mathbb{1}(d_2 = R).$$

But unlike in the dynamic model, if the committee transitions an applicant in Round 1, it gets $P(y = S|x, t, \bar{k})$, or the applicant's success probability unconditional on z , which approximates the expected Round 2 value in the dynamic model (see Appendix D.1) and may differ from $P(y = S|x, z, t, \bar{k})$. Meanwhile, if the committee rejects the applicant, it gets $P(y_1^* = S|t)$, or the success probability of t 's marginal transitioned applicant, whose success probability is conditional on x only. $P(y_1^* = S|t)$ can be less or greater than $P(y_2^* = S|t)$ depending on how much the committee values its time vs. admitting the best possible applicants. The Round 1 value function is:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (16)$$

$$u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R).$$

The restricted CCPs are the same as in the dynamic model:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \tilde{d}_r)/\sigma)}. \quad (17)$$

Rather than write out the specifications for the unrestricted CCPs, success probabilities, and cutoff probabilities, I reference Equations (9)–(11). That said, there are two differences from the dynamic framework. First, there are two estimated success probabilities (P_{y_1}, P_{y_2}), and second, the Round 1 cutoff is now a probability rather than a level. The unrestricted likelihood is then:

$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A) \mathbb{1}(d_{2,n}=A) (1 - P_{d_{2,n}}(\theta_A)) \mathbb{1}(d_{2,n}=R) \left[P_{y_{1,n}}(\theta_{S_1}) P_{y_{2,n}}(\theta_{S_2}) \right]^{\mathbb{1}(y_n=S)} \\ & \left[(1 - P_{y_{1,n}}(\theta_{S_1})) (1 - P_{y_{2,n}}(\theta_{S_2})) \right]^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N (1 - P_{d_{1,n}}(\theta_T)) P_{y_{1,n}}(\theta_{S_1}) \mathbb{1}(y_n=S) (1 - P_{y_{1,n}}(\theta_{S_1})) \mathbb{1}(y_n=F). \end{aligned} \quad (18)$$

I present results from the static model in the main body of the paper because the likelihood-ratio tests and cutoff probabilities are cleaner than in the dynamic model. In Table 3, I present the unrestricted parameter estimates; the simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success at the 1%, 5%, and 1% levels, respectively.¹⁰ Table 4 contains the restricted parameter estimates. As expected, the likelihood

¹⁰*Program Rank*'s sign is negative because the best rank is 1, and the worst rank is 150.

ratio test does not reject optimality for the optimal dataset, but it rejects optimality for the suboptimal dataset at the 1% level. Also as expected, the restricted AIC is better than the unrestricted AIC for the optimal dataset, but the unrestricted AIC is better for the suboptimal dataset.

Although the data are simulated, Table 4 contains an interesting result. There are two years in each of the simulated datasets: one in which the committee transitions 45–50% of the applicants to Round 2, and one in which it only transitions 25–30%. Essentially, in *Year Transition More*, the committee is more concerned with admitting a strong class than saving time and effort.¹¹ And in that year, as expected, the Round 1 cutoff probability is lower than the Round 2 cutoff probability. That is, it is easier to move on to Round 2 than it ultimately is to get into the program. However, in the other year, which is represented by the intercept term, the Round 1 cutoff is greater than the Round 2 cutoff, meaning the $P(\text{Success}|x, t, \bar{k})$ of the marginal applicant rejected in Round 1 is greater than the $P(\text{Success}|x, z, t, \bar{k})$ of the marginal applicant accepted into the program.

How can this be? Sometimes when a committee transitions a seemingly strong applicant to Round 2, the applicant’s z reveals they are weaker than previously thought, e.g. a bad recommendation letter. If the committee transitions too few applicants to Round 2, it can end up in an unfortunate situation where it threw out so many files that it now has to accept those weaker applicants. As a simple example, suppose there are three applicants: A, B, and C. The committee has time to read two applicants’ files and space to accept one applicant. Suppose that overall, A is the best applicant and C is the worst, but A is the worst in objective variables. Then the committee will reject A in Round 1 and accept B, who is actually worse than A. In the simulated data, the minimum percentage of applications the committee must read in Round 2 to avoid this situation is between 25–30%, where the situation occurs, and 45–50%, where it does not. In a later version of the paper, I will determine whether this situation occurs in any of the actual years, and if so, the minimum percentage of files the committee must read to avoid this situation.

4.2 Portfolio Choice with Matriculation

In the portfolio choice problem in Section 4.1, as well as in Bai et al. (2022), there is an assumption that every accepted applicant will matriculate. But in practice, many accepted applicants do not matriculate and instead attend another university. For committees at the very top universities, matriculation probabilities may be less of a concern, as accepted applicants who do not matriculate likely do so for idiosyncratic reasons (e.g. choosing Harvard over MIT). But at the vast majority of universities, the accepted applicants with the highest success probabilities will often have relatively low matriculation probabilities, as the lower-ranked universities were likely safety options from the outset. The question then is whether the committee should select applicants based on matriculation probabilities as well as success probabilities, or whether it should still accept the applicants with the highest success probabilities irrespective of matriculation.

¹¹In *Year Transition More*, the applicants also have higher matriculation probabilities on average, so this variable is called *Year Transition/Matriculate More* in the matriculation and waitlist models.

Table 3: Unrestricted Static Estimates

(a) Optimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5886*** (1.0232)	-2.2559*** (0.1454)	-13.1494*** (2.0072)	-2.3728*** (0.2886)	-8.1224*** (1.1371)	-1.1660*** (0.1344)	-9.5306*** (2.0699)	-1.4839*** (0.2689)
GRE Quantitative Percentile	0.1235*** (0.0118)	0.0240*** (0.0018)	0.1371*** (0.0223)	0.0247*** (0.0033)	0.1078*** (0.0133)	0.0155*** (0.0015)	0.1158*** (0.0227)	0.0180*** (0.0029)
Gender	0.2262 (0.1464)	0.0440 (0.0283)	0.1290 (0.2502)	0.0233 (0.0451)	0.1396 (0.1738)	0.0200 (0.0249)	0.1262 (0.2698)	0.0197 (0.0419)
Demographic 1	-0.2995 (0.1988)	-0.0583 (0.0386)	-0.2639 (0.3743)	-0.0476 (0.0672)	-0.2529 (0.2358)	-0.0363 (0.0338)	-0.4577 (0.3931)	-0.0713 (0.0608)
Demographic 2	0.1552 (0.2215)	0.0302 (0.0431)	0.1711 (0.3843)	0.0309 (0.0693)	-0.1180 (0.2580)	-0.0169 (0.0371)	-0.2318 (0.4320)	-0.0361 (0.0672)
Advanced Math			0.7733*** (0.2534)	0.1396*** (0.0443)			0.6278** (0.2748)	0.0978** (0.0417)
Committee Score			0.0781 (0.0602)	0.0141 (0.0108)			0.0462 (0.0655)	0.0072 (0.0102)
Year Transition More	0.9837*** (0.1472)	0.1915*** (0.0263)	-0.9252*** (0.2512)	-0.1670*** (0.0427)	-0.0592 (0.1689)	-0.0085 (0.0242)	-0.0239 (0.2762)	-0.0037 (0.0430)
Program Rank					-0.0310*** (0.0019)	-0.0045*** (0.0002)	-0.0282*** (0.0027)	-0.0044*** (0.0003)
Observations	1000							
LL Parameters AIC	-1375.9	30	2811.7					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2297*** (1.0144)	-2.1801*** (0.1474)	-6.7297*** (1.7376)	-1.3296*** (0.3202)	-8.2833*** (1.1393)	-1.1914*** (0.1341)	-9.6070*** (1.9611)	-1.4986*** (0.2508)
GRE Quantitative Percentile	0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1097*** (0.0133)	0.0158*** (0.0015)	0.1163*** (0.0216)	0.0181*** (0.0027)
Gender	0.2576* (0.1469)	0.0500* (0.0283)	0.0097 (0.2368)	0.0019 (0.0468)	0.1355 (0.1741)	0.0195 (0.0250)	0.0584 (0.2669)	0.0091 (0.0416)
Demographic 1	-0.8291*** (0.1966)	-0.1610*** (0.0371)	-0.3339 (0.3394)	-0.0660 (0.0669)	-0.2777 (0.2344)	-0.0399 (0.0337)	-0.5157 (0.3648)	-0.0804 (0.0563)
Demographic 2	0.0798 (0.2135)	0.0155 (0.0414)	0.1762 (0.3346)	0.0348 (0.0659)	-0.1808 (0.2566)	-0.0260 (0.0369)	-0.4742 (0.3857)	-0.0740 (0.0598)
Advanced Math			0.3581 (0.2375)	0.0708 (0.0465)			0.5369** (0.2704)	0.0837** (0.0416)
Committee Score			0.2962*** (0.0592)	0.0585*** (0.0106)			0.0616 (0.0643)	0.0096 (0.0100)
Year Transition More	1.1109*** (0.1479)	0.2157*** (0.0257)	-0.7185*** (0.2397)	-0.1419*** (0.0455)	-0.0766 (0.1688)	-0.0110 (0.0243)	0.1314 (0.2795)	0.0205 (0.0434)
Program Rank					-0.0304*** (0.0019)	-0.0044*** (0.0002)	-0.0279*** (0.0028)	-0.0044*** (0.0003)
Observations	1000							
LL Parameters AIC	-1404.1	30	2868.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 4: Restricted Static Estimates

(a) Optimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.6884*** (0.1830)	66.56% (58.17%, 74.02%)	0.1530 (0.2411)	53.82% (42.08%, 65.15%)	-8.0150*** (1.1629)	-9.7263*** (1.7366)	
GRE Quantitative Percentile					0.1057*** (0.0134)	0.1158*** (0.0193)	
Gender					0.1987* (0.1015)	0.1142 (0.1673)	
Demographic 1					-0.2471* (0.1346)	-0.3338 (0.2403)	
Demographic 2					0.0406 (0.1504)	-0.0047 (0.2423)	
Advanced Math						0.6315*** (0.1766)	
Committee Score						0.0546 (0.0421)	
Year Transition More	-0.8466*** (0.2184)	46.05% (39.60%, 52.64%)	0.7965** (0.3769)	72.10% (58.90%, 82.34%)	-0.0628 (0.1667)	0.0285 (0.2701)	
Program Rank					-0.0310*** (0.0019)	-0.0283*** (0.0027)	
Sigma							0.1913*** (0.0252)
Observations	1000						
LL Parameters AIC	-1376.5	21	2795.0				
LR Statistic DF P-Value	1.3137	9	0.9983				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.7340*** (0.2062)	67.57% (58.17%, 75.73%)	0.1682 (0.2593)	54.20% (41.58%, 66.29%)	-8.0582*** (1.2803)	-6.3578*** (1.7689)	
GRE Quantitative Percentile					0.1079*** (0.0148)	0.0727*** (0.0189)	
Gender					0.2185** (0.1042)	0.0409 (0.1610)	
Demographic 1					-0.5505*** (0.1395)	-0.3739 (0.2380)	
Demographic 2					-0.0151 (0.1505)	-0.0921 (0.2356)	
Advanced Math						0.3897** (0.1686)	
Committee Score						0.1754*** (0.0378)	
Year Transition More	-0.9965*** (0.2396)	43.47% (37.12%, 50.05%)	0.6110 (0.3768)	68.55% (55.51%, 79.20%)	-0.0772 (0.1671)	0.0376 (0.2807)	
Program Rank					-0.0305*** (0.0019)	-0.0281*** (0.0027)	
Sigma							0.2001*** (0.0325)
Observations	1000						
LL Parameters AIC	-1417.0	21	2875.9				
LR Statistic DF P-Value	25.7951	9	0.0022				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

As such, I generalize the portfolio choice problem to incorporate matriculation probabilities. The committee now accepts the subset of applicants with the highest sum of success times matriculation probabilities subject to a capacity constraint on the expected number of matriculants. For each applicant n , the committee makes an acceptance or rejection decision given:

$$\begin{aligned} & \max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A), \\ & \text{such that: } \sum_{n=1}^N P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) \leq N_M, \end{aligned} \quad (19)$$

where each applicant's matriculation decision m is to either matriculate or decline, $P(m = M|x, z)$ is each applicant matriculation probability, and N_M is the maximum number of matriculants. Due to the linearity of expectations, the objective function is equivalent to the expected sum of the success probabilities of the accepted applicants who matriculate. Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(d_1, \dots, d_N; \lambda) &= \sum_{n=1}^N P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) \\ &+ \lambda \left\{ N_M - \left[\sum_{n=1}^N P(m_n = M|x_n, z_n) - \sum_{n=1}^N P(m_n = M|x_n, z_n)\mathbb{1}(d_n = R) \right] \right\} \\ &= \sum_{n=1}^N [P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)]P(m_n = M|x_n, z_n) \\ &+ \lambda \left[N_M - \sum_{n=1}^N P(m_n = M|x_n, z_n) \right]. \end{aligned} \quad (20)$$

From the Lagrangian, it is clear that if there is a cutoff rule solution, the solution is on the success probabilities alone. While applicants with higher matriculation probabilities are more valuable in terms of the objective, they are also more expensive in terms of the capacity constraint on the number of matriculants. Intuitively, if the committee were to prioritize applicants with higher matriculation probabilities, it would not be able to give as many acceptances.

Moreover, the only way to get a cutoff rule solution that incorporates matriculation probabilities would be to have the constraint be on the number of acceptances. That is, the constraint would be $\sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A$, the Lagrangian would equal $\sum_{n=1}^N [P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)] + \lambda(N_A - N)$, the applicants would be ranked relative based on $P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)$, and likely matriculants would receive more acceptances.

Crucially, however, because not every accepted applicant will matriculate, this exact problem does not have a cutoff rule solution. For example, suppose $N = 2$, $N_M = 1$, $P(y_1 = S|w_1) = 0.9$, $P(m_1 = M|w_1) = 0.5$, $P(y_2 = S|w_2) = 0.89$, and $P(m_2 = M|w_2) = 0.89$. Since $0.5 + 0.89 > 1$, the

committee can accept only one applicant. The cutoff rule solution would be to accept Applicant 1, but Applicant 2 would yield a higher value of the objective since $0.89 \times 0.89 > 0.9 \times 0.5$. But fortunately, any exceptions to the cutoff rule solution disappear once N becomes sufficiently large; that is, when there is a continuum of applicants as in Bai et al. (2022)’s setup.

Theorem 3. *For large N , the committee’s portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. See Appendix G.

Given this result, it appears that committees should not factor in matriculation probabilities when making admissions decisions. In a one-round model, that is the case. However, I show in Appendix D.2 that in a two-round portfolio choice model with matriculation, the cutoff rule is such that the committee should consider matriculation probabilities in Round 1. This rule holds in particular for lower-ranked programs. For example, if a lower-ranked program’s committee chooses to read only the files of applicants with perfect GRE scores, then it may have trouble filling its cohort in Round 2, considering that the best applicants are unlikely to matriculate. And even if the committee can fill its cohort, that cohort is likely to be filled with applicants who have weak subjective variables. Whereas if the committee had considered matriculation probabilities in Round 1, it may have transitioned applicants with lower GRE scores but strong subjective variables who will actually matriculate. As such, I present two-round matriculation-dynamic and matriculation-static models in which matriculation probabilities matter in Round 1.

4.2.1 Matriculation Models

Because matriculation probabilities do not matter in Round 2, the matriculation-dynamic and static Round 2 value functions are the same as the previous section’s Round 2 value functions:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (21)$$

$$u_2(x, z, t, \bar{k}, d_2) = P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A) + P(y_2^* = S | t) \mathbb{1}(d_2 = R).$$

However, the Round 1 value function contains matriculation probabilities:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (22)$$

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T) + L(y_1^* = S, m_1^* = M | t) \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{[P(y=S|x,\tilde{z},t,\bar{k})-P(y_2^*=S|t)]/\sigma} + 1 \right) P(m = M | x, \tilde{z}, t) dF(\tilde{z} | x, t).$$

The derivation of the Round 1 value function is in Appendix D.2. The object I denote by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2) | x, t, \bar{k}]$ is the Round 1 transition payoff, as opposed to the literal conditional expectation of v_2 . As such, the resulting social surplus function contains $e^{[P(y=S|x,\tilde{z},t,\bar{k})-P(y_2^*=S|t)]/\sigma} + 1$ instead of the usual $e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma}$. Due to the matriculation term $P(m = M | x, \tilde{z}, t)$,

the (incorrect) latter expression does not result in the same ordering of applicants as the (correct) former expression. In contrast, the matriculation-static Round 1 value function is:

$$v_1(x, t, \bar{k}, \varepsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \varepsilon_1(d_1), \text{ such that:} \quad (23)$$

$$u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})P(m = M|x, t)\mathbb{1}(d_1 = T) + P(y_1^* = S, m_1^* = M|t)\mathbb{1}(d_1 = R).$$

The restricted CCPs are the same as in the dynamic and static models without matriculation:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \tilde{d}_r)/\sigma)}. \quad (24)$$

Rather than write out the specifications for the unrestricted CCPs, success probability, and cutoff probabilities, I reference Equations (9)–(11). But the specification for the matriculation probability is: $P_m = P(m = M|x, z, t; \theta_M) = \frac{\exp(f_M(x, z, t)\theta_M)}{1 + \exp(f_M(x, z, t)\theta_M)}$. The unrestricted likelihood is then:

$$\begin{aligned} \mathcal{L}_N^U(\theta) = & \prod_{n=1}^{N_T} f(z_n|x_n, t_n; \theta_F) P_{d_{1,n}}(\theta_T) \left[P_{d_{2,n}}(\theta_A) P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)} (1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \right]^{\mathbb{1}(d_{2,n}=A)} \\ & (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N f(z_n|x_n, t_n; \theta_F) (1 - P_{d_{1,n}}(\theta_T)). \end{aligned} \quad (25)$$

As with the restricted dynamic likelihood, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R) for the restricted matriculation-dynamic likelihood. Also, a Heckman correction is necessary for $f_{z|x,t}$ because *Committee Score* is missing for the applicants rejected in Round 1.

The matriculation-static estimates are in Tables 5 and 6. As with the static estimates without matriculation, the simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success. The likelihood-ratio test rejects optimality at the 1% level only for the suboptimal dataset, though it rejects optimality at the 5% level for the optimal dataset. The reason is that $P(y = S|x, t, \bar{k})P(m = M|x, t)$ is not a perfect approximation for the committee's Round 1 behavior, as like a typical real-world committee, the simulated committee puts more emphasis on success probabilities than matriculation probabilities. Still, the matriculation-static model's likelihood-ratio statistics are lower than the matriculation-dynamic model's. The matriculation-dynamic estimates are in Tables 27 and 28 of Appendix C. Also, while the Round 1 cutoffs in Table 6 are probabilities, they are not comparable to the Round 2 cutoff probabilities because they are with respect to success times matriculation probabilities.

However, discussions with members of this university's committee indicate that the committee does not consider matriculation probabilities in Round 1 because this university is ranked highly enough that in general, it can fill its cohort with applicants who have strong objective and subjective variables. That is not to say though that this university does not use matriculation probabilities when making admissions decisions; it would ignore matriculation probabilities entirely only in a world of perfect information about each applicant's intention. Suppose, for example,

that every time the committee accepts an applicant, it immediately hears back from that applicant whether the applicant will matriculate. Then in Round 2, the committee would start with the applicant with the highest success probability, immediately hear back, proceed to the applicant with the second-highest success probability, and continue until the cohort is filled.

But in the real world, committees do not immediately hear back from applicants. Therefore, if a committee does not consider matriculation probabilities when giving early non-waitlisted acceptances, by the time it hears back from the accepted applicants (most of which will not matriculate since success and matriculation probabilities are negatively correlated), relatively good applicants who would have matriculated will have moved on to other universities. As a result, the committee may have to settle for weaker applicants than if it had considered matriculation probabilities. This dynamic motivates the models of the waitlist process in this section.¹² Because this dynamic is beyond the scope of the portfolio choice problem to cutoff rule framework, I do not write a three-round portfolio choice problem with this dynamic. Rather, the waitlist models extend the existing Bellman equations to three rounds and incorporate the dynamic by endogenizing the cutoff probabilities to make them a function of matriculation probabilities.

4.2.2 Waitlist Models

An admissions committee does not send out all its acceptances at once, but rather in batches. It starts by accepting its most preferred applicants, and then it puts the others who survived Round 1 on a waitlist. Then provided not all its early acceptances matriculate, it accepts applicants off the waitlist and ultimately moves down the waitlist until it has filled its cohort. I model this process as concisely as possible by increasing the number of rounds to three. As shown in Figure 3, the committee observes only the objective variables in Round 1 and either transitions the applicant to Round 2 or rejects them out of courtesy. But in Round 2, instead of accepting or rejecting the survivors based on the objective and subjective variables, it accepts or waitlists them.

Then in Round 3, it accepts or rejects the waitlisted applicants. Between Rounds 2 and 3, it observes how many of the early acceptances matriculate and makes its Round 3 decisions accordingly. If more early acceptances matriculate, it accepts fewer applicants off the waitlist because there are fewer openings left to fill. Intuitively, this situation is better than the other way around. More early acceptances matriculating means the committee gets more of its top choices. In contrast, if the committee has to accept many applicants off the waitlist, it may have to lower its standards because applicants it could have gotten with early offers may have already committed to other schools. Therefore, it should err on the side of giving Round 2 offers to applicants who are relatively likely to matriculate. That said, it would be incorrect to have the committee value applicants by a combination of their success and matriculation probabilities, as that would be the

¹²In addition, the committee can adjust the number of late acceptances it gives based on how many early matriculants it receives. Doing so helps it attain the target number of matriculants not only in expectation but in reality.

Table 5: Unrestricted Matriculation-Static Estimates

(a) Optimal Committee

	Transition		Accept		Success 1		Success 2		Matriculate 1		Matriculate 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5882*** (1.0232)	-2.2558*** (0.1454)	-13.1577*** (2.0081)	-2.3739*** (0.2886)	-8.1342*** (1.1379)	-1.1674*** (0.1344)	-9.5637*** (2.0728)	-1.4883*** (0.2688)	-9.1629** (4.3152)	-1.4870** (0.6431)	-4.5946 (4.6472)	-0.6112 (0.6186)
GRE Quantitative Percentile	0.1236*** (0.0118)	0.0241*** (0.0018)	0.1372*** (0.0223)	0.0248*** (0.0033)	0.1079*** (0.0133)	0.0155*** (0.0015)	0.1162*** (0.0227)	0.0181*** (0.0029)	0.0766* (0.0463)	0.0124* (0.0071)	0.0592 (0.0459)	0.0079 (0.0061)
Gender	0.2266 (0.1464)	0.0441 (0.0283)	0.1289 (0.2502)	0.0233 (0.0451)	0.1399 (0.1739)	0.0201 (0.0249)	0.1264 (0.2700)	0.0197 (0.0419)	0.4989 (0.4398)	0.0810 (0.0703)	0.4882 (0.5063)	0.0649 (0.0657)
Demographic 1	-0.3013 (0.1988)	-0.0587 (0.0386)	-0.2639 (0.3744)	-0.0476 (0.0672)	-0.2532 (0.2359)	-0.0363 (0.0338)	-0.4607 (0.3933)	-0.0717 (0.0608)	0.3408 (0.5565)	0.0553 (0.0907)	0.3585 (0.7459)	0.0477 (0.1001)
Demographic 2	0.1538 (0.2215)	0.0299 (0.0431)	0.1709 (0.3844)	0.0308 (0.0693)	-0.1179 (0.2580)	-0.0169 (0.0371)	-0.2340 (0.4323)	-0.0364 (0.0672)	-0.1604 (0.6964)	-0.0260 (0.1119)	0.1832 (0.7962)	0.0244 (0.1069)
Advanced Math			0.7737*** (0.2535)	0.1396*** (0.0443)			0.6276** (0.2749)	0.0977** (0.0417)			0.8105 (0.5772)	0.1078 (0.0733)
Committee Score			0.0783 (0.0602)	0.0141 (0.0108)			0.0465 (0.0655)	0.0072 (0.0102)			-0.7403*** (0.2065)	-0.0985*** (0.0235)
Year Transition/Matriculate More	0.9840*** (0.1472)	0.1915*** (0.0263)	-0.9251*** (0.2513)	-0.1669*** (0.0427)	-0.0587 (0.1690)	-0.0084 (0.0242)	-0.0234 (0.2763)	-0.0036 (0.0430)	2.8450*** (0.5222)	0.4617*** (0.0280)	3.3277*** (0.5672)	0.4427*** (0.0331)
Program Rank					-0.0310*** (0.0019)	-0.0045*** (0.0002)	-0.0282*** (0.0027)	-0.0044*** (0.0003)				
Observations	1000											
LL Parameters AIC	-1491.1	44	3070.2									

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Transition		Accept		Success 1		Success 2		Matriculate 1		Matriculate 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2298*** (1.0144)	-2.1801*** (0.1474)	-6.7300*** (1.7376)	-1.3296*** (0.3202)	-8.2841*** (1.1393)	-1.1915*** (0.1341)	-9.6026*** (1.9608)	-1.4981*** (0.2508)	-6.1997* (3.2762)	-0.9843* (0.5037)	-1.0401 (3.6083)	-0.1351 (0.4721)
GRE Quantitative Percentile	0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1097*** (0.0133)	0.0158*** (0.0015)	0.1162*** (0.0216)	0.0181*** (0.0027)	0.0419 (0.0375)	0.0066 (0.0059)	0.0129 (0.0391)	0.0017 (0.0051)
Gender	0.2575* (0.1469)	0.0500* (0.0283)	0.0098 (0.2368)	0.0019 (0.0468)	0.1355 (0.1741)	0.0195 (0.0250)	0.0584 (0.2669)	0.0091 (0.0416)	0.4469 (0.4602)	0.0710 (0.0717)	0.1298 (0.4717)	0.0169 (0.0608)
Demographic 1	-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3340 (0.3394)	-0.0660 (0.0669)	-0.2777 (0.2344)	-0.0399 (0.0337)	-0.5156 (0.3648)	-0.0804 (0.0563)	0.2529 (0.5465)	0.0402 (0.0870)	0.6328 (0.6281)	0.0822 (0.0823)
Demographic 2	0.0797 (0.2135)	0.0155 (0.0414)	0.1764 (0.3346)	0.0348 (0.0659)	-0.1809 (0.2566)	-0.0260 (0.0369)	-0.4743 (0.3856)	-0.0740 (0.0598)	-0.6596 (0.6474)	-0.1047 (0.0966)	-0.2222 (0.6864)	-0.0289 (0.0880)
Advanced Math			0.3580 (0.2375)	0.0707 (0.0465)			0.5366** (0.2704)	0.0837** (0.0416)			0.5876 (0.5321)	0.0763 (0.0674)
Committee Score			0.2962*** (0.0592)	0.0585*** (0.0106)			0.0616 (0.0643)	0.0096 (0.0100)			-0.6092*** (0.1637)	-0.0791*** (0.0176)
Year Transition/Matriculate More	1.1108*** (0.1479)	0.2156*** (0.0257)	-0.7185*** (0.2397)	-0.1420*** (0.0455)	-0.0766 (0.1688)	-0.0110 (0.0243)	0.1315 (0.2794)	0.0205 (0.0434)	2.8800*** (0.5694)	0.4572*** (0.0414)	3.4846*** (0.6046)	0.4526*** (0.0349)
Program Rank					-0.0304*** (0.0019)	-0.0044*** (0.0002)	-0.0279*** (0.0028)	-0.0044*** (0.0003)				
Observations	1000											
LL Parameters AIC	-1521.6	44	3131.1									

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 6: Restricted Matriculation-Static Estimates

(a) Optimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Matriculate 1	Matriculate 2	Sigma
Intercept	-1.0201*** (0.1697)	26.50% (20.54%, 33.46%)	0.1479 (0.2334)	53.69% (42.32%, 64.69%)	-8.0229*** (1.2042)	-8.3237*** (1.7684)	-8.2894*** (2.5032)	-4.6066 (4.6480)	
GRE Quantitative Percentile					0.1071*** (0.0140)	0.1006*** (0.0194)	0.0788*** (0.0269)	0.0593 (0.0459)	
Gender					0.1039 (0.1522)	0.1026 (0.1470)	0.3893* (0.2210)	0.4885 (0.5064)	
Demographic 1					-0.2437 (0.2094)	-0.2769 (0.2145)	-0.0899 (0.3273)	0.3587 (0.7459)	
Demographic 2					-0.0480 (0.2258)	0.0110 (0.2131)	0.0296 (0.3847)	0.1834 (0.7962)	
Advanced Math						0.5492*** (0.1679)		0.8106 (0.5773)	
Committee Score						0.0492 (0.0368)		-0.7402*** (0.2064)	
Year Transition/Matriculate More	0.1323 (0.2222)	29.16% (21.96%, 37.57%)	0.6306* (0.3547)	68.54% (55.89%, 78.93%)	-0.1369 (0.1686)	0.0137 (0.2730)	1.8740*** (0.4069)	3.3280*** (0.5674)	
Program Rank					-0.0311*** (0.0019)	-0.0280*** (0.0026)			
Sigma									0.1554*** (0.0253)
Observations	1000								
LL Parameters AIC	-1501.0	35	3072.1						
LR Statistic DF P-Value	19.8266	9	0.0190						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Matriculate 1	Matriculate 2	Sigma
Intercept	-1.1917*** (0.2112)	23.30% (16.72%, 31.48%)	0.1231 (0.2396)	53.07% (41.42%, 64.40%)	-8.0768*** (1.2382)	-3.7273** (1.6988)	-6.1199*** (2.2306)	-1.0400 (3.6084)	
GRE Quantitative Percentile					0.1084*** (0.0144)	0.0455*** (0.0175)	0.0569** (0.0248)	0.0129 (0.0391)	
Gender					0.1111 (0.1581)	0.0275 (0.1164)	0.3411 (0.2134)	0.1298 (0.4717)	
Demographic 1					-0.3744* (0.2131)	-0.2437 (0.1781)	-0.5399* (0.2762)	0.6329 (0.6281)	
Demographic 2					-0.0814 (0.2176)	-0.0173 (0.1697)	-0.0874 (0.3016)	-0.2225 (0.6864)	
Advanced Math						0.2559* (0.1400)		0.5874 (0.5320)	
Committee Score						0.1375*** (0.0346)		-0.6092*** (0.1637)	
Year Transition/Matriculate More	-0.0312 (0.2089)	22.74% (16.19%, 30.97%)	0.3687 (0.3334)	62.05% (50.06%, 72.73%)	-0.1520 (0.1700)	0.0048 (0.2775)	1.4674*** (0.4102)	3.4846*** (0.6047)	
Program Rank					-0.0305*** (0.0019)	-0.0281*** (0.0026)			
Sigma									0.1279*** (0.0330)
Observations	1000								
LL Parameters AIC	-1547.8	35	3165.6						
LR Statistic DF P-Value	52.5157	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

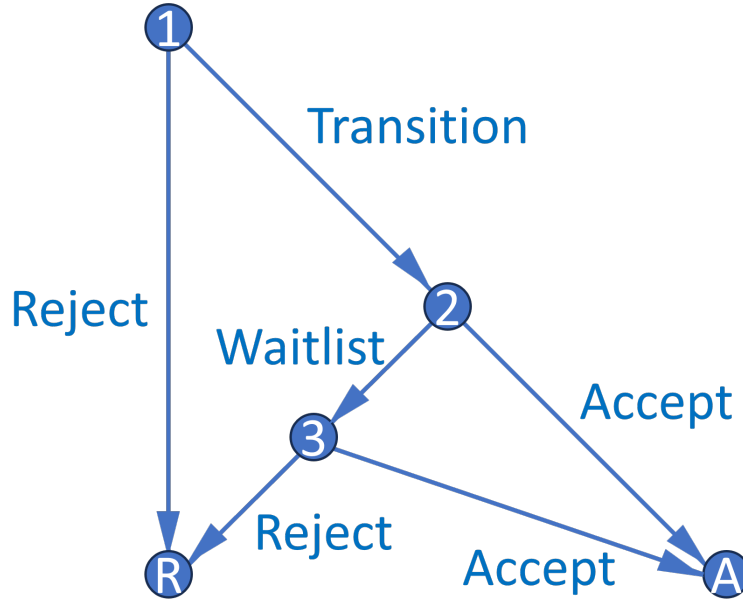


Figure 3: Waitlist Decision Graph

cutoff rule solution to a portfolio choice problem with a constraint on the number of acceptances, not matriculants. The matriculation probabilities must instead appear elsewhere in the objective.

As such, I set up the model by letting the Round 3 cutoff probability be $P(y_3^* = S | \overline{m}_{2A})$ instead of $P(y_3^* = S | t)$. Here, $\overline{m}_{2A}$ is the average early accepted applicant's matriculation probability in a given year, which measures how likely the applicant is to matriculate if accepted. $P(\text{Cutoff})_3$ increases in $\overline{m}_{2A}$ because a larger $\overline{m}_{2A}$ means there will be fewer openings for waitlisted applicants. Given that, what information available at the start of Round 2 determines $\overline{m}_{2A}$? For all applicants in each year, I define $\ddot{m}_2$ as $\overline{m}_2$ but with m_2 excluded, where m_2 is the applicant's own matriculation probability. Within each year, a low m_2 necessarily implies a high $\ddot{m}_2$, since $\overline{m}_2$ is a within-year average. I then estimate $f_{\overline{m}_{2A} | \ddot{m}_2}$ on all transitioned applicants; the parameter estimate should be positive provided there exists variation in $\overline{m}_2$ across years. For example, by decreasing acceptance probabilities, Covid may have made applicants more likely to matriculate if accepted. I assume that the effect of the average matriculation probability of the applicants the committee can accept ($\ddot{m}_2$) on that of the applicants it accepts ($\overline{m}_{2A}$) is stationary across years, and therefore can be used to forecast changes in $\overline{m}_{2A}$ based on changes in $\ddot{m}_2$ resulting from changes in d_2 .

The model implies that the optimal dynamic decision in Round 2 may not be the same as the optimal myopic one. Specifically, a committee may waitlist, as opposed to accept, an applicant with a low m_2 even if their $P(\text{Success})$ is higher than an applicant with a higher m_2 . The reason is that compared with a high m_2 applicant, a low m_2 applicant has a relatively high $\ddot{m}_2$. That implies a high $\overline{F}_{\overline{m}_{2A} | \ddot{m}_2}$, which in turn implies a high Round 3 cutoff probability, Round 3 value function, and payoff for waitlisting the low m_2 applicant in Round 2. Perhaps I will find that in the actual

data, the committee does not think dynamically; that is, removing these “memory” parameters increases the AIC. However, if $\bar{m}_2$ affects $\bar{F}_{\bar{m}_2|\bar{m}_2}$, which in turn affects $\mathbb{E}[P(\text{Cutoff})_3]$, perhaps the committee should. Note that success data for recent years that does not yet exist is unnecessary to identify these two effects because admission data has all the necessary information.

That said, why can't the committee accept the low m_2 applicant in Round 2 and then, if this applicant does not matriculate, fill the additional opening with the high m_2 applicant that it necessarily waitlisted? There are two reasons why such a strategy may not pay off. First, all else equal, matriculation probabilities decline between Rounds 2 and 3 because applicants are more likely to go where they feel wanted. If matriculation probabilities were to decline uniformly for likely vs. unlikely matriculants, the committee could still get away with accepting unlikely matriculants and filling any extra openings with likely matriculants. However, applicants with high matriculation probabilities have more room for those matriculation probabilities to fall if the committee keeps them waiting, meaning the better decision is to accept them in Round 2.

Second, matriculation probabilities can change unpredictably between Rounds 2 and 3 because applicants receive acceptances, waitlists, and rejections from multiple universities, communicate with faculty and graduate students at universities where they were accepted or waitlisted, and attend open houses. These events, which are unknown to the committee at the start of Round 2, make it so that waitlisting a likely matriculant under the belief that they will still matriculate if waitlisted is a risky endeavor; by Round 3, the committee may find out too late that this applicant is now unlikely to matriculate. Therefore, it is safer to accept this applicant in Round 2.¹³

As such, it is clear that accepting applicants who want to matriculate benefits the program. I also discount the expected Round 3 value with $\beta \in (0, 1)$ so that the payoff from accepting an applicant in Round 3 is less than that from accepting them in Round 2. The discount factor is a function of the ratio of expected matriculation probabilities for Round 3 vs. Round 2 acceptances and can be interpreted as the chance of losing a representative applicant to another university. Of course, without a capacity constraint, the committee could give early acceptances to the low and high m_2 applicants, but because of the constraint, it must avoid overshooting its target cohort size. I now present the value functions for the waitlist-dynamic model, starting with Round 3:

$$v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \bar{m}_{2A}, d_3) + \epsilon_3(d_3), \text{ such that:} \quad (26)$$

$$u_3(x, z, t, \bar{k}, \bar{m}_{2A}, d_3) = P(y = S|x, z, t, \bar{k})\mathbb{1}(d_3 = A) + P(y_3^* = S|\bar{m}_{2A})\mathbb{1}(d_3 = R).$$

The Round 2 value function is:

$$v_2(x, z, t, \bar{k}, \bar{m}_2, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \bar{m}_2, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (27)$$

¹³In Appendix E, I present a more general version of the model in which the Round 3 cutoff probability is a function of $\bar{m}_{2A}$ and $\bar{m}_3$, which is the average matriculation probability of the waitlisted applicants and depends on m_3 , the *ex-ante* probability of matriculating if accepted in Round 3. In this model, the effect of matriculation probabilities on Round 2 acceptance probabilities is theoretically ambiguous and depends on the parameter values.

$u_2(x, z, t, \bar{k}, \dot{m}_2, d_2) = P(y = S|x, z, t, \bar{k})\mathbb{1}(d_2 = A) + \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \dot{m}_2]\mathbb{1}(d_2 = W)$, such that:

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \bar{k}, \dot{m}_2] = \int_{\overline{m_{2A}}} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \overline{m_{2A}}, \tilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\dot{m}_2}(\overline{m_{2A}}|\dot{m}_2).$$

The Round 1 value function depends on whether I include a conditional expectation, as in the dynamic model, or whether I do not, as in the static model. I refer to the three-round model without a Round 1 conditional expectation as the waitlist-hybrid model, since it has a static Round 1 and dynamic Round 2. The waitlist-dynamic model's Round 1 value function is:

$$v_1(x, t, \bar{k}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (28)$$

$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \dot{m}_2, \epsilon_2)|x, t, \bar{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R)$, such that:

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \dot{m}_2, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, W\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \dot{m}_2(x, \tilde{z}, t), \tilde{d}_2)/\sigma} \right) dF_{\tilde{z}|x, t}(\tilde{z}|x, t).$$

In contrast, the utility function in the waitlist-hybrid's Round 1 value function is:

$$u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R).$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}. \quad (29)$$

The waitlist-dynamic model's unrestricted CCPs, success probabilities, and cutoff probabilities are $P_{d_1} = P(d_1 = T|x, t; \theta_T)$, $P_{d_2} = P(d_2 = A|x, z, \dot{m}_2; \theta_{A_2})$, $P_{d_3} = P(d_3 = A|x, z, \overline{m_{2A}}; \theta_{A_3})$, $P_y = P(y = S|x, z, t, \bar{k}; \theta_S)$, $L_{y_1^*} = L(y_1^* = S|t; \theta_{S_1^*})$, and $P_{y_3^*} = P(y_3^* = S|\overline{m_{2A}}; \theta_{S_3^*})$. These probabilities have the same specifications as in the dynamic model (see Equations (9)–(11) for the functional forms).¹⁴ Let applicants 1 to N_W be waitlisted, $N_W + 1$ to N_T accepted in Round 2, and $N_T + 1$ to N rejected in Round 1. Then the unrestricted waitlist-dynamic likelihood is:

$$\begin{aligned} \mathcal{L}_{N_1}^U(\theta) &= \prod_{n=1}^{N_W} f_{z|x, t}(z_n|x_n, t_n; \theta_{F_{z|x, t}}) f_{\overline{m_{2A}}|\dot{m}_2}(\overline{m_{2A}, n}|\dot{m}_{2, n}; \theta_{F_{\overline{m_{2A}}|\dot{m}_2}}) P_{d_{1, n}}(\theta_T) (1 - P_{d_{2, n}}(\theta_{A_2})) \\ &\quad P_{d_{3, n}}(\theta_{A_3})^{\mathbb{1}(d_{3, n}=A)} (1 - P_{d_{3, n}}(\theta_{A_3}))^{\mathbb{1}(d_{3, n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \\ &\quad \prod_{n=N_W+1}^{N_T} f_{z|x, t}(z_n|x_n, t_n; \theta_{F_{z|x, t}}) f_{\overline{m_{2A}}|\dot{m}_2}(\overline{m_{2A}, n}|\dot{m}_{2, n}; \theta_{F_{\overline{m_{2A}}|\dot{m}_2}}) P_{d_{1, n}}(\theta_T) P_{d_{2, n}}(\theta_{A_2}) \\ &\quad P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N f_{z|x, t}(z_n|x_n, t_n; \theta_{F_{z|x, t}}) (1 - P_{d_{1, n}}(\theta_T)). \quad (30) \end{aligned}$$

¹⁴ $L_{y_1^*}$ is parameterized as exponential because it can exceed 1. However, in the waitlist-hybrid model, it is denoted as $P_{y_1^*}$ and parameterized as logistic. The waitlist-hybrid model also includes $P_{y_1}(y = S|x, t, \bar{k}; \theta_{S_1})$.

As with the dynamic and matriculation-dynamic likelihoods, a Heckman correction is necessary for $f_{z|x,t}$. I also estimate the parameters of $m_2 = P_{m_2}(\theta_{M_2})$, and therefore $\ddot{m}_2$, in a first-stage likelihood where I restrict the sample to Round 2 acceptances. The dependent variable equals 1 if the applicant matriculates and 0 otherwise, and the independent variables are x , z , and t . Because $\ddot{m}_2$ is a generated regressor in both the unrestricted and restricted likelihoods, I estimate standard errors per Murphy and Topel (1985) and Hardin (2002). The first-stage likelihood is:

$$\mathcal{L}_{N_T - N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_2,n}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)} (1 - P_{m_2,n}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}. \quad (31)$$

To show that the waitlist models' dynamics work as intended, I prove the following theorem:

Theorem 4. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. See Appendix G.

The unrestricted waitlist-hybrid estimates are in Table 7. As expected, the $\ddot{m}_2$ parameter in the $\overline{Fm_{2A}}$ block is positive and significant at the 1% level in both datasets. In addition, the $\ddot{m}_2$ parameter in the *Accept 2* block is negative in both datasets, which is also as expected. The reason is that the committee is more likely to give an early offer to an applicant with a high m_2 and therefore a low $\ddot{m}_2$. Finally, the $\overline{m_{2A}}$ parameter in the *Accept 3* block is negative and significant at the 1% level because a high $\overline{m_{2A}}$ means fewer spots will be open in Round 3 due to more Round 2 matriculants, and so it will be harder to get accepted off the waitlist.

Table 8 contains restricted estimates for the waitlist-hybrid model. In the dataset where the committee is optimizing, the likelihood-ratio test does not reject the model with $p = 0.9136$, whereas in the other dataset, the likelihood-ratio test rejects with $p < 0.0001$. As with the static model, in the year that is not *Year Transition/Matriculate More*, the Round 1 cutoff probability is higher than the Round 3 cutoff probability. As such, the marginal applicant rejected in Round 1's $P(y = S|x, t, \bar{k})$ is greater than the marginal applicant accepted into the program's $P(y = S|x, z, t, \bar{k})$. Moreover, in *Year Transition/Matriculate More*, the Round 3 cutoff probability is higher than the Round 1 cutoff probability not only because there are more applicants who survive Round 1, but also because those applicants have higher matriculation probabilities on average. In Tables 29 and 30 of Appendix C, I present estimates for the waitlist-dynamic model. As with the two-round dynamic model, the likelihood-ratio test rejects this model for both datasets, though as expected, the test statistic is considerably higher for the suboptimal dataset.

Finally, in Appendix F, I present two different infinite-horizon multiyear models. A limitation of the waitlist models is that while they give early acceptances to applicants with higher matriculation probabilities all else equal, matriculation probabilities do not affect the probability of a late acceptance. For this reason, the first multiyear model is constructed so that an applicant's overall acceptance probability is increasing in their matriculation probability, the logic being that more matriculants this year will give the committee the option to be more selective next year.

Table 7: Unrestricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m}_{2A}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0252 (0.0518)	-11.5905*** (1.0234)	-2.2561*** (0.1454)	-11.9661*** (2.5509)	-1.6170*** (0.3154)	-13.2739*** (2.4117)	-1.8259*** (0.3091)	-8.1384*** (1.1381)	-1.2642*** (0.1516)	-9.5601*** (2.0725)	-1.4878*** (0.2688)	-5.8313 (6.4970)	-0.6741 (0.7706)
GRE Quantitative Percentile		0.1236*** (0.0118)	0.0241*** (0.0018)	0.1097*** (0.0271)	0.0148*** (0.0034)	0.1387*** (0.0264)	0.0191*** (0.0034)	0.1079*** (0.0133)	0.0168*** (0.0017)	0.1161*** (0.0227)	0.0181*** (0.0029)	0.0834 (0.0644)	0.0096 (0.0075)
Gender		0.2266 (0.1464)	0.0441 (0.0283)	0.0233 (0.2899)	0.0031 (0.0392)	0.3039 (0.3287)	0.0418 (0.0451)	0.1399 (0.1739)	0.0217 (0.0270)	0.1264 (0.2699)	0.0197 (0.0419)	0.4650 (0.9003)	0.0538 (0.1052)
Demographic 1		-0.3014 (0.1988)	-0.0587 (0.0386)	0.5530 (0.4940)	0.0747 (0.0665)	-0.8834* (0.4622)	-0.1215** (0.0619)	-0.2535 (0.2359)	-0.0394 (0.0366)	-0.4606 (0.3933)	-0.0717 (0.0608)	-0.2627 (0.7744)	-0.0304 (0.0880)
Demographic 2		0.1538 (0.2215)	0.0299 (0.0431)	0.5128 (0.5204)	0.0693 (0.0699)	-0.0526 (0.4528)	-0.0072 (0.0623)	-0.1180 (0.2581)	-0.0183 (0.0401)	-0.2340 (0.4322)	-0.0364 (0.0672)	-0.3966 (1.2105)	-0.0459 (0.1393)
Advanced Math				0.5065* (0.2907)	0.0684* (0.3306)	0.8621*** (0.306)	0.1186*** (0.0451)			0.6273** (0.2749)	0.0976** (0.0417)	1.1399 (0.9839)	0.1318 (0.1080)
Committee Score				0.1061 (0.0689)	0.0143 (0.0093)	0.0378 (0.0672)	0.0052 (0.0092)			0.0464 (0.0655)	0.0072 (0.0102)	-0.9123** (0.3787)	-0.1055*** (0.0317)
Year Transition/Matriculate More		0.9842*** (0.1472)	0.1916*** (0.0263)					-0.0586 (0.1690)	-0.0091 (0.0262)	-0.0233 (0.2763)	-0.0036 (0.0430)	3.3933*** (0.7682)	0.3922*** (0.0697)
Program Rank								-0.0310*** (0.0019)	-0.0048*** (0.0002)	-0.0282*** (0.0027)	-0.0044*** (0.0003)		
m_2	1.0892*** (0.0851)			-1.5983** (0.7869)	-0.2160** (0.1049)								
$\overline{m}_{2A}$						-2.0170*** (0.7696)	-0.2774*** (0.1033)						
$\sqrt{\text{MSE}}$	0.0015*** (0.0005)												
Observations	1000												
LL Parameters AIC	438.6	41	-795.1										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m}_{2A}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0498* (0.0260)	-11.2292*** (1.0143)	-2.1801*** (0.1474)	-6.7111*** (2.0006)	-1.0638*** (0.3043)	-7.1640*** (2.5069)	-0.8321*** (0.2952)	-8.2823*** (1.1392)	-1.2871*** (0.1493)	-9.6011*** (1.9606)	-1.4979*** (0.2508)	-1.4373 (3.8091)	-0.2080 (0.5533)
GRE Quantitative Percentile		0.1226*** (0.0117)	0.0238*** (0.0018)	0.0387* (0.0215)	0.0061* (0.0034)	0.0717*** (0.0278)	0.0083** (0.0033)	0.1097*** (0.0133)	0.0170*** (0.0017)	0.1162*** (0.0216)	0.0181*** (0.0027)	0.0355 (0.0435)	0.0051 (0.0063)
Gender		0.2576* (0.1468)	0.0500* (0.0283)	-0.2763 (0.2746)	-0.0438 (0.0431)	0.4099 (0.3417)	0.0476 (0.0394)	0.1354 (0.1741)	0.0210 (0.0270)	0.0584 (0.2669)	0.0091 (0.0416)	-0.3544 (0.5436)	-0.0513 (0.0788)
Demographic 1		-0.8291*** (0.1966)	-0.1610*** (0.0371)	0.2570 (0.3773)	0.0407 (0.0594)	-1.1547** (0.5110)	-0.1341** (0.0581)	-0.2774 (0.2344)	-0.0431 (0.0364)	-0.5155 (0.3648)	-0.0804 (0.0563)	0.3334 (0.7088)	0.0482 (0.1030)
Demographic 2		0.0799 (0.2135)	0.0155 (0.0414)	0.5360 (0.3885)	0.0850 (0.0605)	-0.3696 (0.4652)	-0.0429 (0.0539)	-0.1806 (0.2566)	-0.0281 (0.0399)	-0.4743 (0.3856)	-0.0740 (0.0598)	-0.7602 (0.7927)	-0.1100 (0.1089)
Advanced Math				0.2361 (0.2639)	0.0374 (0.0417)	0.4867 (0.3566)	0.0565 (0.0417)			0.5367** (0.2704)	0.0837** (0.0416)	0.3982 (0.6445)	0.0576 (0.0922)
Committee Score				0.3736*** (0.0674)	0.0592*** (0.0096)	0.0653 (0.0837)	0.0076 (0.0098)			0.0617 (0.0643)	0.0096 (0.0100)	-0.6725*** (0.1981)	-0.0973*** (0.0217)
Year Transition/Matriculate More		1.1108*** (0.1479)	0.2157*** (0.0257)					-0.0767 (0.1688)	-0.0119 (0.0262)	0.1318 (0.2794)	0.0206 (0.0434)	2.9410*** (0.6722)	0.4256*** (0.0578)
Program Rank								-0.0304*** (0.0019)	-0.0047*** (0.0002)	-0.0279*** (0.0028)	-0.0044*** (0.0003)		
m_2	0.8883*** (0.0489)			-0.5486 (0.5705)	-0.0870 (0.0900)								
$\overline{m}_{2A}$						-2.9216*** (0.9934)	-0.3393*** (0.1141)						
$\sqrt{\text{MSE}}$	0.0010*** (0.0002)												
Observations	1000												
LL Parameters AIC	618.0	41	-1153.9										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 8: Restricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m}_{2A}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer Intercept	-0.0252 (0.0518)	0.6891*** (0.1830)	66.58% (58.19%, 74.03%)	0.4185 (0.4184)	69.29% (54.94%, 80.68%)	-8.0491*** (1.1651)	-9.6599*** (1.7228)	-5.8313 (6.4970)		
GRE Quantitative Percentile						0.1060*** (0.0135)	0.1151*** (0.0191)	0.0834 (0.0644)		
Gender						0.1997** (0.1017)	0.1149 (0.1664)	0.4650 (0.9003)		
Demographic 1						-0.2477* (0.1349)	-0.3286 (0.2390)	-0.2627 (0.7744)		
Demographic 2						0.0405 (0.1507)	-0.0036 (0.2410)	-0.3966 (1.2105)		
Advanced Math							0.6284*** (0.1758)	1.1399 (0.9839)		
Committee Score							0.0548 (0.0419)	-0.9123** (0.3787)		
Year Transition/Matriculate More		-0.8464*** (0.2185)	46.08% (39.62%, 52.67%)		87.77% (63.97%, 96.67%)	-0.0609 (0.1668)	0.0204 (0.2685)	3.3933*** (0.7682)		
Program Rank						-0.0310*** (0.0019)	-0.0283*** (0.0027)			
$\hat{m}_2$	1.0892*** (0.0851)									
$\overline{m}_{2A}$				2.4392* (1.4478)						
$\sqrt{\text{MSE}}$	0.0015*** (0.0005)									
Beta									0.9493*** (0.0396)	
Sigma										0.1917*** (0.0251)
Observations	1000									
LL Parameters AIC	434.1	25	-818.1							
LR Statistic DF P-Value	8.9955	16	0.9136							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m}_{2A}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer Intercept	-0.0498* (0.0260)	0.7428*** (0.2082)	67.76% (58.29%, 75.97%)	0.5736 (0.5852)	74.61% (55.80%, 87.24%)	-8.0749*** (1.2691)	-6.3299*** (1.7373)	-1.4373 (3.8091)		
GRE Quantitative Percentile						0.1082*** (0.0147)	0.0718*** (0.0186)	0.0355 (0.0435)		
Gender						0.2189** (0.1045)	0.0369 (0.1594)	-0.3545 (0.5436)		
Demographic 1						-0.5509*** (0.1397)	-0.3579 (0.2337)	0.3334 (0.7088)		
Demographic 2						-0.0156 (0.1510)	-0.0774 (0.2321)	-0.7602 (0.7927)		
Advanced Math							0.3857** (0.1669)	0.3982 (0.6445)		
Committee Score							0.1748*** (0.0374)	-0.6725*** (0.1981)		
Year Transition/Matriculate More		-1.0055*** (0.2419)	43.47% (37.12%, 50.04%)		93.27% (41.58%, 99.63%)	-0.0797 (0.1668)	0.1064 (0.2766)	2.9410*** (0.6722)		
Program Rank						-0.0306*** (0.0019)	-0.0281*** (0.0027)			
$\hat{m}_2$	0.8883*** (0.0489)									
$\overline{m}_{2A}$				3.6299 (3.2728)						
$\sqrt{\text{MSE}}$	0.0010*** (0.0002)									
Beta									0.8280*** (0.0422)	
Sigma										0.2012*** (0.0325)
Observations	1000									
LL Parameters AIC	596.3	25	-1142.6							
LR Statistic DF P-Value	43.3371	16	0.0002							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Meanwhile, the second multiyear model rewards more diverse applicants, such as those interested in economic theory as opposed to applied research. In this example, by accepting more theory applicants this year, the committee will have to worry less next year about having too many applied students relative to the number of faculty advisors. But this is just one example; the model applies to any form of diversity that increases an applicant's success probability.

5 Optimality Tests

In the previous section, I implemented likelihood-ratio tests comparing the unrestricted and restricted likelihoods to assess the fit of the various models. For the static, matriculation-static, and waitlist-hybrid models, these tests reject the null at the 1% level on only the datasets where the simulated committee was not optimizing, and the static and waitlist-hybrid models do not reject on the optimal dataset at any level. However, for the dynamic, matriculation-dynamic, and waitlist-dynamic models, the tests always reject, though the test statistics are higher on the sub-optimal dataset. But a likelihood-ratio test only determines whether the model fits the data, not whether the committee can make any meaningful improvements by following the model's decision rule as opposed to its own decision rule. To make the latter determination, a different kind of statistical test is necessary, one that compares the model's decision rule to the committee's.

Suppose the committee adopts the dynamic/static, matriculation-dynamic/static, or the waitlist-dynamic/hybrid model's decision rules. Will the average success rate of the set of accepted (or matriculated) applicants increase, and if so, by how much? To answer these questions, I solve the model with $\sigma = 0$ in the conditional choice probabilities (CCPs), compute the yearly success rates of the model's acceptance or matriculation set, and compare those success rates to the success rates of the committee's acceptance or matriculation set. When I solve the model, I plug in the unrestricted success probability parameter estimates, not the restricted estimates. The reason is that the restricted estimates may be biased to rationalize suboptimal behavior (see Anderson et al. 2025). For example, if the committee puts too little weight on the quantitative GRE score, then its restricted parameter estimate will be biased down since the restricted CCPs assume optimality.

For the dynamic model, I rank the applicants in each year by the expected Round 2 social surplus function $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma} \right) dF(\tilde{z}|x, t)$,¹⁵ transition the number of transitioned applicants in the data, rank the survivors by $P(y = S|x, z, t, \bar{k})$, and accept the number of accepted applicants in the data. In contrast, for the static model, I rank the applicants in Round 1 by $P(y = S|x, t, \bar{k})$. After that, the algorithm is the same as the dynamic model's algorithm. Once I have determined which applicants each model accepts, I compute their average success probability and compare it to the average success probability of the applicants that the committee's decision rule selects. Determining which applicants the committee's deci-

¹⁵For the myopic model, the expected social surplus is $\int_{\tilde{z}} \sigma \log \left(e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma} \right) dF(\tilde{z}|t)$.

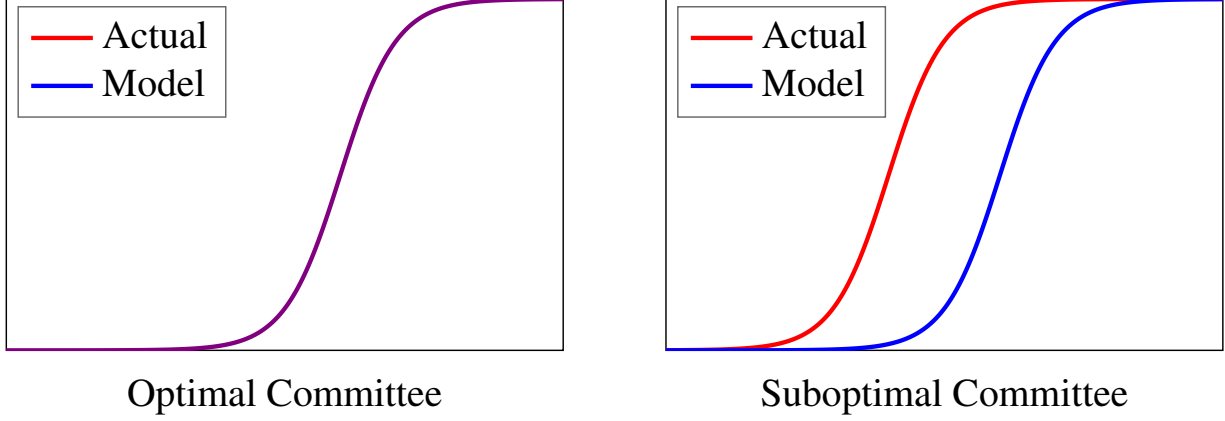


Figure 4: Deviation Gains CDFs

sion rule selects is straightforward. In Round 1, I rank the applicants by $P(d_1 = T|x, t)$, and in Round 2, I rank the survivors by $P(d_2 = A|x, z, t)$. The number of acceptances is always the same.

For the matriculation and waitlist models, the algorithm is more complicated because the purpose of these models is to evaluate the committee's performance based on the applicants who matriculate at the university. A committee that accepts strong applicants who do not matriculate due to getting offers from higher-ranked universities is not doing its job. For the matriculation-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{[P(y=S|x, \tilde{z}, t, \bar{k}) - P(y_2^*=S)]/\sigma} + 1 \right) P(m = M|x, \tilde{z}, t) dF(\tilde{z}|x, t)$, transition the number of transitioned applicants in the data, rank the surviving applicants by $P(y = S|x, z, t, \bar{k})$, and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$ until the number of matriculants in the data matriculate. I then compute the average success probability of the matriculation set across matriculants and across 500 simulations, and compare it to that of the committee's set. For the matriculation-static model, I instead rank the applicants in Round 1 by $P(y = S|x, t, \bar{k})P(m = M|x, t)$.

For the waitlist-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x, z, t, \bar{k}, \ddot{m}_2, \epsilon_2)|x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(e^{P(y=S|x, \tilde{z}, t, \bar{k})/\sigma} + e^{\beta \mathbb{E}[v_3(x, \tilde{z}, t, \bar{k}, \widetilde{m}_{2A}, \epsilon_3)|x, \tilde{z}, t, \bar{k}, \ddot{m}_2(x, \tilde{z}, t)]/\sigma} \right) dF_{\tilde{z}|x, t}(\tilde{z}|x, t)$ and transition the number of transitioned applicants in the data. Next, I rank the surviving applicants by the payoff difference $P(y = S|x, z, t, \bar{k}) - \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \widetilde{m}_{2A}, \epsilon_3)|x, z, t, \bar{k}, \ddot{m}_2] = P(y = S|x, z, t, \bar{k}) - \beta \int_{\widetilde{m}_{2A}} \sigma \log \left(e^{P(y=S|x, z, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{m}_{2A})/\sigma} \right) dF_{\widetilde{m}_{2A}|\ddot{m}_2}(\widetilde{m}_{2A}|\ddot{m}_2)$, accept the number of early acceptances in the data, and waitlist the rest. Each early acceptance matriculates with early matriculation probability $P(m_2 = M|x, z, t)$. After that, I rank the waitlisted applicants by $P(y = S|x, z, t, \bar{k})$ and accept them in descending order, each matriculating with late matriculation probability $P(m_3 = M|x, z, t)$, until the number of early plus late matriculants equals that in the data. Finally, I compute the average success probability of the matriculation set and compare it to the committee's set. For the waitlist-hybrid model, I instead rank the applicants in Round 1 by $P(y = S|x, t, \bar{k})$.

Suppose I find deviation gains. Could committees have incorrect beliefs about which appli-

cants are most likely to succeed? Also, if so, can they do a better job admitting applicants? If there are gains in the dynamic, static, or myopic models, the matriculation rate of the model's (stronger) acceptance set may be less than that of the committee's (weaker) acceptance set. In that case, the estimated gains would be an upper bound. However, the matriculation and waitlist models accept those more likely to matriculate to prevent losing good, interested applicants by either rejecting them before reading their files or making them wait too long for an acceptance.

In addition, the test can credit the committee only for information I observe. The test evaluates both the committee's and the model's acceptance or matriculation set with the same measure, the unrestricted success probability $P(y = S|x, z, t, \bar{k})$, which is estimated from the observables. If the committee acts on a success-predicting variable I do not observe, the measure cannot detect that variable, so any edge the committee's private information provides goes unmeasured.

The protection against this limitation is the comprehensiveness of the data. The application files I observe are the same files the committee read, and from those files, I construct (in addition to the objective variables x) measures of mathematical preparation, the substance of the application essay, the content and tone of the recommendation letters, and the standing of the recommenders. Most importantly, because z includes the committee score, conditioning the success probability on z absorbs much of the residual private information. Nonetheless, no set of observables is guaranteed to span the committee's information set, so the deviation gains test is best read as conditional on the observed variables closely reconstructing what the committee saw.

A separate consideration is sampling error. The models have many parameters, so it is possible for the models to overfit the data. Ideally, I would test the performance of each model's deviation strategy on a holdout set. But given the limited number of observations, it is necessary to use all of them to estimate the parameters. k -fold cross-validation with large k would mitigate this problem, but it would introduce a new computational problem, as the models take far too long to estimate for it to be feasible, even with parallelization across multiple cores. Therefore, I follow Anderson et al. (2025) by using a randomization test to address sampling error.

If I assume for simplicity that the model is a two-round model that chooses the same transition set of applicants as the committee, then the following inequality holds:

Theorem 5. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\begin{aligned} & \frac{1}{N_A} \sum_{n=1}^N P\left(y_n = S \mid x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\} \\ & \geq \frac{1}{N_A} \sum_{n=1}^N P\left(y_n = S \mid x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}. \quad (32) \end{aligned}$$

Proof. See Appendix G.

In this inequality, I am careful to assume that the committee's Round 1 decision rule is the same as the model's. Without this assumption, it is possible in rare cases for the sign to flip if, for

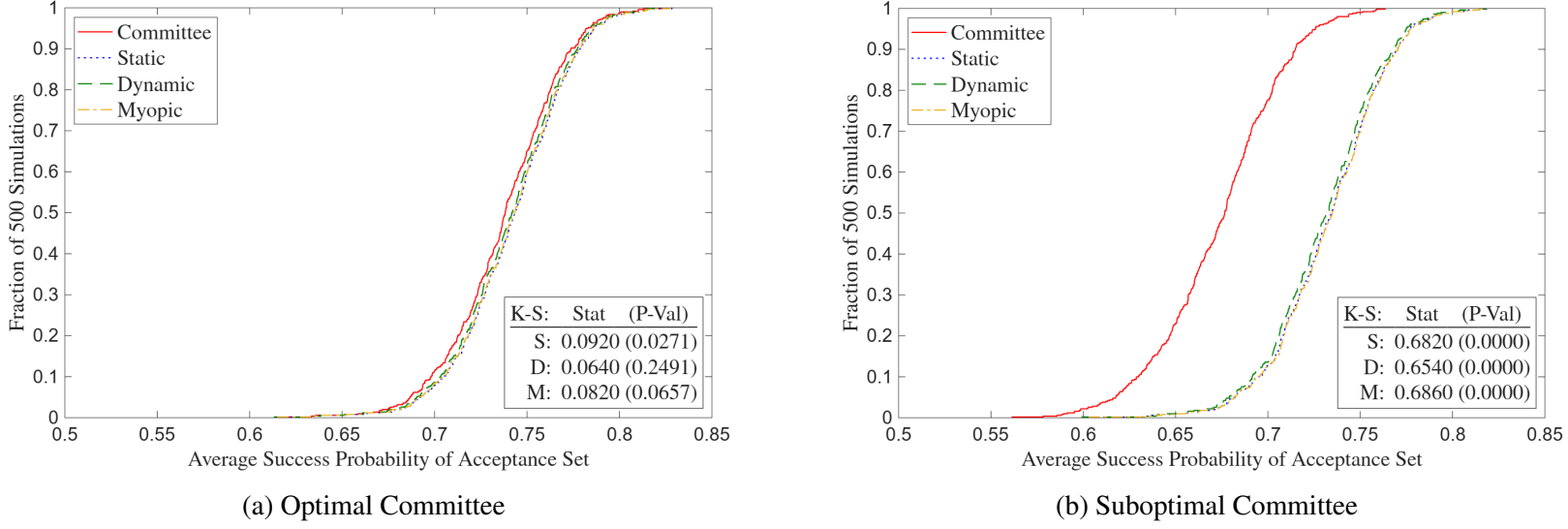


Figure 5: Deviation Gains

example, the committee favors high-upside applicants in Round 1, and it turns out that favoring such applicants ultimately helps it attain a better set of accepted applicants (see Appendix G).

Regardless, for a perturbation $\widetilde{\theta}_{S,t}^U$ about $\hat{\theta}_S^U$'s asymptotic distribution, the sign can flip:

$$\begin{aligned} & \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta}_{S,t}^U) \mathbb{1} \left\{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \right\} \\ & < \frac{1}{N_A} \sum_{n=1}^N P(y_n = S | x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta}_{S,t}^U) \mathbb{1} \left\{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \right\}. \quad (33) \end{aligned}$$

Given that the sign can flip if the committee optimizes or comes close to it, I can use such perturbations of the success parameters to test whether the deviation gains are robust to sampling error. Across $T = 500$ perturbations, I plot the CDFs of the average success probabilities of the acceptance sets of the dynamic, static, and myopic models, as well as of the acceptance set implied by the committee's decision rule. While the acceptance sets are computed with the actual parameter estimates, the average success probabilities are computed with the perturbed estimates.

Figure 4 illustrates what the CDFs will look like in general. When a committee switches from a suboptimal decision rule to the optimal decision rule, the actual decision rule's CDF shifts right to match the model's CDF. Per Figure 5, in the dataset where the committee is optimizing, the models' CDFs are virtually identical to the committee's. But in the dataset where the committee is not optimizing, all the models' CDFs stochastically dominate the committee's CDF. The reason is that the committee is discriminating against *Demographic 1*, putting too much weight on the committee score, and not putting enough weight on whether the applicant took advanced math.

To formally test deviation gains, I perform two-sample Kolmogorov-Smirnov (KS) tests of the equality of each of the models' CDFs and the committee's CDF. For this test, denote $X =$

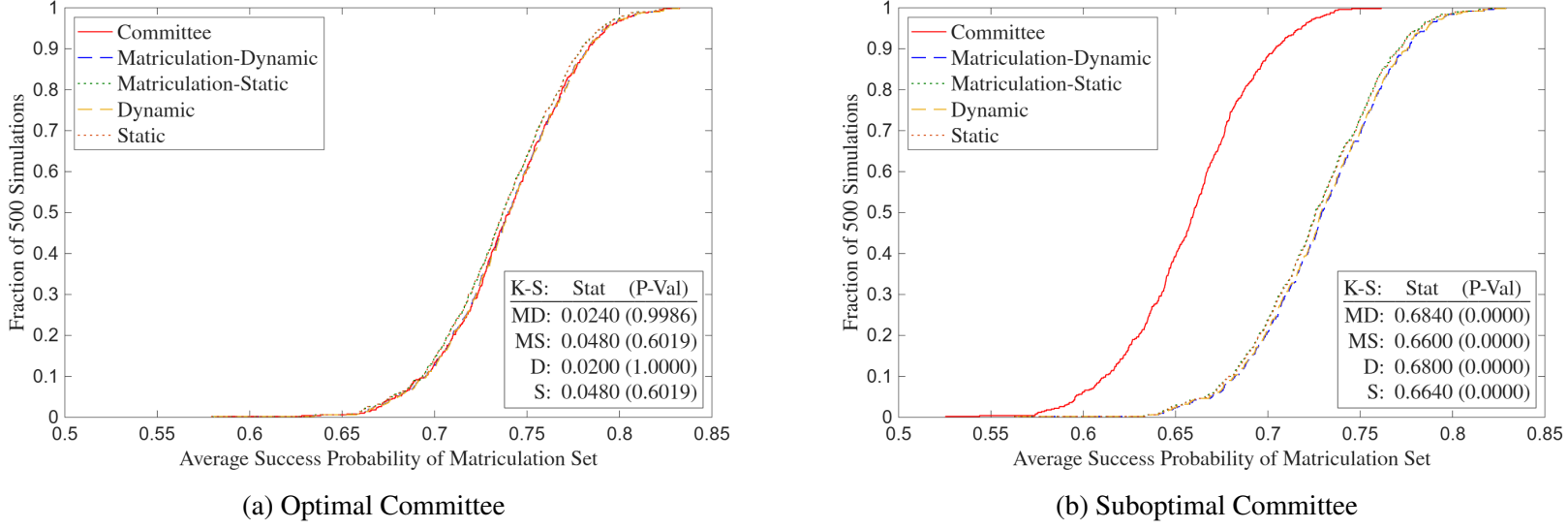
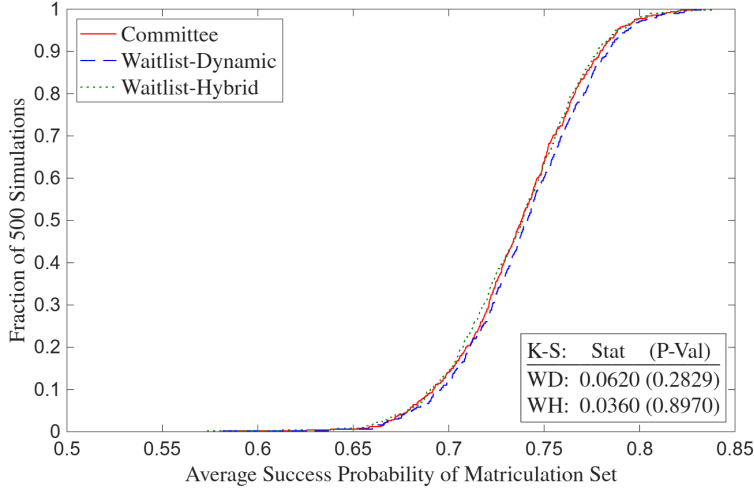


Figure 6: Matriculation Deviation Gains

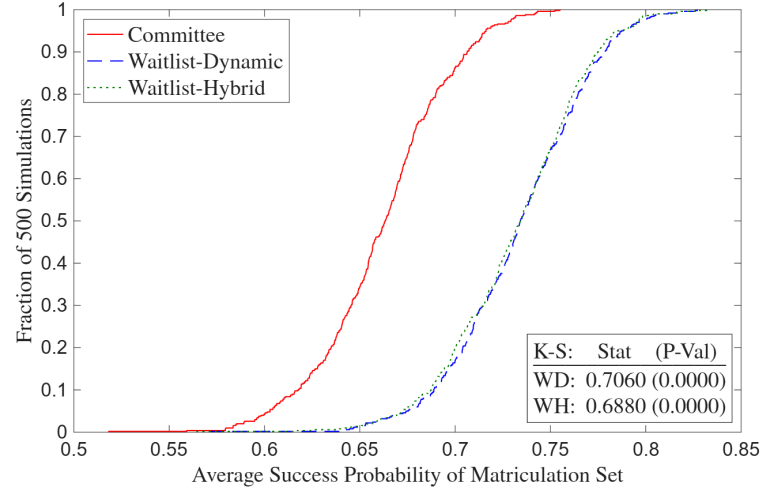
$P_y[d(\hat{\theta}^U), \widetilde{\theta}_S^U]$, and let $F_T(x)$ be X 's CDF over T perturbations. Then the test statistic $D_T = \sup_x |F_T^R(x) - F_T^U(x)|$, and I reject the null of equality at the α level if $D_T > \sqrt{\frac{-\log(\alpha/2)}{T}}$. As the test should, it does not reject or barely rejects equality of the models' CDFs vs. the committee's CDF in the optimal dataset, but it rejects equality of the CDFs with $p < 0.0001$ all three times in the suboptimal dataset. Next, I simulate deviation gains for the matriculation and waitlist models. Analogously to the previous models, I plot CDFs for the matriculation-dynamic, matriculation-static, waitlist-dynamic, and waitlist-hybrid models' decision rules, along with the committee's decision rule. However, as explained earlier in this section, the committee in these frameworks does not ultimately care about the the average success probability of its acceptance set, but rather of its matriculation set. Therefore, the average success probabilities are of the matriculation sets.

The results for the matriculation models are in Figure 6, and the results are essentially the same as for the dynamic, static, and myopic models. For the optimal dataset, the KS tests never reject, and for the suboptimal dataset, they always reject with $p < 0.0001$. In this figure, I also plot the average success probabilities of the matriculation sets for the dynamic and static models without matriculation. It turns out that the final average success probabilities are nearly identical regardless of whether the committee considers matriculation probabilities in Round 1. If this result holds in the actual data, it would justify this university's decision not to consider matriculation probabilities in Round 1. In addition, the results for the waitlist models are in Figure 7. As expected, the KS tests never reject for the optimal dataset, and they always reject with $p < 0.0001$ for the suboptimal dataset. Ultimately however, the main question is whether deviation gains would be borne out in practice. The only way to know would be for a real-world committee to implement the optimal decision rule and see if the result is stronger cohorts across the years.

Finally, it is possible to use models' acceptance or matriculation sets in year t to put a dollar



(a) Optimal Committee



(b) Suboptimal Committee

Figure 7: Waitlist Deviation Gains

value on success in t . Assume that it takes h hours for a committee member to read an application file, and also that each committee member makes an hourly wage of w . Therefore, reading N_T files has a monetary cost of $c = N_T \times h \times w$. If I counterfactually increase N_T , then the average success probability of the acceptance or matriculation set increases because the committee discards fewer applicants in Round 1, who could have a high success probability. As such, I can compute the average success probability for all possible values of N_T , where the lower bound is the number of acceptances N_A , and the upper bound is the number of applicants N . At that point, I can plot c vs. the average success probability and calculate the derivative of the curve numerically. This derivative is the dollar value of a marginal increase in the average success probability, provided the committee is choosing the optimal value of N_T . While the derivative will be positive, it may not be constant, e.g. under diminishing returns to reading more files.

6 Conclusion

This paper studies how decision-makers select among options whose true quality is uncertain and costly to evaluate. Graduate admissions is a natural setting for such a study, as it features a low-cost filtering process followed by a high-cost review of the application files that survive the filter. Because the committee has a target cohort size, and not all accepted applicants matriculate, there is also staggered timing of offers via a waitlist. By structurally modeling the committee's decision problem, I develop the first framework in the economics graduate admissions literature that links application information to admissions decisions and post-program success. This framework makes optimality testable, and gains from deviating to the optimal decision rule quantifiable.

Before estimating structural models, I start with reduced-form regressions and lasso of a bi-

nary acceptance variable on information in application files. I find that relative to the university's other social science programs, economics admissions is more data-driven, as the economics regressions have a higher goodness of fit in and out of sample. I also find evidence of network effects, as having a recommender who wrote letters for other applicants in the sample predicts acceptance. This result holds across the social science programs. Moreover, an omnibus Wald test shows that the determinants of acceptance are significantly different for economics vs. non-economics. This test can also be used to compare the determinants of success across programs.

However, regressions and Wald tests cannot determine whether an admissions committee is behaving optimally. If I were to test whether the average marginal effects (AMEs) of variables in application files, such as GRE scores, are the same in an acceptance regression vs. a success regression, I would not be testing whether the committee is optimizing. As I show at the end of Section 3, equal AMEs are neither necessary nor sufficient for optimization. Therefore, any optimality tests must be based on structural models of the committee's selection problem that account for evaluation costs and multi-stage decision-making. For this reason, I develop and estimate such models of the committee's problem in order of increasing complexity.

I start with two-round dynamic, static, and myopic models. In these models, there is a Round 1 filter based on only objective variables like GRE scores, followed by a Round 2 final decision based on objective and subjective variables, such as those created by textual analysis of recommendation letters. I then develop two-round matriculation-dynamic and matriculation-static models, in which not every accepted applicant matriculates, and the committee considers matriculation probabilities in Round 1. The reason for doing so is to ensure that the committee can fill its cohort with applicants who do not have particularly weak subjective variables.

Finally, I develop three-round waitlist-dynamic and waitlist-hybrid models. Round 1 in these models is the same as in the dynamic and static models without matriculation, but in Round 2, the committee can accept or waitlist the survivors of Round 1. Finally, in Round 3, the committee gives additional acceptances off the waitlist and rejects the remaining waitlisted applicants. In Round 2, the committee is more likely, all else equal, to give early acceptances to likely matriculants. The reason is to ensure that it does not have to give too many late acceptances to weaker applicants from the waitlist. Ideally, the committee will be able to uncover variables, such as a binary variable for attending the open house if accepted or waitlisted, that predict matriculation without also predicting failure. This situation is analogous to the small-market Oakland Athletics baseball team in *Moneyball* using analytics to find good players *they could afford* (Lewis 2004).

After estimating the structural models, I use the estimates to compare each two-round model's decision rule to the committee's decision rule. In a simulated dataset where the committee is optimizing, Kolmogorov-Smirnov (KS) tests consistently fail to reject the null of equal CDFs of the average success probabilities of the models' acceptance or matriculation set vs. the committee's respective set (across perturbations of the estimates about their asymptotic distribution). However, in a simulated dataset where the committee is making mistakes, such as discriminating against a demographic, the model's CDF stochastically dominates the committee's, and the KS

tests consistently reject equal CDFs with $p < 0.0001$. I repeat this process for the three-round waitlist models, comparing the average success probabilities of the matriculation sets. In the suboptimal dataset, the deviation gains are several times larger than in the optimal dataset.

In addition to its contributions to the graduate admissions literature and to literatures on dynamic selection and costly evaluation, this paper contributes to the literature on synthetic data and privacy protection. Recent economics papers use synthetic data provided by the U.S. Census Bureau, but this paper is the first to design and implement a comprehensive synthetic data protocol from the construction of the synthetic data to the final estimation results. As a result, it serves as a template for researchers who wish to work with confidential data that they are not allowed to access directly, which can help them make discoveries they could not have otherwise.

More broadly, this paper provides a tractable framework for studying selection under costly evaluation. By doing so, it advances the literature from identifying variables that predict success to testing whether decision rules are optimal conditional on time and effort constraints. This framework can uncover sources of potential discrimination and efficiency losses from misweighted variables. For example, in the simulated dataset where the committee was undervaluing advanced math and discriminating against a demographic, the likelihood-ratio and deviation gains tests reject the null of optimality. Moreover, by computing the counterfactual of following the optimal decision rule, I can determine how much stronger the acceptance and matriculation sets would be if not for the misweightings. I can also check which parameters in the committee's decision rule appear misaligned with the optimal decision rule and recompute the committee's decision rule with different values of those parameters. If the committee's CDF then aligns with the optimal CDF, I will have determined which variables the committee is over/undervaluing.

This study is based on a single institution, so any specific empirical results reflect that setting. However, the framework is designed to apply broadly to environments where due to evaluation costs, decision-makers must employ a multi-stage process to select among their available options. Such environments range from hiring committees to clinical trials of medicines. That said, admissions committees may pursue objectives beyond maximizing success probabilities, such as diversity or research field balance. The framework can be extended to incorporate such objectives directly into the committee's problem. In addition, a worthwhile avenue for future research would be collecting data from multiple universities to model the applicant's decision problem and compute equilibrium admissions outcomes for applicants across universities.

Ultimately, the models in this paper can help decision-makers make better selection decisions based on *ex-ante* information when faced with uncertainty and costly evaluation. Crucially, decision-makers can improve selection outcomes without having to increase evaluation effort. Instead, they can use the paper's structural framework and optimality tests to refine how they use that information, and consequently do the best possible job selecting from their available options.

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Appendices

A Variables, Rankings, and Dictionaries

This table contains a list of the variables in this paper. Variables available only for the economics program are bold, and those available for all but economics are italic. *#Math Courses* and *Advanced Math* became objective variables starting in 2020 due to a change in the university’s online application system. PhD program rankings are from U.S. News, and placement rankings are from IDEAS/RePEc. The dictionaries for the subjective independent variables are from Madera, Hebl, and Martin (2009), Young and Soroka (2012), and Bai et al. (2022). When using the dictionaries, I implement the Porter stemmer to reduce words to their roots.

Table 9: Variables

(a) Dependent

Admission	Success
1(Transition)	1(Complete Program)
1(Accept)	1(Academic Placement)
1(Waitlist)	1(Government Placement)
1(Open House)	1(Consulting Placement)
1(Matriculate)	1(Tech Placement)
1(Attend Program)	1(Other Placement)
Program Rank	1(No Placement)
	Placement Rank
	1(Top 100 Placement)

(b) Objective Independent

Demographic	Performance
Year	GRE Verbal Percentile
1(Covid Year)	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
1(Female)	Undergraduate GPA
1(U.S.)	TOEFL/IELTS
1(U.S. African/Hispanic/Native)	1(Elite U.S. University)
1(U.S. Asian)	1(Elite U.S. Liberal Arts College)
1(U.S. Other)	1(Other U.S. University/College)
1(International Asian)	1(Elite Foreign University)
1(International Other)	1(Other Foreign University)
	1(Graduate Degree)
	1(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	<i>Concentration Dummy Variables</i>

(c) Subjective Independent

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	#Math Courses
Letters Length	1(Advanced Math)
Positive Terms (%) (YS 2012)	CV Length
Negative Terms (%) (YS 2012)	CV Coding
Standout Terms (%) (BEMS 2022)	CV 1(Publication/Working Paper)
Ability Terms (%) (BEMS 2022)	Essay Length
Research Terms (%) (BEMS 2022)	Essay 1(Specific Professor)
Grindstone Terms (%) (BEMS 2022)	Essay Economics Terms (%)
Teaching Terms (%) (BEMS 2022)	Essay Research Terms (%)
Communal Terms (%) (BEMS 2022)	Essay Positive Terms (%)
Agentic Terms (%) (MHM 2009)	Essay Negative Terms (%)
	Field Dummy Variables

Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)

- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Academia-Equivalent Rankings (Consulting/Technology/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Dictionaries (Recommendation Letters)

- **Positive/Negative Words:** Young and Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai et al. (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera et al. 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop. (Note: I constructed this dictionary because it is unavailable in Bai et al. 2022.)

Dictionaries (Application Essays/CVs)

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

B Additional Reduced-Form Results

Table 10: Logistic Regression Results (Synthetic)

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997 (0.1633)	-0.0323 (0.2215)	0.2028 (0.3078)	-0.4791 (0.3477)	-0.5946 (0.4520)
Winsorized Age	-0.0509** (0.0201)	-0.0198 (0.0174)	0.0576* (0.0307)	-0.0164 (0.0357)	0.1203*** (0.0400)
Female	0.1941 (0.1402)	0.0923 (0.2004)	0.1389 (0.2526)	0.2595 (0.3255)	0.2119 (0.4310)
U.S. African/Hispanic/Native	0.3053 (0.3724)	-0.1457 (0.4303)	-0.2379 (0.4670)	-1.6401 (1.0720)	0.5413 (0.5306)
U.S. Asian	0.5663 (0.3642)	0.6016 (0.4452)	-0.7934 (1.1675)	-0.4305 (0.7936)	0.7122 (0.8863)
International Asian	-0.1428 (0.2169)	0.1630 (0.2803)	0.5780 (0.3740)	-0.4965 (0.4275)	0.1868 (0.6866)
International Other	-0.1231 (0.2311)	0.1395 (0.3046)	0.1097 (0.3237)	-0.1602 (0.3832)	-1.0807 (0.7583)
GRE Verbal Percentile	0.0167*** (0.0048)	0.0247*** (0.0117)	0.0164 (0.0127)	-0.0029 (0.0100)	0.0132 (0.0184)
GRE Quantitative Percentile	0.0233*** (0.0090)	-0.0027 (0.0051)	0.0026 (0.0088)	0.0304** (0.0137)	0.0097 (0.0099)
GRE Analytic Percentile	0.0042 (0.0031)	0.0004 (0.0059)	-0.0073 (0.0092)	-0.0042 (0.0101)	0.0069 (0.0103)
Undergraduate GPA	0.2292 (0.4745)	0.6327 (0.4985)	1.3552** (0.5796)	-1.3171** (0.5584)	-0.1002 (0.6152)
TOEFL/IELTS	-0.4673** (0.2157)	0.5141 (0.4185)	-0.7648 (0.6814)	0.2446 (0.7498)	-0.7354 (0.8025)
Elite U.S. University	-0.4850* (0.2560)	-0.2243 (0.3258)	0.0572 (0.3421)	0.4764 (0.4788)	0.3941 (0.4782)
Elite U.S. Liberal Arts College	0.2429 (0.3049)	0.1759 (0.4042)	0.2358 (0.4964)	0.4860 (0.9631)	0.8652 (0.7261)
Elite Foreign University	-0.1622 (0.3015)	-1.1929 (0.7484)	0.7404 (0.5668)	0.4801 (0.7841)	3.0530*** (0.8673)
Other Foreign University	-0.1087 (0.1979)	0.2043 (0.2732)	0.2721 (0.3391)	0.1731 (0.4292)	0.3023 (0.5926)
Graduate Degree	-0.0198 (0.1624)	0.1770 (0.2439)	-0.1601 (0.2765)	-0.8698*** (0.3320)	-0.2769 (0.3814)
Work Experience	0.2525* (0.1441)	-0.0391 (0.2396)	-0.0736 (0.2629)	-0.2650 (0.3417)	-1.1936** (0.4805)
#Professor Recommenders	-0.1059 (0.0909)	0.0911 (0.1138)	-0.1828 (0.1473)	0.2181 (0.1817)	-0.4923** (0.1956)
#Prolific Recommenders	0.0039 (0.0656)	0.2189** (0.1108)	-0.1743 (0.1723)	0.0673 (0.1642)	0.5134** (0.2213)
#Math Courses	0.0366 (0.0421)				
Advanced Math	0.1961 (0.1416)				
Missing GRE	-0.1993 (0.4328)	-0.2029 (0.2195)	-0.0011 (0.2689)	-0.0433 (0.3503)	-0.6040 (0.4017)
Missing Undergraduate GPA	0.1531 (0.1548)	0.0895 (0.2291)	-0.2455 (0.3015)	0.6298** (0.3125)	-0.3574 (0.4342)
Missing TOEFL/IELTS	-0.2950* (0.1640)	0.4898 (0.3480)	0.5752 (0.5466)	0.1346 (0.4345)	-2.0781*** (0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.0524	0.0366	0.0753	0.1046	0.2230
CV Pseudo-R ²	0.0139	-0.0189	-0.0887	0.0055	-0.0014

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204 (0.0167)	-0.0014 (0.0099)	0.0192 (0.0291)	-0.0298 (0.0216)	-0.0345 (0.0262)	-0.0089 (0.0086)
Winsorized Age	-0.0052** (0.0021)	-0.0009 (0.0008)	0.0054* (0.0029)	-0.0010 (0.0022)	0.0070*** (0.0023)	0.0004 (0.0008)
Female	0.0198 (0.0143)	0.0041 (0.0089)	0.0131 (0.0239)	0.0162 (0.0204)	0.0123 (0.0251)	0.0099 (0.0078)
U.S. African/Hispanic/Native	0.0312 (0.0381)	-0.0065 (0.0192)	-0.0225 (0.0441)	-0.1021 (0.0673)	0.0314 (0.0306)	-0.0137 (0.0147)
U.S. Asian	0.0579 (0.0372)	0.0268 (0.0199)	-0.0750 (0.1101)	-0.0268 (0.0495)	0.0413 (0.0516)	0.0241 (0.0179)
International Asian	-0.0146 (0.0222)	0.0073 (0.0125)	0.0546 (0.0353)	-0.0309 (0.0263)	0.0108 (0.0399)	0.0075 (0.0112)
International Other	-0.0126 (0.0236)	0.0062 (0.0136)	0.0104 (0.0306)	-0.0100 (0.0237)	-0.0627 (0.0443)	-0.0014 (0.0106)
GRE Verbal Percentile	0.0017*** (0.0005)	0.0011** (0.0005)	0.0015 (0.0012)	-0.0002 (0.0009)	0.0008 (0.0010)	0.0008** (0.0004)
GRE Quantitative Percentile	0.0024*** (0.0009)	-0.0001 (0.0002)	0.0003 (0.0008)	0.0019** (0.0009)	0.0006 (0.0006)	0.0003 (0.0002)
GRE Analytic Percentile	0.0004 (0.0003)	0.0000 (0.0003)	-0.0007 (0.0009)	-0.0003 (0.0006)	0.0004 (0.0006)	0.0000 (0.0002)
Undergraduate GPA	0.0234 (0.0485)	0.0282 (0.0223)	0.1281** (0.0553)	-0.0820** (0.0358)	-0.0058 (0.0357)	0.0130 (0.0158)
TOEFL/IELTS	-0.0478** (0.0221)	0.0229 (0.0187)	-0.0723 (0.0645)	0.0152 (0.0466)	-0.0427 (0.0469)	0.0050 (0.0187)
Elite U.S. University	-0.0496* (0.0261)	-0.0100 (0.0145)	0.0054 (0.0324)	0.0297 (0.0297)	0.0229 (0.0274)	0.0036 (0.0108)
Elite U.S. Liberal Arts College	0.0248 (0.0312)	0.0078 (0.0180)	0.0223 (0.0469)	0.0303 (0.0602)	0.0502 (0.0418)	0.0217 (0.0160)
Elite Foreign University	-0.0166 (0.0308)	-0.0532 (0.0337)	0.0700 (0.0537)	0.0299 (0.0490)	0.1771*** (0.0484)	0.0158 (0.0179)
Other Foreign University	-0.0111 (0.0202)	0.0091 (0.0122)	0.0257 (0.0321)	0.0108 (0.0267)	0.0175 (0.0344)	0.0116 (0.0103)
Graduate Degree	-0.0020 (0.0166)	0.0079 (0.0109)	-0.0151 (0.0261)	-0.0542*** (0.0206)	-0.0161 (0.0220)	-0.0093 (0.0085)
Work Experience	0.0258* (0.0148)	-0.0017 (0.0107)	-0.0070 (0.0248)	-0.0165 (0.0213)	-0.0692** (0.0275)	-0.0093 (0.0086)
#Professor Recommenders	-0.0108 (0.0093)	0.0041 (0.0051)	-0.0173 (0.0140)	0.0136 (0.0114)	-0.0286*** (0.0110)	-0.0027 (0.0042)
#Prolific Recommenders	0.0004 (0.0067)	0.0098* (0.0050)	-0.0165 (0.0164)	0.0042 (0.0102)	0.0298** (0.0128)	0.0083* (0.0045)
#Math Courses	0.0037 (0.0043)					
Advanced Math	0.0201 (0.0145)					
Missing GRE	-0.0204 (0.0443)	-0.0090 (0.0098)	-0.0001 (0.0254)	-0.0027 (0.0218)	-0.0350 (0.0239)	-0.0098 (0.0079)
Missing Undergraduate GPA	0.0157 (0.0158)	0.0040 (0.0102)	-0.0232 (0.0285)	0.0392* (0.0201)	-0.0207 (0.0254)	0.0019 (0.0083)
Missing TOEFL/IELTS	-0.0302* (0.0168)	0.0218 (0.0156)	0.0544 (0.0517)	0.0084 (0.0272)	-0.1205*** (0.0434)	0.0063 (0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic DF P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 11: Means by Program

(a) All Applicants

(b) Accept = 1

	Economics	Government	History	Linguistics	Psychology		Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535	Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004	Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230	Female	0.4850	0.5000	0.5588	0.5800	0.6154
Female	0.4201	0.4182	0.4232	0.6133	0.7490	U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728	U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576	International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523	International Other	0.2575	0.2292	0.3235	0.2000	0.1538
International Other	0.2517	0.2967	0.2203	0.3256	0.1584	GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192	GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000	GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077	Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695	TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158	Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840	Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535	Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206	Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852	Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259	Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593	#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831	#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128	#Math Courses	7.2575				
#Math Courses	6.6396					Advanced Math	0.6781				
Advanced Math	0.4827					Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650	Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490	Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827	Observations	233	96	68	50	26
Observations	2078	2231	690	737	486						

Table 12: Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Table 13: Elastic-Net Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3323	-0.3013	-0.3652	-0.2642	-0.0428
Winsorized Age	-0.0786	0.0231	0.0235		
Female	0.3316	0.4214	0.4373		-0.2716
U.S. African/Hispanic/Native	0.2622	0.7009	0.6038		-0.2580
U.S. Asian	-0.2741	-0.4137	-0.3622	0.1731	0.4670
International Asian	-0.6634	-0.1850	0.1309	0.1086	-0.2639
International Other	0.4732	0.1832	0.5739	-0.1280	0.0824
GRE Verbal Percentile	0.0140	0.0187	0.0188		0.0236
GRE Quantitative Percentile	0.1324	0.0117	0.0166	0.0136	
GRE Analytic Percentile	0.0165	0.0157	0.0055	0.0089	
Undergraduate GPA	1.9971	0.5294	0.7165	0.3104	0.3833
TOEFL/IELTS	0.4029	0.4179	0.6717	0.5298	
Elite U.S. University	0.6894	0.3109	0.3740		0.5669
Elite U.S. Liberal Arts College	0.8906	0.5376	0.6277		0.7887
Elite Foreign University	0.3738	0.0332	0.3675	0.5020	0.3558
Other Foreign University		0.0233	0.1382		
Graduate Degree		0.0315	0.1873		0.1018
Work Experience	0.2822	0.3230	0.3192		0.5851
#Professor Recommenders		-0.0179	0.3250	0.2333	
#Prolific Recommenders	0.2238	0.2164	0.1665	0.2212	0.1970
#Math Courses	0.1489				
Advanced Math	0.2309				
Missing GRE		-0.3089	0.1123	-0.0864	-0.5616
Missing Undergraduate GPA		-0.1040	-0.1046		
Missing TOEFL/IELTS	-0.2369	0.2950	0.2603		-0.1782
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2640	0.1422	0.1626	0.1519	0.1653
CV Pseudo-R ²	0.2297	0.0999	0.0786	0.0859	0.0582
α^*	0.9	0.0	0.0	0.4	0.1

Table 14: Post-Lasso Logistic Regression Results

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3960** (0.1996)	-0.4191 (0.2609)	-0.5330 (0.3435)	-0.5737 (0.3824)	
Winsorized Age	-0.0977*** (0.0333)	0.0473* (0.0248)	0.0440 (0.0325)		
Female	0.4106** (0.1670)	0.6570*** (0.2281)	0.6797** (0.2932)		-0.3204 (0.4908)
U.S. African/Hispanic/Native	0.4355 (0.5232)	0.9687*** (0.3521)	1.1594*** (0.4187)		
U.S. Asian	-0.5139 (0.4149)	-0.5533 (0.6435)		0.6970 (0.6906)	0.7934 (0.6164)
International Asian	-0.7960*** (0.2665)	-0.0979 (0.3808)		0.3796 (0.4458)	
International Other	0.5490* (0.2838)	0.3749 (0.3238)	1.1102** (0.4378)	-0.2036 (0.4436)	
GRE Verbal Percentile	0.0155*** (0.0059)	0.0335* (0.0180)	0.0423** (0.0188)		0.0498*** (0.0185)
GRE Quantitative Percentile	0.1549*** (0.0184)	0.0156* (0.0091)	0.0255** (0.0106)	0.0157 (0.0120)	
GRE Analytic Percentile	0.0177*** (0.0041)	0.0232** (0.0101)		0.0159 (0.0121)	
Undergraduate GPA	2.2336*** (0.6626)	0.7028 (0.5465)	1.3004** (0.5090)	0.9096 (0.9856)	
TOEFL/IELTS	0.3900 (0.2878)	0.6701 (0.4545)	1.0010* (0.5503)	0.8625* (0.5186)	
Elite U.S. University	0.8292*** (0.2312)	0.3801 (0.2728)	0.5964* (0.3491)		0.7688 (0.4721)
Elite U.S. Liberal Arts College	1.0523*** (0.3373)	0.6501* (0.3553)	1.0178** (0.5124)		1.3415 (0.8540)
Elite Foreign University	0.4772* (0.2702)		0.5710 (0.5770)	0.9748 (0.5963)	
Graduate Degree			0.2774 (0.3193)		
Work Experience	0.3826** (0.1881)	0.4139 (0.2987)	0.5055 (0.3290)		1.6031** (0.6992)
#Professor Recommenders			0.5752** (0.2597)	0.4712* (0.2697)	
#Prolific Recommenders	0.2727*** (0.0840)	0.3028** (0.1229)	0.2113 (0.1577)	0.3359** (0.1456)	0.4341 (0.2885)
#Math Courses	0.1715*** (0.0593)				
Advanced Math	0.2350 (0.1777)				
Missing GRE	0.1827 (0.7660)			-0.0909 (0.3941)	-0.8985 (0.5569)
Missing TOEFL/IELTS	-0.3983* (0.2266)	0.5313 (0.4700)	0.5689 (0.5289)		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R ²	0.2671	0.1500	0.1847	0.1722	0.1844
CV Pseudo-R ²	0.2168	0.0865	0.0715	0.0104	0.0445

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.0303** (0.0152)	-0.0159 (0.0100)	-0.0404 (0.0261)	-0.0315 (0.0212)	
Winsorized Age	-0.0075*** (0.0025)	0.0018* (0.0010)	0.0033 (0.0025)		
Female	0.0314** (0.0128)	0.0249*** (0.0087)	0.0515** (0.0220)		-0.0144 (0.0222)
U.S. African/Hispanic/Native	0.0333 (0.0400)	0.0367*** (0.0136)	0.0878*** (0.0318)		
U.S. Asian	-0.0394 (0.0318)	-0.0210 (0.0245)		0.0383 (0.0382)	0.0357 (0.0270)
International Asian	-0.0610*** (0.0203)	-0.0037 (0.0144)			0.0208 (0.0245)
International Other	0.0420* (0.0217)	0.0142 (0.0123)	0.0841** (0.0333)	-0.0112 (0.0245)	
GRE Verbal Percentile	0.0012*** (0.0005)	0.0013* (0.0007)	0.0032** (0.0015)		0.0022** (0.0009)
GRE Quantitative Percentile	0.0119*** (0.0014)	0.0006* (0.0004)	0.0019** (0.0008)	0.0009 (0.0007)	
GRE Analytic Percentile	0.0014*** (0.0003)	0.0009** (0.0004)		0.0009 (0.0007)	
Undergraduate GPA	0.1710*** (0.0506)	0.0267 (0.0208)	0.0985** (0.0397)	0.0499 (0.0542)	
TOEFL/IELTS	0.0299 (0.0219)	0.0254 (0.0173)	0.0758* (0.0420)	0.0474 (0.0290)	
Elite U.S. University	0.0635*** (0.0176)	0.0144 (0.0104)	0.0452* (0.0262)		0.0346 (0.0214)
Elite U.S. Liberal Arts College	0.0806*** (0.0255)	0.0247* (0.0136)	0.0771** (0.0387)		0.0604 (0.0398)
Elite Foreign University	0.0365* (0.0207)		0.0432 (0.0434)	0.0535 (0.0331)	
Graduate Degree			0.0210 (0.0243)		
Work Experience	0.0293*** (0.0143)	0.0157 (0.0113)	0.0383 (0.0250)		0.0722** (0.0335)
#Professor Recommenders			0.0436** (0.0195)	0.0259* (0.0147)	
#Prolific Recommenders	0.0209*** (0.0064)	0.0115** (0.0047)	0.0160 (0.0119)	0.0184** (0.0080)	0.0195 (0.0127)
#Math Courses	0.0131*** (0.0045)				
Advanced Math	0.0180 (0.0136)				
Missing GRE	0.0140 (0.0587)			-0.0050 (0.0216)	-0.0404 (0.0253)
Missing TOEFL/IELTS	-0.0305* (0.0175)	0.0202 (0.0179)	0.0431 (0.0399)		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 15: Economics Means (Matriculation)

Mean	All Observations	Matriculate = 1
Matriculate	0.2814	
Covid Year	0.1905	0.2000
Winsorized Age	25.0303	25.1692
Female	0.4848	0.4154
International	0.6710	0.7385
GRE Verbal Percentile	84.5065	79.7385
GRE Quantitative Percentile	91.9784	91.2308
GRE Analytic Percentile	69.1775	65.4615
Undergraduate GPA	3.7160	3.6868
TOEFL/IELTS	7.6394	7.5953
Elite Undergraduate	0.4892	0.4000
Graduate Degree	0.5801	0.6308
Work Experience	0.6320	0.5231
#Professor Recommenders	2.5498	2.5846
#Prolific Recommenders	1.2727	1.1692
Advanced Math	0.6753	0.5077
Missing Undergraduate GPA	0.4892	0.5077
Missing TOEFL/IELTS	0.6537	0.6000
Observations	231	65

Table 16: Correlation Matrix (Matriculation)

	Covid	Age	Fem	Int	GREV	GREQ	GREA	GPA	T/I	Elite	Grad	Work	Prof	Pro	Math	MGPA	MT/I
Covid Year	1.00																
Winsorized Age	0.03	1.00															
Female	0.04	-0.09	1.00														
International	-0.04	0.11	0.02	1.00													
GRE Verbal Percentile	-0.04	-0.14	-0.04	-0.34	1.00												
GRE Quantitative Percentile	0.03	-0.03	-0.15	0.26	0.04	1.00											
GRE Analytic Percentile	0.08	-0.09	-0.01	-0.50	0.44	-0.19	1.00										
Undergraduate GPA	-0.00	-0.37	0.10	-0.40	0.21	-0.13	0.23	1.00									
TOEFL/IELTS	-0.06	-0.05	-0.03	0.24	0.23	0.15	0.02	-0.21	1.00								
Elite Undergraduate	-0.03	-0.17	0.11	-0.37	0.33	-0.00	0.27	0.19	-0.18	1.00							
Graduate Degree	-0.06	0.42	-0.12	0.43	-0.25	0.12	-0.25	-0.48	0.17	-0.38	1.00						
Work Experience	-0.04	0.34	0.00	-0.19	0.08	-0.17	0.30	-0.02	-0.03	0.05	-0.07	1.00					
#Professor Recommenders	-0.06	-0.24	-0.01	0.23	-0.05	0.09	-0.11	0.11	0.02	-0.07	0.05	-0.30	1.00				
#Prolific Recommenders	0.04	-0.02	0.14	0.14	-0.02	0.11	-0.08	-0.11	0.10	-0.11	0.21	-0.02	0.22	1.00			
Advanced Math	0.01	-0.17	0.01	-0.21	0.24	0.06	0.17	0.15	-0.12	0.22	-0.25	0.01	0.04	0.14	1.00		
Missing Undergraduate GPA	0.01	0.28	-0.12	0.65	-0.34	0.18	-0.39	-0.64	0.32	-0.58	0.57	-0.04	0.03	0.19	-0.26	1.00	
Missing TOEFL/IELTS	0.05	-0.20	0.12	-0.51	0.27	-0.08	0.38	0.45	-0.46	0.49	-0.36	0.12	0.01	-0.00	0.23	-0.71	1.00
Observations	231																

Table 17: Logistic Regression Results (Matriculation)

Dependent Variable = Matriculate	Coefficient	AME	Lasso	Post-Lasso Coefficient	Post-Lasso AME
Covid Year	0.1384 (0.4306)	0.0232 (0.0724)			
Winsorized Age	0.0261 (0.0900)	0.0044 (0.0152)			
Female	-0.6244* (0.3570)	-0.1049* (0.0589)	-0.3914	-0.6158* (0.3471)	-0.1037* (0.0573)
International	0.7372 (0.5175)	0.1239 (0.0856)	0.1687	0.6690 (0.4693)	0.1127 (0.0779)
GRE Verbal Percentile	-0.0224* (0.0130)	-0.0038* (0.0021)	-0.0194	-0.0215* (0.0123)	-0.0036* (0.0020)
GRE Quantitative Percentile	-0.0609* (0.0348)	-0.0102* (0.0058)	-0.0425	-0.0631* (0.0341)	-0.0106* (0.0056)
GRE Analytic Percentile	0.0034 (0.0091)	0.0006 (0.0015)			
Undergraduate GPA	-4.1670*** (1.6083)	-0.7001*** (0.2570)	-2.2674	-4.1144*** (1.4501)	-0.6932*** (0.2319)
TOEFL/IELTS	-0.3597 (0.5858)	-0.0604 (0.0984)	-0.1628	-0.2931 (0.5263)	-0.0494 (0.0887)
Elite Undergraduate	-0.9465* (0.4954)	-0.1590* (0.0827)	-0.3943	-0.9245* (0.4844)	-0.1558* (0.0816)
Graduate Degree	-0.2158 (0.4760)	-0.0363 (0.0799)			
Work Experience	-0.8483** (0.4006)	-0.1425** (0.0666)	-0.6164	-0.7994** (0.3458)	-0.1347** (0.0572)
#Professor Recommenders	0.0296 (0.2316)	0.0050 (0.0389)			
#Prolific Recommenders	0.0268 (0.1948)	0.0045 (0.0328)			
Advanced Math	-1.0856*** (0.3627)	-0.1824*** (0.0574)	-0.8427	-1.0371*** (0.3463)	-0.1747*** (0.0548)
Missing Undergraduate GPA	-2.1263*** (0.8113)	-0.3573*** (0.1304)	-0.9377	-2.0496*** (0.7114)	-0.3453*** (0.1135)
Missing TOEFL/IELTS	-0.1722 (0.6110)	-0.0289 (0.1026)			
Observations	231	231	231	231	231
Pseudo-R ²	0.1469		0.1288	0.1445	
CV Pseudo-R ²	-0.0928		0.0112	0.0485	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 18: Simulated Wald Test Summary Statistics

(a) Same Determinants

	Economics		Non-Economics 1		Non-Economics 2	
	Mean	SD	Mean	SD	Mean	SD
Accept	0.1335	0.3401	0.0900	0.2862	0.0786	0.2691
Success	0.3000	0.4583	0.4057	0.4910	0.1157	0.3199
GRE Quantitative Percentile	84.8390	7.5405	65.1343	7.4495	65.6114	7.6706
Gender	0.3935	0.4885	0.3952	0.4889	0.5914	0.4916
Demographic 1	0.5005	0.5000	0.3091	0.4621	0.2600	0.4386
Demographic 2	0.2505	0.4333	0.2957	0.4563	0.3414	0.4742
Advanced Math	0.5095	0.4999				
Committee Score	4.9325	1.9877	6.0191	2.0205	6.0243	2.0038
Observations	2000		2300		700	

(b) Different Determinants

	Economics		Non-Economics 1		Non-Economics 2	
	Mean	SD	Mean	SD	Mean	SD
Accept	0.1335	0.3401	0.0452	0.2078	0.0543	0.2266
Success	0.3000	0.4583	0.3126	0.4636	0.1314	0.3379
GRE Quantitative Percentile	84.8390	7.5405	65.1343	7.4495	65.6114	7.6706
Gender	0.3935	0.4885	0.3952	0.4889	0.5914	0.4916
Demographic 1	0.5005	0.5000	0.3091	0.4621	0.2600	0.4386
Demographic 2	0.2505	0.4333	0.2957	0.4563	0.3414	0.4742
Advanced Math	0.5095	0.4999				
Committee Score	4.9325	1.9877	6.0191	2.0205	6.0243	2.0038
Observations	2000		2300		700	

Table 19: Simulated Wald Test Correlation Matrix

(a) Same Determinants

	Acc	Suc	GRE	Gen	Dem1	Dem2	Com	NEcon1	NEcon2
Accept	1								
Success	0.1603	1							
GRE Quantitative Percentile	0.2772	0.1665	1						
Gender	0.0484	-0.0581	-0.0795	1					
Demographic 1	-0.0233	0.0199	0.3377	-0.0136	1				
Demographic 2	0.0487	0.0148	-0.0876	-0.0209	-0.4918	1			
Committee Score	0.2045	0.1486	-0.0770	0.0980	-0.1567	-0.0359	1		
Non-Economics 1	-0.0474	0.1636	-0.6017	-0.0501	-0.1326	0.0238	0.1930	1	
Non-Economics 2	-0.0357	-0.1787	-0.2472	0.1384	-0.0988	0.0514	0.0854	-0.3724	1
Observations	5000								

(b) Different Determinants

	Acc	Suc	GRE	Gen	Dem1	Dem2	Com	NEcon1	NEcon2
Accept	1								
Success	0.0885	1							
GRE Quantitative Percentile	0.2376	0.1167	1						
Gender	0.0582	-0.0220	-0.0795	1					
Demographic 1	0.0152	-0.0004	0.3377	-0.0136	1				
Demographic 2	0.0078	0.0003	-0.0876	-0.0209	-0.4918	1			
Committee Score	0.1437	0.1500	-0.0770	0.0980	-0.1567	-0.0359	1		
Non-Economics 1	-0.1232	0.0624	-0.6017	-0.0501	-0.1326	0.0238	0.1930	1	
Non-Economics 2	-0.0405	-0.1352	-0.2472	0.1384	-0.0988	0.0514	0.0854	-0.3724	1
Observations	5000								

Table 20: Simulated Wald Test Results

(a) Same Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848*** (1.1315)	-1.5822*** (0.0956)	-28.5524*** (1.4963)	-1.3677*** (0.0500)	-10.4586*** (0.7236)	-1.9105*** (0.1070)	-10.2610*** (0.4815)	-1.8514*** (0.0628)
GRE Quantitative Percentile	0.1509*** (0.0131)	0.0146*** (0.0012)	0.3245*** (0.0179)	0.0155*** (0.0007)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.1154*** (0.0070)	0.0208*** (0.0011)
Gender	0.4714*** (0.1475)	0.0455*** (0.0140)	0.7880*** (0.1696)	0.0377*** (0.0081)	-0.1384 (0.1095)	-0.0253 (0.0200)	-0.0497 (0.0895)	-0.0090 (0.0161)
Demographic 1	-0.6287*** (0.2088)	-0.0607*** (0.0200)	-1.8488*** (0.2445)	-0.0886*** (0.0115)	-0.3486** (0.1467)	-0.0637** (0.0267)	-0.3079*** (0.1123)	-0.0556*** (0.0202)
Demographic 2	0.1531 (0.2072)	0.0148 (0.0200)	0.3703* (0.1929)	0.0177* (0.0091)	0.1756 (0.1558)	0.0321 (0.0284)	0.0282 (0.1069)	0.0051 (0.0193)
Advanced Math	0.2889* (0.1612)	0.0279* (0.0156)			0.3267*** (0.1112)	0.0597*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0216*** (0.0039)	0.3994*** (0.0506)	0.0191*** (0.0023)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.0757*** (0.0228)	0.0137*** (0.0041)
Non-Economics 1			0.8376*** (0.2297)	0.0401*** (0.0107)			1.9483*** (0.1236)	0.3515*** (0.0205)
Observations Pseudo-R ²	5000	0.3251			5000	0.1431		
LL Parameters AIC	-1130.4	14	2288.8		-2691.4	14	5410.8	
Wald Statistic DF P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Different Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291*** (1.1274)	-1.5784*** (0.0955)	-8.4753*** (0.8673)	-0.3642*** (0.0412)	-10.4581*** (0.7237)	-1.9104*** (0.1070)	-3.2236*** (0.4018)	-0.5997*** (0.0728)
GRE Quantitative Percentile	0.1502*** (0.0131)	0.0145*** (0.0012)	0.0402*** (0.0128)	0.0017*** (0.0006)	0.1058*** (0.0088)	0.0193*** (0.0014)	0.0034 (0.0062)	0.0006 (0.0012)
Gender	0.4702*** (0.1473)	0.0454*** (0.0140)	0.5942*** (0.1820)	0.0255*** (0.0079)	-0.1372 (0.1095)	-0.0251 (0.0200)	0.0375 (0.0868)	0.0070 (0.0161)
Demographic 1	-0.6246*** (0.2085)	-0.0604*** (0.0200)	-0.2541 (0.2393)	-0.0109 (0.0103)	-0.3544** (0.1467)	-0.0647** (0.0267)	-0.1152 (0.1103)	-0.0214 (0.0205)
Demographic 2	0.1555 (0.2070)	0.0150 (0.0200)	0.0081 (0.2109)	0.0003 (0.0091)	0.1713 (0.1557)	0.0313 (0.0284)	-0.0562 (0.1030)	-0.0105 (0.0192)
Advanced Math	0.2886* (0.1611)	0.0279* (0.0156)			0.3284*** (0.1112)	0.0600*** (0.0202)		
Committee Score	0.2241*** (0.0413)	0.0217*** (0.0039)	0.3817*** (0.0528)	0.0164*** (0.0025)	0.1026*** (0.0287)	0.0187*** (0.0052)	0.1808*** (0.0228)	0.0336*** (0.0041)
Non-Economics 1			-0.0452 (0.1995)	-0.0019 (0.0086)			1.1393*** (0.1219)	0.2120*** (0.0219)
Observations Pseudo-R ²	5000	0.1488			5000	0.0758		
LL Parameters AIC	-1155.7	14	2339.5		-2747.2	14	5522.4	
Wald Statistic DF P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

C Additional Structural Results

Table 21: Simulated Summary Statistics

(a) Optimal Committee			(b) Suboptimal Committee		
	Mean	SD		Mean	SD
Transition	0.3740	0.4839	Transition	0.3850	0.4866
Waitlist	0.3040	0.4600	Waitlist	0.2960	0.4565
Accept	0.1260	0.3318	Accept	0.1320	0.3385
Matriculate	0.0540	0.2260	Matriculate	0.0480	0.2138
Success	0.3080	0.4617	Success	0.3070	0.4612
GRE Quantitative Percentile	84.9310	7.5286	GRE Quantitative Percentile	84.9310	7.5286
Gender	0.3970	0.4893	Gender	0.3970	0.4893
Demographic 1	0.5380	0.4986	Demographic 1	0.5380	0.4986
Demographic 2	0.2580	0.4375	Demographic 2	0.2580	0.4375
Advanced Math	0.5060	0.5000	Advanced Math	0.5060	0.5000
Committee Score	5.0240	2.0638	Committee Score	5.0240	2.0638
Year Transition/Matriculate More	0.5000	0.5000	Year Transition/Matriculate More	0.5000	0.5000
Program Rank	77.3340	53.8839	Program Rank	77.1650	53.8030
Observations	1000		Observations	1000	

Table 22: Simulated Correlation Matrix

(a) Optimal Committee														
	Trans	Wait	Acc	Mat	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year	Rank	
Transition	1													
Waitlist	0.8550	1												
Accept	0.4912	0.1159	1											
Matriculate	0.3091	0.1018	0.6292	1										
Success	0.0797	0.0253	0.1514	0.1952	1									
GRE Quantitative Percentile	0.3387	0.2134	0.3353	0.2296	0.3433	1								
Gender	0.0233	0.0147	0.0307	0.0141	0.0076	-0.0398	1							
Demographic 1	0.0157	-0.0198	0.0255	0.0262	0.0621	0.2726	-0.0147	1						
Demographic 2	0.0449	0.0475	0.0309	0.0108	-0.0171	-0.0817	0.0027	-0.6363	1					
Advanced Math	0.0238	-0.0079	0.0859	0.0856	0.1696	0.1015	-0.0608	0.0994	-0.0436	1				
Committee Score	0.0221	-0.0024	0.0467	-0.0478	0.0898	0.0440	-0.0292	-0.0670	0.0673	0.2325	1			
Year Transition/Matriculate More	0.1860	0.2044	-0.0181	0.1416	-0.0477	-0.0410	-0.0225	-0.0361	0.0457	0.0120	0.0329	1		
Program Rank	-0.0484	0.0114	-0.1432	-0.1788	-0.4839	-0.2223	-0.0013	-0.0508	0.0029	-0.1054	-0.0424	0.0451	1	
Observations	1000													

(b) Suboptimal Committee														
	Trans	Wait	Acc	Mat	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year	Rank	
Transition	1													
Waitlist	0.8195	1												
Accept	0.4929	0.0254	1											
Matriculate	0.2838	-0.0124	0.5758	1										
Success	0.0749	0.0386	0.0927	0.1345	1									
GRE Quantitative Percentile	0.3032	0.2166	0.2198	0.1468	0.3433	1								
Gender	0.0301	0.0425	0.0157	-0.0005	0.0050	-0.0398	1							
Demographic 1	-0.0830	-0.0670	-0.0712	0.0017	0.0602	0.2726	-0.0147	1						
Demographic 2	0.1018	0.0632	0.0874	0.0066	-0.0208	-0.0817	0.0027	-0.6363	1					
Advanced Math	0.0254	-0.0122	0.0662	0.0628	0.1633	0.1015	-0.0608	0.0994	-0.0436	1				
Committee Score	0.0446	-0.0585	0.1515	0.0133	0.0921	0.0440	-0.0292	-0.0670	0.0673	0.2325	1			
Year Transition/Matriculate More	0.2158	0.1928	0.0236	0.1684	-0.0499	-0.0410	-0.0225	-0.0361	0.0457	0.0120	0.0329	1		
Program Rank	-0.0608	-0.0029	-0.1116	-0.1676	-0.4782	-0.2140	0.0057	-0.0400	-0.0021	-0.1029	-0.0463	0.0401	1	
Observations	1000													

Table 23: Unrestricted Dynamic Estimates

(a) Optimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.8856*** (0.5961)	-2.2380*** (0.1481)	-1.8387** (0.7553)	-0.4508** (0.1830)	1.1272 (16.2398)	-11.5951*** (1.0237)	-2.2567*** (0.1454)	-13.1774*** (2.0106)	-2.3765*** (0.2887)	-9.5941*** (2.0756)	-1.4923*** (0.2688)
GRE Quantitative Percentile	0.0734*** (0.0069)	0.0239*** (0.0018)	0.0200** (0.0090)	0.0049** (0.0022)	0.0347 (0.1464)	0.1236*** (0.0118)	0.0241*** (0.0018)	0.1373*** (0.0223)	0.0248*** (0.0033)	0.1166*** (0.0228)	0.0181*** (0.0029)
Gender	0.1239 (0.0876)	0.0403 (0.0283)	-0.2344* (0.1309)	-0.0575* (0.0319)	-0.0191 (0.3331)	0.2273 (0.1464)	0.0442 (0.0283)	0.1326 (0.2503)	0.0239 (0.0451)	0.1266 (0.2701)	0.0197 (0.0419)
Demographic 1	-0.1835 (0.1204)	-0.0597 (0.0390)	0.3700** (0.1732)	0.0907** (0.0421)	-0.4213 (0.4246)	-0.3016 (0.1988)	-0.0587 (0.0386)	-0.2534 (0.3751)	-0.0457 (0.0674)	-0.4621 (0.3935)	-0.0719 (0.0608)
Demographic 2	0.1038 (0.1371)	0.0338 (0.0446)	0.0915 (0.1910)	0.0224 (0.0468)	-0.1378 (0.3798)	0.1541 (0.2215)	0.0300 (0.0431)	0.1815 (0.3852)	0.0327 (0.0694)	-0.2358 (0.4325)	-0.0367 (0.0672)
Advanced Math					0.6844*** (0.2024)			0.7755*** (0.2536)	0.1399*** (0.0443)	0.6262** (0.2751)	0.0974** (0.0417)
Committee Score								0.0786 (0.0602)	0.0142 (0.0108)	0.0464 (0.0656)	0.0072 (0.0102)
Year Transition More	0.5856*** (0.0896)	0.1903*** (0.0271)	0.0650 (0.1280)	0.0159 (0.0314)	0.3547 (1.1695)	0.9842*** (0.1473)	0.1916*** (0.0263)	-0.9247*** (0.2514)	-0.1668*** (0.0427)	-0.0245 (0.2764)	-0.0038 (0.0430)
Program Rank										-0.0282*** (0.0027)	-0.0044*** (0.0003)
Inverse Mills			-0.0117 (0.0876)	-0.0029 (0.0215)	0.6840 (3.0216)						
√MSE					1.9475*** (0.0675)						
Observations	1000										
LL Parameters AIC	-2980.4 45 6050.9										

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.6689*** (0.5819)	-2.1611*** (0.1472)	-1.9352** (0.7570)	-0.4741*** (0.1830)	2.4660 (13.2168)	-11.2317*** (1.0146)	-2.1804*** (0.1474)	-6.7297*** (1.7378)	-1.3294*** (0.3202)	-9.6076*** (1.9611)	-1.4988*** (0.2508)
GRE Quantitative Percentile	0.0728*** (0.0068)	0.0236*** (0.0018)	0.0211** (0.0090)	0.0052** (0.0022)	0.0241 (0.1207)	0.1226*** (0.0117)	0.0238*** (0.0018)	0.0550*** (0.0191)	0.0109*** (0.0036)	0.1163*** (0.0216)	0.0181*** (0.0027)
Gender	0.1429 (0.0878)	0.0463 (0.0283)	-0.2331* (0.1310)	-0.0571* (0.0319)	-0.0487 (0.3253)	0.2576* (0.1469)	0.0500* (0.0283)	0.0094 (0.2368)	0.0019 (0.0468)	0.0582 (0.2669)	0.0091 (0.0416)
Demographic 1	-0.4938*** (0.1210)	-0.1600*** (0.0382)	0.3684** (0.1733)	0.0903** (0.0421)	-0.6607 (0.8101)	-0.8297*** (0.1966)	-0.1611*** (0.0371)	-0.3342 (0.3394)	-0.0660 (0.0669)	-0.5149 (0.3648)	-0.0803 (0.0563)
Demographic 2	0.0593 (0.1383)	0.0192 (0.0448)	0.0925 (0.1911)	0.0227 (0.0468)	-0.2616 (0.2999)	0.0795 (0.2135)	0.0154 (0.0414)	0.1766 (0.3346)	0.0349 (0.0659)	-0.4742 (0.3857)	-0.0740 (0.0598)
Advanced Math					0.7204*** (0.1970)			0.3577 (0.2376)	0.0707 (0.0465)	0.5362** (0.2704)	0.0836** (0.0416)
Committee Score								0.2963*** (0.0592)	0.0585*** (0.0106)	0.0616 (0.0643)	0.0096 (0.0100)
Year Transition More	0.6627*** (0.0892)	0.2148*** (0.0264)	0.0672 (0.1281)	0.0165 (0.0314)	0.2348 (1.1150)	1.1109*** (0.1479)	0.2156*** (0.0257)	-0.7196*** (0.2397)	-0.1422*** (0.0455)	0.1322 (0.2795)	0.0206 (0.0434)
Program Rank										-0.0279*** (0.0028)	-0.0044*** (0.0003)
Inverse Mills			0.0217 (0.0877)	0.0053 (0.0215)	0.5369 (2.5822)						
√MSE					1.9192*** (0.0682)						
Observations	1000										
LL Parameters AIC	-3022.8 45 6135.6										

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 24: Restricted Dynamic Estimates

(a) Optimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-6.8785*** (0.5931)	-2.1460*** (0.7578)	1.6943 (16.0475)	-0.3317*** (0.0799)	0.7177 (0.6137, 0.8394)	-0.0013 (0.2112)	49.97% (39.77%, 60.17%)	-9.4681*** (1.9363)	
GRE Quantitative Percentile	0.0733*** (0.0069)	0.0238*** (0.0090)	0.0301 (0.1447)					0.1129*** (0.0214)	
Gender	0.1239 (0.0875)	-0.2142 (0.1305)	-0.0162 (0.3308)					0.2242* (0.1228)	
Demographic 1	-0.1830 (0.1206)	0.3587** (0.1725)	-0.4137 (0.4217)					-0.3289* (0.1830)	
Demographic 2	0.1032 (0.1366)	0.0928 (0.1906)	-0.1487 (0.3764)					0.0191 (0.1770)	
Advanced Math			0.6902*** (0.2026)					0.5665*** (0.1716)	
Committee Score								0.0494 (0.0376)	
Year Transition More	0.5862*** (0.0890)	0.0735 (0.1278)	0.3078 (1.1597)	-0.0264 (0.0929)	0.6990 (0.5943, 0.8222)	0.5684* (0.3209)	63.81% (52.90%, 73.46%)	0.0551 (0.2667)	
Program Rank								-0.0283*** (0.0027)	
Inverse Mills		0.0059 (0.0873)	0.5449 (2.9855)						
$\sqrt{\text{MSE}}$			1.9477*** (0.0675)						
Sigma									0.1364*** (0.0247)
Observations	1000								
LL Parameters AIC	-3007.4	36	6086.9						
LR Statistic DF P-Value	53.9671	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-6.6818*** (0.5799)	-2.1644*** (0.7594)	4.2962 (12.7315)	-0.3502*** (0.1027)	0.7046 (0.5761, 0.8617)	-0.0924 (0.2223)	47.69% (37.10%, 58.50%)	-7.3142*** (2.2678)	
GRE Quantitative Percentile	0.0730*** (0.0068)	0.0241*** (0.0090)	0.0100 (0.1164)					0.0830*** (0.0242)	
Gender	0.1429 (0.0878)	-0.2188* (0.1307)	-0.0564 (0.3142)					0.1543 (0.1202)	
Demographic 1	-0.4963*** (0.1208)	0.3435** (0.1730)	-0.5701 (0.7888)					-0.4873** (0.2057)	
Demographic 2	0.0526 (0.1367)	0.1010 (0.1907)	-0.2640 (0.2871)					-0.0116 (0.1751)	
Advanced Math			0.7383*** (0.1977)					0.3463** (0.1685)	
Committee Score								0.1702*** (0.0363)	
Year Transition More	0.6650*** (0.0892)	0.0694 (0.1278)	0.0120 (1.0816)	-0.0581 (0.0979)	0.6648 (0.5489, 0.8052)	0.5363 (0.3410)	60.92% (49.31%, 71.41%)	0.1172 (0.2764)	
Program Rank								-0.0280*** (0.0027)	
Inverse Mills		0.0303 (0.0875)	0.0036 (2.4835)						
$\sqrt{\text{MSE}}$			1.9216*** (0.0687)						
Sigma									0.1427*** (0.0392)
Observations	1000								
LL Parameters AIC	-3067.2	36	6206.4						
LR Statistic DF P-Value	88.8580	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 25: Unrestricted Myopic Estimates

(a) Optimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.8922*** (0.5855)	-2.2397*** (0.1455)	0.0000 (0.0894)	0.0000 (0.0224)	4.6343*** (0.3851)	-11.5884*** (1.0233)	-2.2558*** (0.1454)	-13.1808*** (2.0107)	-2.3771*** (0.2886)	-9.5620*** (2.0727)	-1.4880*** (0.2688)
GRE Quantitative Percentile	0.0735*** (0.0068)	0.0239*** (0.0018)				0.1236*** (0.0118)	0.0241*** (0.0018)	0.1374*** (0.0223)	0.0248*** (0.0033)	0.1161*** (0.0227)	0.0181*** (0.0029)
Gender	0.1231 (0.0878)	0.0400 (0.0284)				0.2270 (0.1464)	0.0442 (0.0283)	0.1287 (0.2504)	0.0232 (0.0451)	0.1256 (0.2700)	0.0196 (0.0419)
Demographic 1	-0.1833 (0.1207)	-0.0596 (0.0392)				-0.3013 (0.1988)	-0.0587 (0.0386)	-0.2625 (0.3746)	-0.0473 (0.0673)	-0.4572 (0.3933)	-0.0711 (0.0608)
Demographic 2	0.0998 (0.1325)	0.0324 (0.0430)				0.1537 (0.2215)	0.0299 (0.0431)	0.1729 (0.3847)	0.0312 (0.0693)	-0.2293 (0.4323)	-0.0357 (0.0672)
Advanced Math					0.6765*** (0.2007)			0.7742*** (0.2536)	0.1396*** (0.0443)	0.6294** (0.2750)	0.0979** (0.0417)
Committee Score								0.0786 (0.0602)	0.0142 (0.0108)	0.0465 (0.0655)	0.0072 (0.0102)
Year Transition More	0.5878*** (0.0874)	0.1910*** (0.0265)	0.0480 (0.1265)	0.0120 (0.0316)	0.1112 (0.2267)	0.9844*** (0.1472)	0.1916*** (0.0263)	-0.9246*** (0.2514)	-0.1667*** (0.0427)	-0.0256 (0.2763)	-0.0040 (0.0430)
Program Rank										-0.0282*** (0.0027)	-0.0044*** (0.0003)
Inverse Mills			-0.0117 (0.0873)	-0.0029 (0.0218)	0.0310 (0.3086)						
$\sqrt{\text{MSE}}$					1.9512*** (0.0675)						
Observations	1000										
LL Parameters AIC	-2990.9	37	6055.8								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME		Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.6745*** (0.5802)	-2.1620*** (0.1471)	0.0001 (0.0894)	0.0000 (0.0224)	4.7861*** (0.3886)	-11.2475*** (1.0157)	-2.1826*** (0.1474)	-6.7350*** (1.7379)	-1.3306*** (0.3202)	-9.4454*** (1.9475)	-1.4773*** (0.2511)
GRE Quantitative Percentile	0.0729*** (0.0067)	0.0236*** (0.0018)				0.1228*** (0.0117)	0.0238*** (0.0018)	0.0551*** (0.0191)	0.0109*** (0.0036)	0.1146*** (0.0214)	0.0179*** (0.0027)
Gender	0.1436 (0.0882)	0.0465 (0.0284)				0.2579* (0.1469)	0.0500* (0.0283)	0.0099 (0.2368)	0.0019 (0.0468)	0.0571 (0.2663)	0.0089 (0.0416)
Demographic 1	-0.5025*** (0.1224)	-0.1628*** (0.0388)				-0.8300*** (0.1967)	-0.1611*** (0.0371)	-0.3342 (0.3394)	-0.0660 (0.0669)	-0.5108 (0.3638)	-0.0799 (0.0563)
Demographic 2	0.0498 (0.1303)	0.0161 (0.0422)				0.0798 (0.2136)	0.0155 (0.0414)	0.1763 (0.3346)	0.0348 (0.0659)	-0.4742 (0.3845)	-0.0742 (0.0598)
Advanced Math					0.7042*** (0.1959)			0.3576 (0.2375)	0.0707 (0.0465)	0.5350** (0.2698)	0.0837** (0.0416)
Committee Score								0.2963*** (0.0592)	0.0585*** (0.0106)	0.0603 (0.0641)	0.0094 (0.0100)
Year Transition More	0.6650*** (0.0876)	0.2154*** (0.0259)	0.0479 (0.1265)	0.0120 (0.0316)	0.0131 (0.2328)	1.1114*** (0.1480)	0.2157*** (0.0257)	-0.7177*** (0.2397)	-0.1418*** (0.0455)	0.1269 (0.2787)	0.0198 (0.0434)
Program Rank										-0.0279*** (0.0028)	-0.0044*** (0.0003)
Inverse Mills			0.0215 (0.0874)	0.0054 (0.0218)	-0.0260 (0.3194)						
$\sqrt{\text{MSE}}$					1.9289*** (0.0683)						
Observations	1000										
LL Parameters AIC	-3034.5	37	6143.0								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 26: Restricted Myopic Estimates

(a) Optimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-6.8898*** (0.5858)	0.0138 (0.0889)	4.6976*** (0.3827)	-0.3379*** (0.0805)	0.7133 (0.6092, 0.8352)	-0.0114 (0.2109)	49.72% (39.54%, 59.92%)	-9.5277*** (1.9476)	
GRE Quantitative Percentile	0.0735*** (0.0068)							0.1136*** (0.0215)	
Gender	0.1236 (0.0878)							0.2123* (0.1216)	
Demographic 1	-0.1869 (0.1208)							-0.3182* (0.1803)	
Demographic 2	0.0984 (0.1325)							0.0218 (0.1747)	
Advanced Math			0.6791*** (0.2009)					0.5421*** (0.1700)	
Committee Score								0.0501 (0.0376)	
Year Transition More	0.5880*** (0.0875)	0.0534 (0.1261)	0.0953 (0.2262)	-0.0200 (0.0935)	0.6992 (0.5938, 0.8232)	0.5846* (0.3224)	63.95% (52.95%, 73.66%)	0.0680 (0.2666)	
Program Rank								-0.0283*** (0.0027)	
Inverse Mills		0.0059 (0.0871)	-0.0157 (0.3085)						
$\sqrt{\text{MSE}}$			1.9515*** (0.0675)						
Sigma									0.1360*** (0.0247)
Observations	1000								
LL Parameters AIC	-3018.6	28	6093.2						
LR Statistic DF P-Value	55.4004	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-6.6479*** (0.5823)	0.0117 (0.0892)	5.0956*** (0.3911)	-0.3313*** (0.1040)	0.7180 (0.5857, 0.8803)	-0.0989 (0.2260)	47.53% (36.77%, 58.52%)	-8.0392*** (2.3887)	
GRE Quantitative Percentile	0.0727*** (0.0068)							0.0909*** (0.0257)	
Gender	0.1419 (0.0884)							0.1542 (0.1261)	
Demographic 1	-0.5158*** (0.1227)							-0.5370** (0.2151)	
Demographic 2	0.0409 (0.1308)							-0.0343 (0.1874)	
Advanced Math			0.7098*** (0.1966)					0.3509** (0.1755)	
Committee Score								0.1785*** (0.0371)	
Year Transition More	0.6650*** (0.0877)	0.0496 (0.1263)	-0.0867 (0.2322)	-0.0615 (0.0974)	0.6752 (0.5561, 0.8197)	0.5725 (0.3489)	61.62% (49.67%, 72.32%)	0.1281 (0.2799)	
Program Rank								-0.0279*** (0.0027)	
Inverse Mills		0.0310 (0.0874)	-0.2695 (0.3211)						
$\sqrt{\text{MSE}}$			1.9314*** (0.0689)						
Sigma									0.1530*** (0.0410)
Observations	1000								
LL Parameters AIC	-3079.6	28	6215.1						
LR Statistic DF P-Value	90.0832	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 27: Unrestricted Matriculation-Dynamic Estimates

(a) Optimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	AME		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.8597*** (0.5924)	-2.2316*** (0.1477)	-1.9410** (0.7570)	-0.4755*** (0.1830)	1.6728 (16.2360)	-11.5715*** (1.0220)	-2.2534*** (0.1454)	-13.1576*** (2.0082)	-2.3740*** (0.2886)	-9.5642*** (2.0729)	-1.4883*** (0.2688)	-4.6191 (4.6546)	-0.6137 (0.6188)
GRE Quantitative Percentile	0.0731*** (0.0069)	0.0238*** (0.0018)	0.0212** (0.0090)	0.0052*** (0.0022)	0.0297 (0.1463)	0.1234*** (0.0117)	0.0240*** (0.0018)	0.1372*** (0.0223)	0.0248*** (0.0033)	0.1162*** (0.0227)	0.0181*** (0.0029)	0.0594 (0.0459)	0.0079 (0.0061)
Gender	0.1236 (0.0875)	0.0402 (0.0283)	-0.2327* (0.1310)	-0.0570* (0.0319)	-0.0267 (0.3325)	0.2263 (0.1463)	0.0441 (0.0283)	0.1299 (0.2502)	0.0234 (0.0451)	0.1265 (0.2700)	0.0197 (0.0419)	0.4891 (0.5072)	0.0650 (0.0657)
Demographic 1	-0.1822 (0.1204)	-0.0593 (0.0391)	0.3688** (0.1733)	0.0904** (0.0421)	-0.4086 (0.4230)	-0.3009 (0.1987)	-0.0586 (0.0386)	-0.2641 (0.3744)	-0.0477 (0.0672)	-0.4623 (0.3933)	-0.0719 (0.0608)	0.3647 (0.7467)	0.0485 (0.1001)
Demographic 2	0.1038 (0.1374)	0.0338 (0.0447)	0.0928 (0.1911)	0.0227 (0.0468)	-0.1425 (0.3777)	0.1537 (0.2214)	0.0299 (0.0431)	0.1715 (0.3844)	0.0310 (0.0693)	-0.2346 (0.4323)	-0.0365 (0.0672)	0.1894 (0.7969)	0.0252 (0.1069)
Advanced Math					0.6847*** (0.2024)			0.7729*** (0.2535)	0.1394*** (0.0443)	0.6277** (0.2750)	0.0977** (0.0417)	0.8131 (0.5781)	0.1080 (0.0733)
Committee Score								0.0784 (0.0602)	0.0141 (0.0108)	0.0465 (0.0655)	0.0072 (0.0102)	-0.7419*** (0.2070)	-0.0986*** (0.0235)
Year Transition/Matriculate More	0.5856*** (0.0890)	0.1905*** (0.0269)	0.0672 (0.1281)	0.0165 (0.0314)	0.3191 (1.1744)	0.9839*** (0.1472)	0.1916*** (0.0263)	-0.9248*** (0.2512)	-0.1669*** (0.0427)	-0.0234 (0.2763)	-0.0036 (0.0430)	3.3336*** (0.5689)	0.4429*** (0.0331)
Program Rank										-0.0282*** (0.0027)	-0.0044*** (0.0003)		
Inverse Mills			-0.0117 (0.0877)	-0.0029 (0.0215)	0.5822 (3.0278)								
√MSE					1.9473*** (0.0674)								
Observations	1000												
LL Parameters AIC	-3032.8	53	6171.6										

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Selection		Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	AME		Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.6176*** (0.5820)	-2.1485*** (0.1480)	-1.9502** (0.7572)	-0.4778*** (0.1830)	0.3577 (13.3153)	-11.2235*** (1.0139)	-2.1794*** (0.1474)	-6.6880*** (1.7336)	-1.3228*** (0.3201)	-9.6417*** (1.9640)	-1.5035*** (0.2508)	-1.0305 (3.6053)	-0.1340 (0.4722)
GRE Quantitative Percentile	0.0722*** (0.0068)	0.0234*** (0.0018)	0.0213** (0.0090)	0.0052** (0.0022)	0.0434 (0.1212)	0.1226*** (0.0117)	0.0238*** (0.0018)	0.0546*** (0.0190)	0.0108*** (0.0036)	0.1166*** (0.0216)	0.0182*** (0.0027)	0.0128 (0.0391)	0.0017 (0.0051)
Gender	0.1427 (0.0877)	0.0463 (0.0283)	-0.2313* (0.1310)	-0.0567* (0.0319)	-0.0078 (0.3301)	0.2574* (0.1468)	0.0500* (0.0283)	0.0095 (0.2365)	0.0019 (0.0468)	0.0588 (0.2670)	0.0092 (0.0416)	0.1302 (0.4712)	0.0169 (0.0608)
Demographic 1	-0.4889*** (0.1207)	-0.1587*** (0.0382)	0.3677** (0.1733)	0.0901** (0.0421)	-0.7897 (0.8107)	-0.8293*** (0.1966)	-0.1610*** (0.0371)	-0.3319 (0.3389)	-0.0657 (0.0669)	-0.5180 (0.3650)	-0.0808 (0.0563)	0.6342 (0.6274)	0.0825 (0.0823)
Demographic 2	0.0638 (0.1365)	0.0207 (0.0443)	0.0923 (0.1911)	0.0226 (0.0468)	-0.2510 (0.3147)	0.0807 (0.2134)	0.0157 (0.0414)	0.1752 (0.3341)	0.0347 (0.0659)	-0.4739 (0.3858)	-0.0739 (0.0598)	-0.2186 (0.6856)	-0.0284 (0.0880)
Advanced Math					0.7183*** (0.1971)			0.3562 (0.2372)	0.0704 (0.0465)	0.5362** (0.2705)	0.0836** (0.0416)	0.5857 (0.5314)	0.0762 (0.0674)
Committee Score								0.2953*** (0.0591)	0.0584*** (0.0106)	0.0621 (0.0643)	0.0097 (0.0100)	-0.6090*** (0.1636)	-0.0792*** (0.0176)
Year Transition/Matriculate More	0.6590*** (0.0897)	0.2140*** (0.0266)	0.0673 (0.1281)	0.0165 (0.0314)	0.4092 (1.1213)	1.1086*** (0.1478)	0.2153*** (0.0257)	-0.7151*** (0.2395)	-0.1414*** (0.0455)	0.1339 (0.2796)	0.0209 (0.0434)	3.4794*** (0.6030)	0.4524*** (0.0349)
Program Rank										-0.0279*** (0.0028)	-0.0043*** (0.0003)		
Inverse Mills			0.0216 (0.0878)	0.0053 (0.0215)	0.9527 (2.6242)								
√MSE					1.9199*** (0.0682)								
Observations	1000												
LL Parameters AIC	-3076.7	53	6259.5										

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 28: Restricted Matriculation-Dynamic Estimates

(a) Optimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.8765*** (0.5987)	-2.1894*** (0.7583)	1.9972 (15.9460)	-1.8367*** (0.2144)	0.1593 (0.1047, 0.2425)	-0.0481 (0.2300)	48.80% (37.78%, 59.93%)	-9.5908*** (1.9907)	-9.4588** (4.0585)	
GRE Quantitative Percentile	0.0733*** (0.0070)	0.0243*** (0.0090)	0.0244 (0.1437)					0.1149*** (0.0219)	0.1145*** (0.0394)	
Gender	0.1238 (0.0876)	-0.2213* (0.1308)	-0.0397 (0.3269)					0.1539 (0.1420)	0.7340 (0.4500)	
Demographic 1	-0.1827 (0.1205)	0.3599** (0.1729)	-0.3846 (0.4163)					-0.3116 (0.2118)	0.1025 (0.7169)	
Demographic 2	0.1038 (0.1371)	0.0944 (0.1907)	-0.1432 (0.3732)					0.0187 (0.2063)	0.0660 (0.7706)	
Advanced Math			0.6463*** (0.2034)					0.6107*** (0.1752)	1.0353* (0.5821)	
Committee Score								0.0258 (0.0396)	-0.6806*** (0.1995)	
Year Transition/Matriculate More	0.5862*** (0.0880)	0.0697 (0.1280)	0.4096 (1.1542)	-1.1177*** (0.2803)	0.0521 (0.0280, 0.0971)	0.6524* (0.3415)	64.66% (53.27%, 74.60%)	0.1064 (0.2725)	3.0310*** (0.5329)	
Program Rank								-0.0281*** (0.0027)		
Inverse Mills		0.0076 (0.0876)	0.6513 (2.9723)							
$\sqrt{\text{MSE}}$			1.9465*** (0.0673)							
Sigma										0.1438*** (0.0288)
Observations	1000									
LL Parameters AIC	-3079.7	44	6247.4							
LR Statistic DF P-Value	93.8647	9	0.0000							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.6694*** (0.5818)	-2.0782*** (0.7577)	1.8331 (12.8996)	-1.5897*** (0.3135)	0.2040 (0.1103, 0.3771)	0.0447 (0.2623)	51.12% (38.48%, 63.62%)	-7.9747*** (2.4202)	-6.5042* (3.3836)	
GRE Quantitative Percentile	0.0728*** (0.0068)	0.0230** (0.0090)	0.0290 (0.1175)					0.0912*** (0.0263)	0.0775** (0.0363)	
Gender	0.1432 (0.0878)	-0.2278* (0.1310)	-0.0392 (0.3207)					0.0809 (0.1626)	0.3895 (0.4607)	
Demographic 1	-0.4925*** (0.1216)	0.3537** (0.1730)	-0.6819 (0.7846)					-0.5146** (0.2582)	0.0520 (0.6528)	
Demographic 2	0.0610 (0.1386)	0.0927 (0.1909)	-0.2556 (0.3020)					-0.1236 (0.2554)	-0.4287 (0.6996)	
Advanced Math			0.7068*** (0.1966)					0.4391** (0.1972)	0.7410 (0.5529)	
Committee Score								0.1685*** (0.0389)	-0.5421*** (0.1608)	
Year Transition/Matriculate More	0.6636*** (0.0885)	0.0722 (0.1281)	0.3404 (1.0933)	-1.3758*** (0.3777)	0.0515 (0.0205, 0.1293)	0.5862 (0.4102)	65.27% (50.62%, 77.51%)	0.0925 (0.2912)	3.1433*** (0.5896)	
Program Rank								-0.0281*** (0.0027)		
Inverse Mills		0.0280 (0.0878)	0.7046 (2.5325)							
$\sqrt{\text{MSE}}$			1.9150*** (0.0677)							
Sigma										0.1973*** (0.0592)
Observations	1000									
LL Parameters AIC	-3144.2	44	6376.3							
LR Statistic DF P-Value	134.8755	9	0.0000							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 29: Unrestricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	Selection		Advanced Math		Committee Score		Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.8204*** (0.5916)	-2.2220*** (0.1482)	-1.8778*** (0.7560)	-0.4603*** (0.1830)	1.4110 (16.4452)	-0.0252 (0.0518)	-11.5747*** (1.0223)	-2.2538*** (0.1454)	-11.9723*** (2.5516)	-1.6176*** (0.3154)	-13.2745*** (2.4105)	-1.8267*** (0.3092)	-9.5648*** (2.0734)	-1.4881*** (0.2688)	-5.8313 (6.4970)	-0.6741 (0.7706)
GRE Quantative Percentile	0.0727*** (0.0069)	0.0237*** (0.0018)	0.0205** (0.0090)	0.0050** (0.0022)	0.0320 (0.1479)		0.1234*** (0.0117)	0.0240*** (0.0018)	0.1098*** (0.0271)	0.0148*** (0.0034)	0.1387*** (0.0264)	0.0191*** (0.0034)	0.1162*** (0.0227)	0.0181*** (0.0029)	0.0834 (0.0644)	0.0096 (0.0075)
Gender	0.1223 (0.0874)	0.0399 (0.0284)	-0.2335* (0.1309)	-0.0572* (0.0319)	-0.0212 (0.3347)		0.2266 (0.1463)	0.0441 (0.0283)	0.0238 (0.2899)	0.0032 (0.0392)	0.3047 (0.3286)	0.0419 (0.0451)	0.1266 (0.2700)	0.0197 (0.0419)	0.4650 (0.9003)	0.0538 (0.1052)
Demographic 1	-0.1818 (0.1208)	-0.0592 (0.0392)	0.3698** (0.1732)	0.0906** (0.0421)	-0.4101 (0.4258)		-0.3005 (0.1987)	-0.0585 (0.0386)	0.5480 (0.4930)	0.0740 (0.0664)	-0.8825* (0.4620)	-0.1214** (0.0619)	-0.4583 (0.3934)	-0.0713 (0.0608)	-0.2627 (0.7744)	-0.0304 (0.0880)
Demographic 2	0.1033 (0.1379)	0.0337 (0.0449)	0.0914 (0.1910)	0.0224 (0.0468)	-0.1355 (0.3816)		0.1539 (0.2214)	0.0300 (0.0431)	0.5077 (0.5194)	0.0686 (0.0698)	-0.0528 (0.4527)	-0.0073 (0.0623)	-0.2336 (0.4325)	-0.0363 (0.0672)	-0.3966 (1.2105)	-0.0459 (0.1393)
Advanced Math					0.6850*** (0.2024)				0.5092* (0.2908)	0.0688* (0.0390)	0.8607*** (0.3304)	0.1184*** (0.0451)	0.6272** (0.2750)	0.0976** (0.0417)	1.1399 (0.9839)	0.1318 (0.1080)
Committee Score									0.1059 (0.0689)	0.0143 (0.0093)	0.0368 (0.0672)	0.0051 (0.0092)	0.0467 (0.0656)	0.0073 (0.0102)	-0.9123** (0.3787)	-0.1055*** (0.0317)
Year Transition/Matriculate More	0.5840*** (0.0888)	0.1903*** (0.0269)	0.0657 (0.1281)	0.0161 (0.0314)	0.3398 (1.1922)		0.9840*** (0.1472)	0.1916*** (0.0263)					-0.0208 (0.2765)	-0.0032 (0.0430)	3.3933*** (0.7682)	0.3922*** (0.0697)
Program Rank													-0.0282*** (0.0027)	-0.0044*** (0.0003)		
η_2							1.0892*** (0.0851)			-1.6009** (0.7877)	-0.2163** (0.1049)					
$\overline{m}_{24}$												-2.0053*** (0.7676)	-0.2760*** (0.1031)			
Inverse Mills			-0.0118 (0.0877)	-0.0029 (0.0215)	0.6352 (3.0830)											
$\sqrt{\text{MSE}}$					1.9474*** (0.0674)	0.0015*** (0.0005)										
Observations	1000															
LL Parameters AIC	-1166.0		56		2444.0											
Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.																

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Selection		Advanced Math		Committee Score		$\overline{m}_{24}$		Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.6742*** (0.5822)	-2.1625*** (0.1472)	-1.8209*** (0.7551)	-0.4465** (0.1830)	2.3332 (13.2737)	-0.0498* (0.0260)	-11.2295*** (1.0144)	-2.1801*** (0.1474)	-6.7108*** (2.0006)	-1.0636*** (0.3043)	-7.1632*** (2.5070)	-0.8319*** (0.2951)	-9.6070*** (1.9611)	-1.4986*** (0.2508)	-1.4373 (3.8091)	-0.2080 (0.5533)		
GRE Quantative Percentile	0.0729*** (0.0068)	0.0236*** (0.0018)	0.0198** (0.0090)	0.0049** (0.0022)	0.0253 (0.1212)		0.1226*** (0.0117)	0.0238*** (0.0018)	0.0387* (0.0215)	0.0061* (0.0034)	0.0717*** (0.0278)	0.0083** (0.0033)	0.1163*** (0.0216)	0.0181*** (0.0027)	0.0355 (0.0435)	0.0051 (0.0063)		
Gender	0.1431 (0.0878)	0.0463 (0.0283)	-0.2349* (0.1309)	-0.0576* (0.0319)	-0.0461 (0.3263)		0.2576* (0.1469)	0.0500* (0.0283)	-0.2767 (0.2747)	-0.0439 (0.0431)	0.4105 (0.3417)	0.0583 (0.0394)	0.0091 (0.2669)	-0.3544 (0.0416)	-0.0513 (0.5436)	-0.0513 (0.0788)		
Demographic 1	-0.4942*** (0.1216)	-0.1601*** (0.0384)	0.3710** (0.1732)	0.0910** (0.0421)	-0.6644 (0.8123)		-0.8289*** (0.1966)	-0.1609*** (0.0371)	0.2571 (0.3773)	0.0407 (0.0594)	-1.1569*** (0.5110)	-0.1344** (0.0581)	-0.5154 (0.3648)	-0.0804 (0.0563)	0.3334 (0.7088)	0.0482 (0.1030)		
Demographic 2	0.0587 (0.1391)	0.0190 (0.0451)	0.0914 (0.1910)	0.0224 (0.0468)	-0.2575 (0.3005)		0.0800 (0.2135)	0.0155 (0.0414)	0.5376 (0.3886)	0.0852 (0.0605)	-0.3713 (0.4651)	-0.0431 (0.0539)	-0.4750 (0.3857)	-0.0741 (0.0598)	-0.7602 (0.7927)	-0.1100 (0.1089)		
Advanced Math					0.7199*** (0.1970)				0.2362 (0.2639)	0.0374 (0.0417)	0.4873 (0.3567)	0.0566 (0.0417)	0.5371** (0.2704)	0.0838** (0.0416)	0.3982 (0.6445)	0.0576 (0.0922)		
Committee Score									0.3738*** (0.0674)	0.0592*** (0.0096)	0.0657 (0.0837)	0.0076 (0.0098)	0.0619 (0.0643)	0.0097 (0.0100)	-0.6725*** (0.1981)	-0.0973*** (0.0217)		
Year Transition/Matriculate More	0.6631*** (0.0893)	0.2149*** (0.0264)	0.0645 (0.1280)	0.0158 (0.0314)	0.2447 (1.1193)		1.1108*** (0.1479)	0.2156*** (0.0257)					0.1318 (0.2795)	0.0206 (0.0434)	2.9410*** (0.6722)	0.4256*** (0.0578)		
Program Rank													-0.0279*** (0.0028)	-0.0044*** (0.0003)				
η_2							0.8883*** (0.0489)			-0.5479 (0.5705)	-0.0868 (0.0900)							
$\overline{m}_{24}$												-2.9231*** (0.9938)	-0.3395*** (0.1142)					
Inverse Mills			0.0216 (0.0877)	0.0053 (0.0215)	0.5617 (2.5920)													
$\sqrt{\text{MSE}}$					1.9193*** (0.0682)	0.0010*** (0.0002)												
Observations	1000																	
LL Parameters AIC	-1000.8		56		2113.6													

Murphy–Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 30: Restricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	Selection	Advanced Math	Committee Score	$\overline{m}_{2A}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-6.8934*** (0.5867)	-2.1481*** (0.7577)	6.8150 (15.9113)	-0.0252 (0.0518)	-0.2242*** (0.0862)	0.7992 (0.6750, 0.9462)	0.1909 (0.3049)	60.10% (49.32%, 69.97%)	-9.3017*** (1.9470)	-5.8313 (6.4970)		
GRE Quantative Percentile	0.0736*** (0.0068)	0.0238*** (0.0090)	-0.0161 (0.1439)						0.1111*** (0.0215)	0.0834 (0.0644)		
Gender	0.1233 (0.0876)	-0.2140 (0.1305)	-0.0994 (0.3249)						0.2193* (0.1213)	0.4650 (0.9003)		
Demographic 1	-0.1870 (0.1204)	0.3580** (0.1725)	-0.3079 (0.4307)						-0.3208* (0.1819)	-0.2627 (0.7744)		
Demographic 2	0.0950 (0.1381)	0.0935 (0.1906)	-0.2041 (0.3486)						0.0196 (0.1747)	-0.3966 (1.2105)		
Advanced Math			0.6950*** (0.2026)						0.5545*** (0.1699)	1.1399 (0.9839)		
Committee Score									0.0495 (0.0370)	-0.9123** (0.3787)		
Year Transition/Matriculate More	0.5884*** (0.0881)	0.0733 (0.1278)	-0.0583 (1.1568)		-0.0226 (0.0821)	0.7813 (0.6543, 0.9329)		74.06% (59.51%, 84.72%)	0.0598 (0.2658)	3.3933*** (0.7682)		
Program Rank									-0.0282*** (0.0027)			
m_2				1.0892*** (0.0851)								
$\overline{m}_{2A}$							1.3481* (0.7973)					
Inverse Mills		0.0059 (0.0873)	-0.4103 (2.9426)									
$\sqrt{\text{MSE}}$			1.9465*** (0.0673)	0.0015*** (0.0005)								
Beta											0.9750*** (0.0336)	
Sigma												0.1342*** (0.0248)
Observations	1000											
LL Parameters AIC	-1197.1	40	2474.2									
LR Statistic DF P-Value	62.2160	16	0.0000									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Selection	Advanced Math	Committee Score	$\overline{m}_{2A}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-6.6832*** (0.5773)	-2.1667*** (0.7596)	7.5299 (12.8480)	-0.0498* (0.0260)	-0.2877** (0.1208)	0.7500 (0.5919, 0.9502)	0.2342 (0.3534)	61.63% (47.37%, 74.14%)	-6.8552*** (2.2794)	-1.4373 (3.8091)		
GRE Quantative Percentile	0.0731*** (0.0067)	0.0241*** (0.0090)	-0.0196 (0.1177)						0.0779*** (0.0242)	0.0355 (0.0435)		
Gender	0.1417 (0.0878)	-0.2190* (0.1307)	-0.1202 (0.3135)						0.1414 (0.1163)	-0.3545 (0.5436)		
Demographic 1	-0.5019*** (0.1204)	0.3470** (0.1730)	-0.3722 (0.8108)						-0.4508** (0.1994)	0.3334 (0.7088)		
Demographic 2	0.0438 (0.1386)	0.0999 (0.1908)	-0.2738 (0.2786)						0.0013 (0.1677)	-0.7602 (0.7927)		
Advanced Math			0.7394*** (0.1977)						0.3279** (0.1648)	0.3982 (0.6445)		
Committee Score									0.1647*** (0.0367)	-0.6725*** (0.1981)		
Year Transition/Matriculate More	0.6666*** (0.0886)	0.0685 (0.1279)	-0.2625 (1.0988)		-0.0385 (0.0869)	0.7216 (0.5845, 0.8910)		77.06% (53.85%, 90.63%)	0.1669 (0.2714)	2.9410*** (0.6722)		
Program Rank									-0.0281*** (0.0027)			
m_2				0.8883*** (0.0489)								
$\overline{m}_{2A}$							1.7264 (1.1486)					
Inverse Mills		0.0303 (0.0875)	-0.6289 (2.5011)									
$\sqrt{\text{MSE}}$			1.9215*** (0.0688)	0.0010*** (0.0002)								
Beta											0.8713*** (0.0429)	
Sigma												0.1365*** (0.0403)
Observations	1000											
LL Parameters AIC	-1055.8	40	2191.7									
LR Statistic DF P-Value	110.1173	16	0.0000									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

D Two-Round Portfolio Choice

D.1 Without Matriculation

In Section 4.1, I show that the committee's portfolio choice problem reduces to a cutoff rule problem, keeping it to one round for simplicity. Then in Sections 4.1.1 and 4.1.2, I write out dynamic and static two-round cutoff rule problems with Round 1 filtering and Round 2 final selections. In this appendix, I solve a two-round portfolio choice problem by backward induction, showing that each round's solution takes the cutoff rule form in the paper. I start with Round 2:

$$\max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A. \quad (34)$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{2,n} = A$, which means that the constraint binds. The Lagrangian with multiplier λ_2 is:

$$\begin{aligned} \mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) &= \sum_{n=1}^{N_T} P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \left\{ N_A - \left[N_T - \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N_T} [P(y_n = S | x_n, z_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] + \lambda_2 (N_A - N_T) \end{aligned} \quad (35)$$

By the same logic in Section 4.1, the Round 2 solution is as follows with set-identified cutoff probability $\lambda_2^* = P(y_2^* = S) \in [P(y_{N_A+1} = S | x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S | x_{N_A}, z_{N_A})]$:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(x_n, z_n | \lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S | x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S | x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S | x_n, z_n) < \lambda_2^*. \end{cases} \quad (36)$$

When the committee solves Round 1, it must anticipate λ_2^* , which is a function of the $(\mathbf{x}, \mathbf{z})$ of the entire transitioned pool. I make two assumptions so that this anticipation is tractable. First, conditional on $\mathbf{x}$, I assume that $\mathbf{z}$ is independent across applicants. Second, I assume that N is large. Under these assumptions, the same Glivenko-Cantelli argument as in the proof of Theorem 1, with $\mathbf{z}$ in place of the preference shocks, makes λ_2^* deterministic for large N .

Denote $\hat{G}_{N_T}(c) = \frac{1}{N_T} \sum_{n=1}^{N_T} \mathbb{1}\{P(y_n = S | x_n, z_n) \leq c\} \mathbb{1}\{d_{1,n} = T\}$ as the empirical distribution of the success probabilities of the N_T transitioned applicants. The cutoff λ_2^* is the $(1 - N_A/N_T)$ quantile of this distribution, since the committee accepts the N_A such applicants with the highest success probabilities. Because only $\mathbf{x}$ is observed in Round 1, transition depends on x_n alone; the Round 1 cutoff rule derived below transitions applicant n when x_n lies in transition region $\mathcal{X}_T$. For large N , the cutoff λ_1^* converges to a constant by the same argument, so $\mathcal{X}_T$ is deterministic.

Since the pairs (x_n, z_n) are iid and membership in $\mathcal{X}_T$ depends only on x_n , the transitioned success probabilities are themselves iid, with conditional distribution $G(c) = P(P(y = S | x, z) \leq$

$c|x \in \mathcal{X}_T$). So by the Glivenko-Cantelli theorem, $\hat{G}_{N_T}$ converges almost surely to G for large N_T . Therefore, as $N_A/N_T \rightarrow q_2$, $\hat{G}_{N_T}$'s $(1 - N_A/N_T)$ quantile converges to G 's deterministic $(1 - q_2)$ quantile. In turn, the set-identified interval for λ_2^* collapses to a point, denoted $\bar{\lambda}_2^*$. As in the proof of Theorem 1, independence is essential; a common component across applicants would slide $\hat{G}_{N_T}$ in unison and leave λ_2^* random in the limit. This convergence is a fixed point because the transition region $\mathcal{X}_T$ is pinned down by the Round 1 cutoff λ_1^* , which ranks applicants by their anticipated Round 2 payoff, which itself embeds $\bar{\lambda}_2^*$. The pair $(\lambda_1^*, \bar{\lambda}_2^*)$ is jointly determined. Because $\bar{\lambda}_2^*$ is deterministic, no individual applicant's signal moves the cutoff, the committee's anticipated Round 2 payoff separates across applicants, and Round 1 reduces to a cutoff rule.

Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n, z_n), \bar{\lambda}_2^*\} | x_n] = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{\tilde{z}|x}(\tilde{z}_n | x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T. \quad (37)$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) &= \sum_{n=1}^N EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^N \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^N \{ EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \} + \lambda_1 (N_T - N). \end{aligned} \quad (38)$$

By the same logic in Section 4.1, the Round 1 solution is as follows with set-identified cutoff level $\lambda_1^* = L(y_1^* = S) \in [EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T})]$:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \quad (39)$$

There is one discrepancy between $\lambda_1^* = L(y_1^* = S)$ in this appendix vs. Section 4.1.1. Here, λ_1^* is bounded above by 1 because for all n , it is the case that $\max\{P(y_n = S|x_n, \tilde{z}_n), \bar{\lambda}_2^*\} \leq 1$ pointwise in $\tilde{z}_n$. Therefore, $EV_{2,n}(x_n) \leq \int_{\tilde{z}_n} dF_{\tilde{z}|x}(\tilde{z}_n | x_n) = 1$. But in Section 4.1.1, the type 1 extreme value preference shocks increase this upper bound to $1 + \sigma \log 2$, where I now denote $\bar{\lambda}_2^*$ by $P(y_2^* = S)$:

$$\begin{aligned} \mathbb{E}[v_2(x_n, z_n, \epsilon_{2,n}) | x_n] &= \int_{\tilde{z}_n} \sigma \log \left(e^{P(y_n=S|x_n, \tilde{z}_n)/\sigma} + e^{P(y_2^*=S)/\sigma} \right) dF(\tilde{z}_n | x_n) \\ &\leq \int_{\tilde{z}_n} \sigma \log \left(2e^{\max\{P(y_n=S|x_n, \tilde{z}_n), P(y_2^*=S)\}/\sigma} \right) dF(\tilde{z}_n | x_n) \\ &= \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), P(y_2^* = S)\} dF(\tilde{z}_n | x_n) + \sigma \log 2 \\ &\leq 1 + \sigma \log 2. \end{aligned} \quad (40)$$

In the static version of the two-round portfolio choice model, the Round 1 problem is:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S|x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T. \quad (41)$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) &= \sum_{n=1}^N P(y_n = S|x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^N \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^N [P(y_n = S|x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N). \end{aligned} \quad (42)$$

By the same logic in Section 4.1, the solution is as follows with set-identified cutoff probability $\lambda_1^* = P(y_1^* = S) \in [P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})]$:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^*. \end{cases} \quad (43)$$

D.2 With Matriculation

In Section 4.2, I show that because the committee's capacity constraint is on the number of matriculations rather than acceptances, a one-round portfolio choice problem with matriculation reduces to a cutoff rule problem on success probabilities. However, a two-round portfolio choice problem with matriculation is less straightforward because matriculation ends up mattering in Round 1. In this appendix, I present such a model with Round 1 filtering and Round 2 final selections. For readability, I keep finite-sum notation rather than the integrals in the proof of Theorem 3 in Appendix G. But I invoke that theorem's large- N framework throughout because it removes the finite- N edge cases and, as in Appendix D.1, collapses λ_2^* to the deterministic $\bar{\lambda}_2^*$. Working backward, I start with Round 2, where for brevity, I denote $w = (x, z)$:

$$\begin{aligned} \max_{(d_{2,1}, \dots, d_{2,N_T}) \in \{A, R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n) P(m_n = M|w_n) \mathbb{1}(d_{2,n} = A), \\ \text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n) \mathbb{1}(d_{2,n} = A) \leq N_M. \end{aligned} \quad (44)$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{2,n} = A$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{2,1}, \dots, d_{2,N_T}; \lambda_2) &= \sum_{n=1}^{N_T} P(y_n = S|w_n) P(m_n = M|w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \left\{ N_M - \left[\overline{N_M} - \sum_{n=1}^{N_T} P(m_n = M|w_n) \mathbb{1}(d_{2,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N_T} [P(y_n = S|w_n) \mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)] P(m_n = M|w_n) + \lambda_2 (N_M - \overline{N_M}), \end{aligned} \quad (45)$$

where $\overline{N}_M = \sum_{n=1}^{N_T} P(m_n = M|w_n)$ is the expected number of matriculants if everyone is accepted.

Edge cases aside, the Round 2 solution, with cutoff probability $\lambda_2^* = P(y_2^* = S)$, is:

$$\forall n \in \{1, \dots, N_T\}, d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \quad (46)$$

Let $EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n) - \overline{\lambda}_2^*, 0\} P(m_n = M|x_n, \tilde{z}_n) dF_{\tilde{z}|x}(\tilde{z}_n|x_n)$. This netted form, and not $\int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), \overline{\lambda}_2^*\} P(m_n = M|x_n, \tilde{z}_n) dF_{\tilde{z}|x}(\tilde{z}_n|x_n)$, is the correct transition payoff. To see why, evaluate the Round 2 Lagrangian at the cutoff rule solution with deterministic cutoff $\overline{\lambda}_2^*$, where $P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)$ becomes $\max\{P(y_n = S|w_n), \overline{\lambda}_2^*\}$:

$$\mathcal{L}(d_2^*; \overline{\lambda}_2^*) = \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n), \overline{\lambda}_2^*\} P(m_n = M|w_n) + \overline{\lambda}_2^*(N_M - \overline{N}_M). \quad (47)$$

By $\max\{P(y_n = S|w_n), \overline{\lambda}_2^*\} = \max\{P(y_n = S|w_n) - \overline{\lambda}_2^*, 0\} + \overline{\lambda}_2^*$, and $\overline{N}_M = \sum_{n=1}^{N_T} P(m_n = M|w_n)$:

$$\begin{aligned} \mathcal{L}(d_2^*; \overline{\lambda}_2^*) &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda}_2^*, 0\} P(m_n = M|w_n) + \overline{\lambda}_2^* \overline{N}_M + \overline{\lambda}_2^*(N_M - \overline{N}_M) \\ &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda}_2^*, 0\} P(m_n = M|w_n) + \overline{\lambda}_2^* N_M. \end{aligned} \quad (48)$$

Although $\overline{N}_M$ varies with the Round 1 transition decisions, the $\overline{\lambda}_2^* \overline{N}_M$ terms cancel, leaving the constant $\overline{\lambda}_2^* N_M$. The Round 1 transition payoff is therefore the per-applicant netted term $\max\{P(y_n = S|w_n) - \overline{\lambda}_2^*, 0\} P(m_n = M|w_n)$, whose expectation over $\tilde{z}_n$ defines $EV_{2,n}$.

The netting distinguishes the matriculation case from the case without it. Without matriculation, the netted and non-netted forms differ by $\int_{\tilde{z}_n} \overline{\lambda}_2^* dF_{\tilde{z}|x}(\tilde{z}_n|x_n) = \overline{\lambda}_2^*$, a constant across applicants that leaves the Round 1 ordering unchanged. With matriculation, they differ by $\int_{\tilde{z}_n} \overline{\lambda}_2^* P(m_n = M|x_n, \tilde{z}_n) dF_{\tilde{z}|x}(\tilde{z}_n|x_n)$, which varies with x_n . The non-netted form would therefore order applicants differently, and incorrectly. The Round 1 portfolio choice problem is:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T. \quad (49)$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) &= \sum_{n=1}^N EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^N \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^N [EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N). \end{aligned} \quad (50)$$

Edge cases aside, the Round 1 solution with cutoff level $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \quad (51)$$

The Round 1 transition payoff with Type 1 extreme value shocks is therefore:

$$\mathbb{E}[v_2(x_n, z_n, \epsilon_{2,n}) | x_n] = \int_{\tilde{z}_n} \sigma \log \left(e^{[P(y_n=S|x_n, \tilde{z}_n) - P(y_2^*=S)]/\sigma} + 1 \right) P(m_n = M | x_n, \tilde{z}_n) dF(\tilde{z}_n | x_n). \quad (52)$$

In the static version of the model with matriculation, the Round 1 problem is:

$$\max_{(d_{1,1}, \dots, d_{1,N}) \in \{T, R\}^N} \sum_{n=1}^N P(y_n = S | x_n) P(m_n = M | x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^N \mathbb{1}(d_{1,n} = T) \leq N_T. \quad (53)$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1}, \dots, d_{1,N}; \lambda_1) &= \sum_{n=1}^N P(y_n = S | x_n) P(m_n = M | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^N \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^N [P(y_n = S | x_n) P(m_n = M | x_n) \mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)] + \lambda_1 (N_T - N). \end{aligned} \quad (54)$$

Edge cases aside, the solution is as follows with cutoff probability $\lambda_1^* = P(y_1^* = S, m_1^* = M)$:

$$\forall n \in \{1, \dots, N\}, d_{1,n}(x_n | \lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S | x_n) P(m_n = M | x_n) > \lambda_1^* \\ \{T, R\} & \text{if } P(y_n = S | x_n) P(m_n = M | x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S | x_n) P(m_n = M | x_n) < \lambda_1^*. \end{cases} \quad (55)$$

The committee prefers to transition likely matriculants so that it can be more selective in Round 2. As a simple example, suppose that $N = 4$, $N_T = 2$, and $N_M = 1$, and suppose that recommendation letters matter more for success than GRE scores. Suppose also that Applicant 1 has a high GRE and strong recommendation letter, Applicant 2 has a high GRE and weak letter, Applicant 3 has a low GRE and strong letter, and Applicant 4 has a low GRE and weak letter. If the committee ignores matriculation probabilities in Round 1, it will transition Applicants 1 and 2 and then accept Applicant 1. But suppose that because Applicant 1 is so strong, they choose a higher-ranked university. Then the committee has to settle for Applicant 2. Whereas if the committee had transitioned Applicants 3 and 4, it could have accepted and landed Applicant 3.

However, discussions with committee members at the university featured in this paper indicate that the committee does not consider matriculation probabilities in Round 1. The reason is that this university is ranked highly enough that it can fill its cohort with applicants who have

strong objective and subjective variables, leaving Round 2 with the slack to be selective on success regardless of Round 1's matriculation weighting. If this university were in the top 10, it could, in keeping with the existing graduate admissions literature that features primarily top 10 programs, ignore matriculation probabilities altogether and solve the dynamic portfolio choice problem without matriculation. But it is not, so it takes a middle ground and defers matriculation considerations to after Round 1, where richer signals from application files, such as mentioning specific professors in the application essay, predict matriculation without predicting failure.¹⁶

Therefore, in Section 4.2.2, I introduce the waitlist and expand the cutoff rule framework from two rounds to three. Success probabilities equal, the committee favors likely matriculants in Round 2 to avoid losing high-quality applicants who would have matriculated by suboptimally keeping those applicants waiting until Round 3 for an acceptance. I also make the simplifying assumption that matriculation probabilities do not enter the Round 1 objective. In the waitlist-dynamic model, I base the transition payoff on the portfolio choice problem without matriculation: $EV_{2,n}(x_n) = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), \bar{\lambda}_2^*\} dF_{z|x}(\tilde{z}_n|x_n)$, which yields the same ordering of applicants as $\int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n) - \bar{\lambda}_2^*, 0\} dF_{z|x}(\tilde{z}_n|x_n)$. As such, the transition payoff is:

$$\mathbb{E}[v_2(x, z, \ddot{m}_2, \varepsilon_2)|x, t] = \int_{\tilde{z}} \sigma \log \left(e^{P(y=S|x, \tilde{z})/\sigma} + e^{\beta \mathbb{E}[v_3(x, z, \overline{m}_{2A}, \varepsilon_3)|x, \tilde{z}, \ddot{m}_2(x, \tilde{z}, t)]/\sigma} \right) dF_{z|x}(\tilde{z}|x), \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, \overline{m}_{2A}, \varepsilon_3)|x, z, \ddot{m}_2] = \int_{\overline{m}_{2A}} \sigma \log \left(e^{P(y=S|x, z)/\sigma} + e^{P(y_3^*=S|\overline{m}_{2A})/\sigma} \right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2). \quad (56)$$

Analogously, in the waitlist-hybrid model, I specify the transition payoff as $P(y = S|x)$.

¹⁶Per Table 17 of Appendix B, the variables in x that predict matriculation in economics are being male, having low verbal and quantitative GRE scores, having a low undergraduate GPA, not attending an elite undergraduate institution, having no work experience, and not having taken advanced math. Except for gender, all these variables are indicative of a lower-quality applicant. In addition, the only variables that might predict success that do not predict non-matriculation are having more professor and prolific recommenders, and it is unclear whether those variables actually predict success (though I will answer that question in a future version of this paper).

E Symmetric Waitlist Models

In Section 4.2.2, I present waitlist models in which success probabilities equal, the committee is more likely to give Round 2 acceptances to likely matriculants. Intuitively, this primary channel holds because likely matriculants decrease the expected number of spots to fill in Round 3, letting the committee be more selective. But there is also a secondary channel: if the committee accepts likely matriculants in Round 2, then the unlikely matriculants it waitlists end up in the Round 3 pool. As such, while there are fewer expected spots to fill, it is easier to fill those spots. For tractability, I do not estimate this secondary channel in the main specification. But tractability aside, there are two economic reasons why the primary channel is dominant. First, matriculation probabilities decline between Rounds 2 and 3, and for likely matriculants, they have more room to fall. Second, matriculation probabilities can change unpredictably between those rounds.

Still, I present symmetric waitlist models that account for both the primary and secondary channels. There are two additional states: $\bar{m}_3$, which is a given year's average waitlisted applicant matriculation probability, and m_3 , which is the *ex-ante* probability that an applicant will matriculate if waitlisted. By *ex-ante*, I mean that m_3 depends only on information in the application files and not events that happen between Rounds 2 and 3. Therefore, the committee knows m_3 in Round 2. As with $\bar{m}_{2A}$, the Round 3 cutoff probability is increasing in $\bar{m}_3$. In addition, $\bar{m}_3$ is increasing in m_3 , analogously to how $\bar{m}_{2A}$ is increasing in $\bar{m}_2$. Whether the committee will be more or less likely to accept likely matriculants in Round 2 is theoretically ambiguous and depends on the parameter values. The equations are as follows, starting with Round 3:

$$v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, \epsilon_3) = \max_{d_3 \in \{A, R\}} u_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, d_3) + \epsilon_3(d_3), \text{ such that:} \quad (57)$$

$$u_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, d_3) = P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_3 = A) + P(y_3^* = S | \bar{m}_{2A}, \bar{m}_3) \mathbb{1}(d_3 = R).$$

The Round 2 value function is:

$$v_2(x, z, t, \bar{k}, \bar{m}_2, m_3, \epsilon_2) = \max_{d_2 \in \{A, W\}} u_2(x, z, t, \bar{k}, \bar{m}_2, m_3, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (58)$$

$$u_2(x, z, \bar{m}_2, m_3, d_2) = P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A) + \beta \mathbb{E}[v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, \epsilon_3) | x, z, t, \bar{k}, \bar{m}_2, m_3] \mathbb{1}(d_2 = W), \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, \epsilon_3) | x, z, t, \bar{k}, \bar{m}_2, m_3] = \int_{\bar{m}_{2A}} \int_{\bar{m}_3} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A, R\}} e^{u_3(x, z, t, \bar{k}, \bar{m}_{2A}, \bar{m}_3, \tilde{d}_3) / \sigma} \right) dF_{\bar{m}_3 | m_3}(\bar{m}_3 | m_3) dF_{\bar{m}_{2A} | \bar{m}_2}(\bar{m}_{2A} | \bar{m}_2).$$

Finally, the Round 1 value function is:

$$v_1(x, t, \bar{k}) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{k}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (59)$$

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \bar{m}_2, m_3, \epsilon_2) | x, t, \bar{k}] \mathbb{1}(d_1 = T) + L(y_1^* = S | t) \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \bar{m}_2, m_3, \epsilon_2) | x, t, \bar{k}] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \bar{k}, \bar{m}_2(x, \tilde{z}, t), m_3(x, \tilde{z}, t), \tilde{d}_2) / \sigma} \right) dF_{\tilde{z} | x, t}(\tilde{z} | x, t).$$

In the symmetric waitlist-hybrid model, the Round 1 utility function is instead:

$$u_1(x, t, \bar{k}, d_1) = P(y = S|x, t, \bar{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R). \quad (60)$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(\text{states}, \tilde{d}_r)/\sigma)}. \quad (61)$$

The unrestricted CCPs, success probabilities, and cutoff level/probability are $P_{d_1} = P(d_1 = T|x, t; \theta_T)$, $P_{d_2} = P(d_2 = A|x, z, m_2, m_3; \theta_{A_2})$, $P_{d_3} = P(d_3 = A|x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3})$, $P_y = P(y = S|x, z, t, \bar{k}; \theta_S)$, $L_{y_1^*} = L(y_1^* = S|t; \theta_{S^*})$, and $P_{y_3^*} = P(y_3^* = S|\overline{m_{2A}}, \overline{m_3}; \theta_{S^*})$. They have the same specifications as the probabilities and levels earlier in the paper; see Equations (9)–(11) in Section 4.1.2.

Let applicants 1 to N_W be waitlisted, $N_W + 1$ to N_T be accepted in Round 2, and $N_T + 1$ to N be rejected in Round 1. Then the unrestricted symmetric waitlist-dynamic likelihood is:

$$\begin{aligned} \mathcal{L}_{N_1}^U(\theta) &= \prod_{n=1}^{N_W} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_{2A}}|m_2}(\overline{m_{2A}}, n; \theta_{F_{\overline{m_{2A}}|m_2}}) f_{\overline{m_3}|m_3}(\overline{m_3}, n; \theta_{F_{\overline{m_3}|m_3}}) \\ &P_{d_{1,n}}(\theta_T)(1 - P_{d_{2,n}}(\theta_{A_2}))P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \\ &\prod_{n=N_W+1}^{N_T} f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}}) f_{\overline{m_{2A}}|m_2}(\overline{m_{2A}}, n; \theta_{F_{\overline{m_{2A}}|m_2}}) f_{\overline{m_3}|m_3}(\overline{m_3}, n; \theta_{F_{\overline{m_3}|m_3}}) \\ &P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^N f_{z|x,t}(z_n|x_n, t_n; \theta_{F_{z|x,t}})(1 - P_{d_{1,n}}(\theta_T)). \quad (62) \end{aligned}$$

Finally, m_2 and m_3 are parameterized as follows: $m_2 = P_{m_2} = P(m_2 = M|x, z, t; \theta_{M_2})$ and $m_3 = P_{m_3} = P(m_3 = M|x, z, t; \theta_{M_3})$. To estimate θ_{M_2} and θ_{M_3} , I would use the first-stage likelihoods: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1 - P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$ and $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A, m_{3,n}=M)}(1 - P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A, m_{3,n}=D)}$. The samples for estimating m_2 and m_3 are applicants accepted in Rounds 2 and 3 respectively. Since m_2 and m_3 are generated regressors, the standard errors are those from Murphy and Topel (1985) and Hardin (2002).

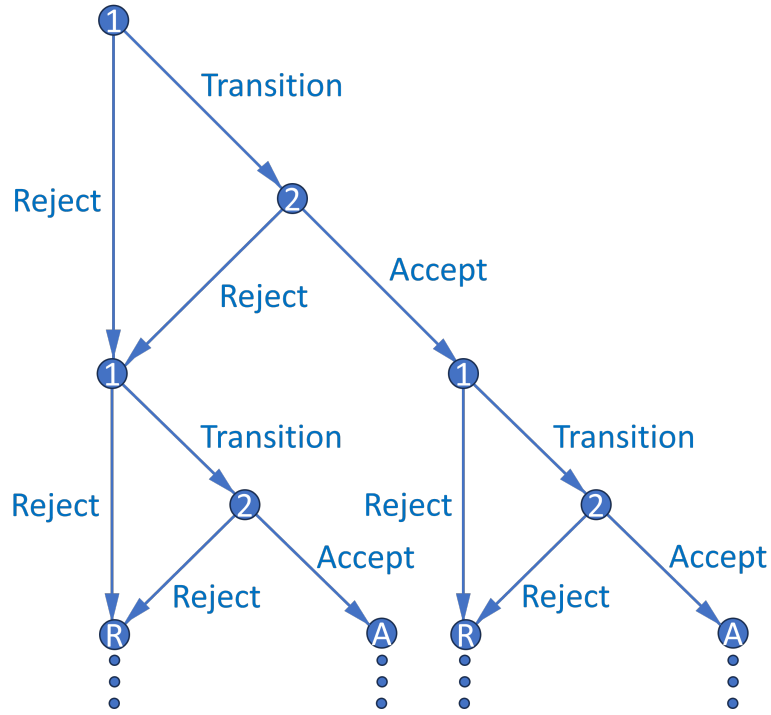


Figure 8: Multiyear Decision Graph

F Multiyear Models

There are two different multiyear models. In the first model, the memory variable is the size of the previous year's cohort. The committee accepts applicants more likely to matriculate because it wants to ensure that next year's memory variable, which is this year's cohort size, is sufficiently large that it can avoid accepting too many relatively weak applicants next year. In the second model, the memory is a diversity variable that increases an applicant's success probability; for example, the previous year's cohort's main area of interest. Applicants with a diverse area of interest may be more likely to succeed because their advisor, with fewer other students to advise, can give them more individualized attention. The committee will therefore accept applicants with a less common area of interest to help balance out the program for next year's class.

F.1 Matriculation Model

In the waitlist models in Section 4.2.2, the committee benefits from giving early acceptances to applicants who are more likely to matriculate. Doing so helps prevent the program from losing out on good applicants who would likely have matriculated if given an early offer but matriculate elsewhere due to events that occurred while on the waitlist. However, the waitlist models do not imply that the committee should also give more late acceptances to likely matriculants. To model that phenomenon, a multiyear model is necessary, as there must be a period after the final round in

any given year for dynamics to matter in that round. Figure 8 is a decision graph of the simplest possible multiyear model, in which each year has only two rounds, as in the dynamic model. This model is infinite horizon, and future years are discounted with factor $\beta \in (0, 1)$. To avoid the considerations of two-stage estimation, I suppose for simplicity that matriculation probabilities m come from the committee's assessment of each applicant in Round 2. I also suppose that success probabilities depend only on applicant characteristics and not program rank.

In each year's Round 2 value function, I specify the cutoff probability as $P(y_2^* = S | c_{-1})$ instead of $P(y_2^* = S | t)$, where c_{-1} is the previous year's cohort size. The cutoff probability increases in c_{-1} because the committee must keep the total number of students in the program across years relatively constant. So if c_{-1} is large, the committee must now accept fewer students.

Given that, what information available at the start of Round 2 determines c ? If and only if the applicant is accepted, their own matriculation probability m does. Now suppose the committee maximizes discounted "lifetime" cohort success by solving an infinite-horizon dynamic programming problem across years. As in the waitlist models, the optimal dynamic decision may not match the optimal myopic one. Instead, the committee may accept an applicant with a high m even if their expected success probability is less than what the Round 2 cutoff level (which need not be between 0 and 1) would otherwise be. The reason is that accepting a high m applicant increases c 's distribution, which increases next year's Round 2 cutoff and value.

Intuitively, even if the committee can accept more applicants next year to make up for a smaller cohort this year, they will do so out of necessity, for example, to avoid future TA shortages. This situation will be undesirable provided next year's marginal applicant is of lower quality than this year's above-marginal applicant. Also, as in the waitlist models, even if the data show that the committee does not think dynamically, if c_{-1} affects the Round 2 cutoff, perhaps it should. Finally, estimating this model is more computationally challenging since it does not have a closed-form solution. Nested fixed point algorithms suffer from the curse of dimensionality, and the number of state variables in application files, in particular those from textual documents, is large. But principal component analysis can reduce the size of the state space.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x, z, t, m, c_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, m, c_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (63)$$

$$u_2(x, z, t, m, c_{-1}, d_2) = \{P(y = S | x, z, t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1') | m]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S | c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', c, \epsilon_1') | (m)] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1' \in \{T, R\}} e^{u_1(x', t', c, d_1') / \sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c | (m)) \\ = \int_c \int_{t_1'} \cdots \int_{t_{J_t'}} \int_{x_1'} \cdots \int_{x_{J_x'}} \left[\cdot \right] dF_{x,1}(x_1' | x_2', \dots, x_{J_x}') \cdots dF_{x,J_x}(x_{J_x}') dF_{t,1}(t_1' | t_2', \dots, t_{J_t}') \cdots dF_{t,J_t}(t_{J_t}') dF_{c|(m)}(c | (m)).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x, z, t, m, c_{-1}) = \frac{\exp(u_2(x, z, t, m, c_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, m, c_{-1}, \tilde{d}_2)/\sigma)}. \quad (64)$$

Each year's Round 1 value function is:

$$v_1(x, t, c_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, c_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (65)$$

$$u_1(x, t, c_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}] \mathbb{1}(d_1 = T) + \{L(y_1^* = S|t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon_1')]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\begin{aligned} \mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}] &= \int_{\tilde{m}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A, R\}} e^{u_2(x, \tilde{z}, t, \tilde{m}, c_{-1}, \tilde{d}_2)/\sigma} \right) dF_{z|x, t}(\tilde{z}|x, t) dF_{m|x, t}(\tilde{m}|x, t) \\ &= \int_{\tilde{m}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_z} \left[\cdot \right] dF_{z|x, t, 1}(\tilde{z}_1|\tilde{z}_2, \dots, \tilde{z}_z, x, t) \cdots dF_{z|x, t, J_z}(\tilde{z}_{J_z}|x, t) dF_{m|x, t}(\tilde{m}|x, t). \end{aligned}$$

And as such, Round 1's conditional choice probabilities (CCPs) are:

$$P(d_1|x, t, c_{-1}) = \frac{\exp(u_1(x, t, c_{-1}, d_1)/\sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, c_{-1}, \tilde{d}_1)/\sigma)}. \quad (66)$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P(d_1 = T|x, t, c_{-1}; \theta_T)$, $P_{d_2} = P(d_2 = A|x, z, m, c_{-1}; \theta_A)$, $P_y = P(y = S|x, z, t; \theta_S)$, $L_{y_1^*} = L(y_1^* = S|t; \theta_{S_1^*})$, and $L_{y_2^*} = L(y_2^* = S|c_{-1}; \theta_{S_2^*})$, which aside from the cutoffs have the usual logistic specification. In this model, the cutoffs are not in a function that restricts their range. Given that, the unrestricted likelihood is:

$$\begin{aligned} \mathcal{L}_N^U(\theta) &= \prod_{n=1}^{N_T} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x, t}(z_n|x_n, t_n; \theta_{F_{z|x, t}}) f_{m|x, t}(m_n|x_n, t_n; \theta_{F_{m|x, t}}) f_{c|m}(c_n|m_n; \theta_{F_{c|m}}) \\ &\quad P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \\ &\quad \prod_{n=N_T+1}^N f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x, t}(z_n|x_n, t_n; \theta_{F_{z|x, t}}) f_{m|x, t}(m_n|x_n, t_n; \theta_{F_{m|x, t}}) (1 - P_{d_{1,n}}(\theta_T)). \quad (67) \end{aligned}$$

As with the dynamic models in the paper, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R) . To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 6. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. See Appendix G.

F.2 Diversity Model

I remove the state variable c_{-1} and add in its place two new states a and $\bar{a}_{-1}$. Let $a = 1(\text{Diverse})$, where a diverse characteristic is one whose probability of appearing among next year's applicants

is bounded above by some $p \in (0, 1)$. For example, I could have $a = \mathbb{1}(\text{TheoryField})$, where in most years, theory applicants are less common than applied applicants. Also, let $\bar{a}_{-1}$ equal the percentage of last year's acceptance set that is theory. If $|a - \bar{a}_{-1}|$ shrinks, then the applicant's expected success probability decreases, as their future advisor may have less time for them.¹⁷

As such, the optimal dynamic decision may not match the optimal myopic one. In particular, the committee may accept a diverse applicant it would not have otherwise because doing so increases the success probability of next year's applicant pool. Accepting a diverse applicant increases the distribution of $\bar{a}$ conditional on a , which because diverse applicants are less common increases the distribution of $|a' - \bar{a}|$, which increases next year's expected success probability, which increases next year's expected value. That said, I can also estimate a myopic model with memory variable $\bar{a}_{-1}$. The committee will still prioritize diverse applicants because diversity increases applicants' success probabilities, but there will be no forward-looking dynamic effect.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, a, \bar{a}_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:} \quad (68)$$

$$u_2(x, z, t, a, \bar{a}_{-1}, d_2) = \{P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a]\} \mathbb{1}(d_2 = A) \\ + \{L(y_2^* = S|t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_2 = R), \text{ such that:}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] = \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left(\sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{a}, d'_1)/\sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\tilde{a}|a}(\tilde{a}|a) = \\ \int_{\tilde{a}} \int_{\tilde{t}'_1} \cdots \int_{\tilde{t}'_t} \int_{\tilde{x}'_1} \cdots \int_{\tilde{x}'_x} \left[\cdot \right] F_{x,1}(x'_1|x'_2, \dots, x'_{J_x}) \cdots dF_{x,J_x}(x'_{J_x}) dF_{t,1}(t'_1|t'_2, \dots, t'_{J_t}) \cdots dF_{t,J_t}(t'_{J_t}) dF_{\tilde{a}|a}(\tilde{a}|a).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(u_2(x, z, t, a, \bar{a}_{-1}, d_2)/\sigma)}{\sum_{\tilde{d}_2 \in \{A, R\}} \exp(u_2(x, z, t, a, \bar{a}_{-1}, \tilde{d}_2)/\sigma)}. \quad (69)$$

Each year's Round 1 value function is:

$$v_1(x, t, \bar{a}_{-1}, \epsilon_1) = \max_{d_1 \in \{T, R\}} u_1(x, t, \bar{a}_{-1}, d_1) + \epsilon_1(d_1), \text{ such that:} \quad (70)$$

$$u_1(x, t, \bar{a}_{-1}, d_1) = \mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}] \mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S|t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)]\} \mathbb{1}(d_1 = R), \text{ such that:}$$

¹⁷If the program never gets too imbalanced, this relationship will not be identifiable, in which case estimation will not work for this choice of a . However, there may also be different suitable diversity variables.

$$\begin{aligned}\mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \varepsilon_2) | x, t, \bar{a}_{-1}] &= \int_{\tilde{a}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in A, R} e^{u_2(x, \tilde{z}, t, \tilde{a}, \bar{a}_{-1}, \tilde{d}_2) / \sigma} \right) dF_{z|x, t}(\tilde{z} | x, t) dF_a(\tilde{a}) \\ &= \int_{\tilde{a}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1 | \tilde{z}_2, \dots, \tilde{z}_J, x, t) \cdots dF_J(\tilde{z}_J | x, t) dF_a(\tilde{a})\end{aligned}$$

$$\begin{aligned}\mathbb{E}[v_1(x', t', \bar{a}, \varepsilon'_1)] &= \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log \left(\sum_{d'_1 \in \{T, R\}} e^{u_1(\tilde{x}', \tilde{t}', \tilde{a}, d'_1) / \sigma} \right) dF_{x'}(\tilde{x}') dF_{t'}(\tilde{t}') dF_{\bar{a}}(\tilde{a}) = \\ &\int_{\tilde{a}} \int_{\tilde{t}'_1} \cdots \int_{\tilde{t}'_{J_t}} \int_{\tilde{x}'_1} \cdots \int_{\tilde{x}'_{J_x}} \left[\cdot \right] F_{x,1}(x'_1 | x'_2, \dots, x'_{J_x}) \cdots dF_{x, J_x}(x'_{J_x}) dF_{t,1}(t'_1 | t'_2, \dots, t'_{J_t}) \cdots dF_{t, J_t}(t'_{J_t}) dF_{\bar{a}}(\tilde{a}).\end{aligned}$$

And as such, Round 1's conditional choice probabilities are:

$$P(d_1 | x, t, \bar{a}_{-1}) = \frac{\exp(u_1(x, t, \bar{a}_{-1}, d_1) / \sigma)}{\sum_{\tilde{d}_1 \in \{T, R\}} \exp(u_1(x, t, \bar{a}_{-1}, \tilde{d}_1) / \sigma)}. \quad (71)$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P(d_1 = T | x, t, \bar{a}_{-1}; \theta_T)$, $P_{d_2} = P(d_2 = A | x, z, t, a, \bar{a}_{-1}; \theta_A)$, $P_y = P(y = S | x, z, t, |a - \bar{a}_{-1}|; \theta_S)$, $L_{y_1^*} = L(y_1^* = S | t; \theta_{S_1^*})$, and $L_{y_2^*} = L(y_2^* = S | t; \theta_{S_2^*})$; all but the cutoffs are logistic. Given that, the unrestricted likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) &= \prod_{n=1}^{N_A} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x, t}(z_n | x_n, t_n; \theta_{F_{z|x, t}}) f_a(a) f_{\bar{a}}(\bar{a}) f_{\bar{a}|a}(\bar{a} | a) P_{d_{2,n}} P_{y_n}^{\mathbb{1}(y_n=S)} (1 - P_{y_n})^{\mathbb{1}(y_n=F)} \\ &\prod_{n=N_A+1}^{N_{T,R}} f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x, t}(z_n | x_n, t_n; \theta_{F_{z|x, t}}) f_a(a) f_{\bar{a}}(\bar{a}) (1 - P_{d_{2,n}}) P_{y_n}^{\mathbb{1}(y_n=S)} (1 - P_{y_n})^{\mathbb{1}(y_n=F)} \\ &\prod_{n=N_{T,R}+1}^N f_x(x_n; \theta_{F_x}) f_t(t_n; \theta_{F_t}) f_{z|x, t}(z_n | x_n, t_n; \theta_{F_{z|x, t}}) f_a(a) f_{\bar{a}}(\bar{a}) (1 - P_{d_{1,n}}). \quad (72)\end{aligned}$$

I estimate $f_{\bar{a}|a}(\bar{a} | a)$ on the N_A accepted applicants only (those from $N_A + 1$ to $N_{T,R}$ are rejected in Round 2) because only accepted applicants can influence the acceptance set's area of interest. Also, as with the dynamic models in the paper, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R) . To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 7. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z . Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p , the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. See Appendix G.

G Proofs of Theorems

Theorem 1. *For large N , the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. The portfolio choice problem with type 1 extreme value preference shocks is:

$$\max_{(d_1, \dots, d_N) \in \{A, R\}^N} \sum_{n=1}^N [P(y_n = S|x_n, z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)], \text{ such that } \sum_{n=1}^N \mathbb{1}(d_n = A) \leq N_A. \quad (73)$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n = A$, and so the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(d_1, \dots, d_N; \lambda) &= \sum_{n=1}^N [P(y_n = S|x_n, z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + \varepsilon_n(R)\mathbb{1}(d_n = R)] + \lambda \left\{ N_A - \left[N - \sum_{n=1}^N \mathbb{1}(d_n = R) \right] \right\} \\ &= \sum_{n=1}^N [P(y_n = S|x_n, z_n) + \varepsilon_n(A))\mathbb{1}(d_n = A) + (\lambda + \varepsilon_n(R))\mathbb{1}(d_n = R)] + \lambda(N_A - N). \end{aligned} \quad (74)$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1 = S|x_1, z_1) + \varepsilon_1(A) - \varepsilon_1(R) \geq \dots \geq P(y_N = S|x_N, z_N) + \varepsilon_N(A) - \varepsilon_N(R)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* = \lambda^*(\boldsymbol{\varepsilon}(A), \boldsymbol{\varepsilon}(R)) \in [P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}) + \varepsilon_{N_A+1}(A) - \varepsilon_{N_A+1}(R), P(y_{N_A} = S|x_{N_A}, z_{N_A}) + \varepsilon_{N_A}(A) - \varepsilon_{N_A}(R)]$. That is, in this setup, λ^* is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds λ^* and reject the rest:

$$\forall n \in \{1, \dots, N\}, d_n(x_n, z_n, \varepsilon_n | \lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) + \varepsilon_n(A) > \lambda^* + \varepsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) + \varepsilon_n(A) = \lambda^* + \varepsilon_n(R) \\ R & \text{if } P(y_n = S|x_n, z_n) + \varepsilon_n(A) < \lambda^* + \varepsilon_n(R). \end{cases} \quad (75)$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes λ^* and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require taking expectations across each applicant's shocks, which would be fine except λ^* is a function of each applicant's shocks simultaneously. Suppose for now that λ^* is instead deterministic. Then:

$$\begin{aligned} P(d_n = A|x_n, z_n) &= P(P(y_n = S|x_n, z_n) + \varepsilon_n(A) \geq \lambda^* + \varepsilon_n(R)) \\ &= P(\varepsilon_n(R) - \varepsilon_n(A) \leq P(y_n = S|x_n, z_n) - \lambda^*) \\ &= \frac{\exp(P(y_n = S|x_n, z_n)/\sigma)}{\exp(P(y_n = S|x_n, z_n)/\sigma) + \exp(\lambda^*/\sigma)}, \end{aligned} \quad (76)$$

and the likelihood becomes the product of individual binary logitics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, λ^* is not deterministic for finite N , so the logistic conditional choice probabilities do not hold exactly. But for large N , λ^* converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N} \sum_{n=1}^N \mathbb{1}\{P(y_n = S|x_n, z_n) + \varepsilon_n(A) - \varepsilon_n(R) \leq c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff λ^* is the $(1 - N_A/N)$ quantile of this distribution. Because the shocks are iid, the perturbed indices in $\hat{F}_N$ are iid. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution F for large N . Therefore, as $N_A/N \rightarrow q$, $\hat{F}_N$'s $(1 - N_A/N)$ quantile converges to F 's deterministic $(1 - q)$ quantile. The iid assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving λ^* random in the limit. Independence makes the simultaneous fluctuations of the N shocks cancel, so $\hat{F}_N$ and λ^* converge. With λ^* deterministic, no individual applicant's shock affects λ^* , the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds.

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A . Also, for any $\hat{P}_{y^*}$ and N_A , $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. Supposing one round, the restricted likelihood is:

$$\mathcal{L}^R = \prod_{n=1}^N \left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right]^{\mathbb{1}(d_n=A)} \left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right]^{\mathbb{1}(d_n=R)} \left[\frac{e^{P_{y_n}}}{1 + e^{P_{y_n}}} \right]^{\mathbb{1}(y_n=S)} \left[\frac{1}{1 + e^{P_{y_n}}} \right]^{\mathbb{1}(y_n=F)} \quad (77)$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in P_{y^*} , providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in P_{y^*} , providing upward pressure.

Since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = \mathbb{1}(y_n = S) + \mathbb{1}(y_n = F) = 1$, the restricted log-likelihood is:

$$\begin{aligned} \ell^R &= \sum_{n=1}^N \left[\mathbb{1}(d_n = A) \log \left(\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) + \mathbb{1}(d_n = R) \log \left(\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}} \right) \right. \\ &\quad \left. + \mathbb{1}(y_n = S) \log \left(\frac{e^{P_{y_n}}}{1 + e^{P_{y_n}}} \right) + \mathbb{1}(y_n = F) \log \left(\frac{1}{1 + e^{P_{y_n}}} \right) \right] \\ &= \sum_{n=1}^N \left[\frac{1}{\sigma} (\mathbb{1}(d_n = A) P_{y_n} + \mathbb{1}(d_n = R) P_{y^*}) - \log(e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}) + \mathbb{1}(y_n = S) P_{y_n} - \log(1 + e^{P_{y_n}}) \right]. \end{aligned} \quad (78)$$

Since ℓ^R is strictly concave in P_{y^*} , I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma} \left[N - N_A - \sum_{n=1}^N \frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} \right] = 0 \quad \Rightarrow \quad \sum_{n=1}^N \frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \quad (79)$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected N_A . Therefore, $\hat{P}_{y^*}$ is such that the expected N_A equals N_A .

Given that $\hat{P}_{y^*}$ is identified, I will show that ℓ^R , and therefore $\mathcal{L}^R$, are highest when the committee accepts the N_A applicants with the highest P_{y_n} . The only part of ℓ^R that depends on d_n is

$\sum_{n=1}^N [\mathbb{1}(d_n = A)P_{y_n} + \mathbb{1}(d_n = R)P_{y^*}] = NP_{y^*} + \sum_{n=1}^N \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$; this equality holds since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = 1$. As such, the only part depending on d_n is $\sum_{n=1}^N \mathbb{1}(d_n = A)(P_{y_n} - P_{y^*})$.

This expression is highest when $d_n = A$ for the N_A highest values of P_{y_n} , so the likelihood is highest when the committee accepts the N_A highest P_{y_n} . Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher P_{y_n} . However, if the committee engages in positive sorting, the model's fit will improve.

Theorem 3. *For large N , the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. For brevity, let $w = (x, z)$. Then for large N , the portfolio choice problem is:

$$\begin{aligned} \max_{\delta \in \{\delta: \mathbb{R}^K \rightarrow [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw, \\ \text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w)f(w)dw \leq N_M, \end{aligned} \quad (80)$$

where δ is the committee's decision rule, and f is the density of w . Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases in δ , which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(\delta; \lambda) &= \int_{w \in \mathbb{R}^K} P(y = S|w)P(m = M|w)\delta(w)f(w)dw \\ &\quad + \lambda \left\{ N_M - \left[\int_{w \in \mathbb{R}^K} P(m = M|w)f(w)dw - \int_{w \in \mathbb{R}^K} P(m = M|w)(1 - \delta(w))f(w)dw \right] \right\} \quad (81) \\ &= \int_{w \in \mathbb{R}^K} [P(y = S|w)\delta(w) + \lambda(1 - \delta(w))]P(m = M|w)f(w)dw + \lambda \left[N_M - \int_{w \in \mathbb{R}^K} P(m = M|w)f(w)dw \right]. \end{aligned}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept the necessary fraction of applicants such that $\int_{w \in \mathbb{R}^K} P(m = M|w)\delta(w|\lambda^*)f(w)dw = N_M$, which results in a cutoff rule solution on success probabilities alone:

$$\forall w \in \mathbb{R}^K, \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0, 1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^*. \end{cases} \quad (82)$$

The committee accepts all applicants with success probabilities $P(y = S|w)$ strictly greater than the cutoff probability λ^* , rejects all applicants with $P(y = S|w)$ strictly less than λ^* , and may accept or reject any applicants with $P(y = S|w)$ equal to λ^* , if such a mass of applicants exists. All that remains is to identify λ^* . Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned} M^H(\lambda) &= \int_{\{w \in \mathbb{R}^K | P(y=S|w) \geq \lambda\}} P(m = M|w)f(w)dw \\ M^L(\lambda) &= \int_{\{w \in \mathbb{R}^K | P(y=S|w) > \lambda\}} P(m = M|w)f(w)dw. \end{aligned} \quad (83)$$

The upper bound function M^H is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some λ , and the lower bound function M^L is the fraction whose success probabilities are strictly greater than λ . Both functions are between 0 and 1, both are decreasing in λ , and for any λ , it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_{\lambda} \{ \lambda | M^L(\lambda) \leq N_M \}$. Intuitively, λ^* is the largest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y = S|w) > \lambda$ does not exceed the maximum number of matriculants N_M . If there is a positive mass of matriculation-probability-scaled applicants with $P(y = S|w) = \lambda^*$ (that is, if $M^H(\lambda^*) - M^L(\lambda^*) > 0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0, 1]$ of that mass of applicants such that $M^L(\lambda^*) + \alpha^*[M^H(\lambda^*) - M^L(\lambda^*)] = N_M$. That is, the committee will accept every applicant with $P(y = S|w) > \lambda^*$ and accept just enough applicants with $P(y = S|w) = \lambda^*$ to fill the cohort.

Theorem 4. *In the waitlist-dynamic and hybrid models, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L , such that $P(y = S|x^H, z^H, t, \bar{k}) = P(y = S|x^L, z^L, t, \bar{k})$, but $m_2^H > m_2^L$. That is, the applicants have equal success probabilities, but H has the higher matriculation probability. Since $\dot{m}_2$ is the average matriculation probability of all applicants in t with m_2 excluded, $\dot{m}_2^H < \dot{m}_2^L$ by construction. Given this inequality, I will show $P(d_2 = A|x^H, z^H, t, \bar{k}, \dot{m}_2^H) > P(d_2 = A|x^L, z^L, t, \bar{k}, \dot{m}_2^L)$. It is the case that:

$$P(d_2 = A|x, z, t, \bar{k}, \dot{m}_2) = \frac{\exp(P(y = S|x, z, t, \bar{k})/\sigma)}{\exp(P(y = S|x, z, t, \bar{k})/\sigma) + \exp(\beta \mathbb{E}[v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \epsilon_3)|x, z, t, \bar{k}, \dot{m}_2]/\sigma)}, \text{ where}$$

$$\mathbb{E}[v_3(x, z, t, \bar{k}, \bar{m}_{2A}, \epsilon_3)|x, z, t, \bar{k}, \dot{m}_2] = \int_{\bar{m}_{2A}} \sigma \log \left(e^{P(y=S|x, z, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right) dF_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2). \quad (84)$$

The only place $\dot{m}_2$ appears in $P(d_2 = A|x, z, t, \bar{k}, \dot{m}_2)$ is in the distribution of $\bar{m}_{2A}$ conditional on $\dot{m}_2$. I parameterize $\bar{m}_{2A}|\dot{m}_2 \sim \mathcal{N}(\dot{m}_2 \beta_{\dot{m}_2}, \sigma_{\dot{m}_2}^2)$, where by assumption $\beta_{\dot{m}_2} > 0$. $\beta_{\dot{m}_2}$ is positive because years when $\dot{m}_2$ is high are years when matriculation probabilities are high in general. Using inter-year variation, this relationship results in a greater expected $\bar{m}_{2A}$ in those years. Then:

$$F_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2^H; \beta_{\dot{m}_2}, \sigma_{\dot{m}_2}^2) = \Phi \left(\frac{\bar{m}_{2A} - \dot{m}_2^H \beta_{\dot{m}_2}}{\sigma_{\dot{m}_2}} \right) > \Phi \left(\frac{\bar{m}_{2A} - \dot{m}_2^L \beta_{\dot{m}_2}}{\sigma_{\dot{m}_2}} \right) = F_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2^L; \beta_{\dot{m}_2}, \sigma_{\dot{m}_2}^2). \quad (85)$$

Therefore, $\bar{m}_{2A}|\dot{m}_2^L$ first-order stochastically dominates $\bar{m}_{2A}|\dot{m}_2^H$. To apply this stochastic dominance, I must show that $\sigma \log \left(e^{P(y=S|x, z, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right)$ is strictly increasing in $\bar{m}_{2A}$. But the only place $\bar{m}_{2A}$ appears in this expression is in $P(y_3^* = S|\bar{m}_{2A}; \theta_{S_3^*}) = \frac{\exp(f_{S_3^*}(\bar{m}_{2A})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\bar{m}_{2A})\theta_{S_3^*})}$, where by assumption $f_{S_3^*}(\cdot)$ is strictly increasing and $\theta_{S_3^*} > 0$. $\theta_{S_3^*}$ is positive because the expected Round 3 cutoff probability is higher when early acceptees are more likely on average to matriculate. As such, I have that $P(y_3^* = S|\bar{m}_{2A})$, and in turn $\sigma \log \left(e^{P(y=S|x, z, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right)$, are strictly

increasing in $\bar{m}_{2A}$. Combining this result with $(\bar{m}_{2A}|\dot{m}_2^L) \succ_{\text{FOSD}} (\bar{m}_{2A}|\dot{m}_2^H)$, I obtain:

$$\begin{aligned}
\mathbb{E}[v_3(x^L, z^L, t, \bar{k}, \bar{m}_{2A}, \epsilon_3)|x^L, z^L, t, \bar{k}, \dot{m}_2^L] &= \int_{\bar{m}_{2A}} \sigma \log \left(e^{P(y=S|x^L, z^L, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right) dF_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2^L) \\
&= \int_{\bar{m}_{2A}} \sigma \log \left(e^{P(y=S|x^H, z^H, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right) dF_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2^L) \\
&> \int_{\bar{m}_{2A}} \sigma \log \left(e^{P(y=S|x^H, z^H, t, \bar{k})/\sigma} + e^{P(y_3^*=S|\bar{m}_{2A})/\sigma} \right) dF_{\bar{m}_{2A}|\dot{m}_2}(\bar{m}_{2A}|\dot{m}_2^H) \\
&= \mathbb{E}[v_3(x^H, z^H, t, \bar{k}, \bar{m}_{2A}, \epsilon_3)|x^H, z^H, t, \bar{k}, \dot{m}_2^H]. \tag{86}
\end{aligned}$$

The second equality holds since $P(y = S|x^H, z^H, t, \bar{k}) = P(y = S|x^L, z^L, t, \bar{k})$. Therefore, by Equation (84) and Inequality (86), I have that $P(d_2 = A|x^H, z^H, t, \bar{k}, \dot{m}_2^H) > P(d_2 = A|x^L, z^L, t, \bar{k}, \dot{m}_2^L)$.

Theorem 5. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\begin{aligned}
&\frac{1}{N_A} \sum_{n=1}^N P(y_n = S|x_n, z_n, t_n; \hat{\theta}_S^U) \mathbb{1} \left\{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \right\} \\
&\geq \frac{1}{N_A} \sum_{n=1}^N P(y_n = S|x_n, z_n, t_n; \hat{\theta}_S^U) \mathbb{1} \left\{ d_{1,n} [P_{y_n}(\hat{\theta}^U)] = T, d_{2,n}^U [P_{y_n}(\hat{\theta}_S^U)] = A \right\} \tag{87}
\end{aligned}$$

Proof. This theorem states that conditional on a fixed Round 1 decision rule $d_1[P_y(\hat{\theta}^U)]$ and corresponding transition set of exogenously-determined size N_T , the sample average success probability of the also-exogenous N_A applicants that the model accepts in Round 2 weakly exceeds the sample average success probability of the N_A applicants that any actual committee accepts in Round 2. The inequality holds since the model's Round 2 decision rule is to accept the N_A applicants with the highest success probabilities. Therefore, the sample average success probability of the model's acceptances must weakly exceed that of any other set of N_A acceptances.

However, the inequality would not always hold if it were written as follows (changes in bold):

$$\begin{aligned}
&\frac{1}{N_A} \sum_{n=1}^N P(y_n = S|x_n, z_n, t_n; \hat{\theta}_S^U) \mathbb{1} \left\{ \mathbf{d}_{1,n}^R [\mathbf{P}_{y_n}(\hat{\theta}_S^U)] = \mathbf{T}, d_{2,n}^R [P_{y_n}(\hat{\theta}_S^U)] = A \right\} \\
&\text{vs. } \frac{1}{N_A} \sum_{n=1}^N P(y_n = S|x_n, z_n, t_n; \hat{\theta}_S^U) \mathbb{1} \left\{ \mathbf{d}_{1,n}^U [\mathbf{P}_{y_n}(\hat{\theta}_T^U)] = \mathbf{T}, d_{2,n}^U [P_{y_n}(\hat{\theta}_A^U)] = A \right\} \tag{88}
\end{aligned}$$

The difference is that the Round 1 transition set is no longer fixed. Now, the model's transition set is based on choosing the N_T applicants in each year with the highest $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}] = \int_{\bar{z}} \sigma \log(e^{P(y=S|x, z, t, \bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}) dF(\bar{z}|x, t)$. In contrast, the committee may have a different Round 1 decision rule. Crucially, although the model is choosing applicants akin to those with the highest expected success probabilities, it is ignoring any value added from upside potential.

For example, suppose $N_A = 1$, and there is an applicant in Round 1 with a low expected success probability but a known high degree of variance in the distribution of z conditional on

their (x_n, t_n) . That is, $F(z|x, t)$ is heteroskedastic. Also suppose that their low expected success probability yields an $\mathbb{E}[v_2(x, z, t, \bar{k}, \epsilon_2)|x, t, \bar{k}]$ too low to survive Round 1, even though their actual success probability ends up being the highest of any applicant once z is realized in Round 2. If the committee's decision rule considers upside potential, then the committee may accept this applicant, while the model will not. In most cases, the model's Round 1 decision rule should lead to a better acceptance set than the committee's rule. But such an outcome is not guaranteed.

Theorem 6. *In the matriculation multiyear model, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L , such that $P(y = S|x^H, z^H, t) = P(y = S|x^L, z^L, t)$, but $m^H > m^L$. That is, H has the higher matriculation probability. I will show that $P(d_2 = A|x^H, z^H, t, m^H, c_{-1}) > P(d_2 = A|x^L, z^L, t, m^L, c_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\begin{aligned}\pi_2^A &\equiv P(y = S|x, z, t) + \beta \mathbb{E}[v_1(x', t', c, \epsilon'_1)|m], & \pi_2^R &\equiv L(y_2^* = S|c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, \epsilon'_1)], \\ \pi_1^A &\equiv \mathbb{E}[v_2(x', z', t', m', c, \epsilon'_2)|x', t', c], & \pi_1^R &\equiv L(y_1^* = S|t') + \mathbb{E}[v_1(x'', t'', c', \epsilon''_1)].\end{aligned}\quad (89)$$

It is the case that:

$$\begin{aligned}P(d_2 = A|x, z, t, m, c_{-1}) &= \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:} \\ \mathbb{E}[v_1(x', t', c, \epsilon'_1)|m] &= \int_c \int_{t'} \int_{x'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_x(x') dF_t(t') dF_{c|m}(c|m), \text{ where:} \\ \mathbb{E}[v_2(x', z', t', m', c, \epsilon'_2)|x', t', c] &= \int_{\tilde{m}'} \int_{\tilde{z}'} \sigma \log(e^{\pi_2^A/\sigma} + e^{\pi_2^R/\sigma}) dF_{z|x,t}(\tilde{z}'|x', t') dF_{m|x,t}(\tilde{m}'|x', t'),\end{aligned}\quad (90)$$

where π_2^A and π_2^R are defined analogously to Equation (89) at the next year's states.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only place m appears in $P(d_2 = A|x, z, t, m, c_{-1})$ is in the distribution of c conditional on m . I parameterize $c|m \sim \mathcal{N}(m\beta_m, \sigma_m^2)$, where by assumption $\beta_m > 0$. β_m is positive because higher matriculation probabilities lead to a greater expected cohort size. Multiple years of data are necessary to identify β_m , since cohort sizes vary only across years. Then:

$$F_{c|m}(c|m^H; \beta_m, \sigma_m^2) = \Phi\left(\frac{c - m^H \beta_m}{\sigma_m}\right) < \Phi\left(\frac{c - m^L \beta_m}{\sigma_m}\right) = F_{c|m}(c|m^L; \beta_m, \sigma_m^2). \quad (91)$$

Therefore, $c|m^H$ first-order stochastically dominates $c|m^L$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', c, \epsilon'_1)|m]$'s integrand consisting of all but the outermost integral is strictly increasing in c . But the only place c appears there is in $\mathbb{E}[v_2(x', z', t', m', c, \epsilon'_2)|x', t', c]$, and the only place c appears in that expectation is in $L(y_2^* = S|c; \theta_{S_2^*}) = f_{S_2^*}(c)\theta_{S_2^*}$.

By assumption, $f_{S_2^*}(\cdot)$ is strictly increasing and $\theta_{S_2^*} > 0$. $\theta_{S_2^*, c}$ is positive because a greater cohort size this year means that the committee, needing to keep the total number of students in

the program relatively constant across years, can afford to be more selective next year. As such, I have that $P(y_2^* = S|c)$, and in turn $\mathbb{E}[v_1(x', t', c, \epsilon'_1)|m]$'s integrand, are strictly increasing in c . Combining this result with $[c|m^H] \succ_{FOSD} [c|m^L]$, I obtain the inequality:

$$\begin{aligned}\mathbb{E}[v_1(x', t', c, \epsilon'_1)|m^H] &= \int_c \int_{t'} \int_{x'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_x(x') dF_t(t') dF_{c|m}(c|m^H) \\ &> \int_c \int_{t'} \int_{x'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_x(x') dF_t(t') dF_{c|m}(c|m^L) \\ &= \mathbb{E}[v_1(x', t', c, \epsilon'_1)|m^L].\end{aligned}\quad (92)$$

Therefore, by Equation (90), Inequality (92), and $P(y = S|x^H, z^H, t) = P(y = S|x^L, z^L, t)$, I have that $P(d_2 = A|x^H, z^H, t, m^H, c_{-1}) > P(d_2 = A|x^L, z^L, t, m^L, c_{-1})$. Intuitively, if the committee rejects either applicant, their m does not matter. But since their success probabilities are equal, accepting applicant H yields a higher expected payoff than accepting applicant L .

Theorem 7. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z . Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p , the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. Suppose there are two arbitrary applicants in year t , one with $a = 1$ and the other with $a = 0$, but with $P(y = S|x^1, z^1, t, 1 - \bar{a}_{-1}) = P(y = S|x^0, z^0, t, \bar{a}_{-1})$. Unlike in the previous two proofs, these equal success probabilities imply that unless $\bar{a}_{-1} = \frac{1}{2}$, it must be that $(x^1, z^1) \neq (x^0, z^0)$. I will show that $P(d_2 = A|x^1, z^1, t, 1, \bar{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \bar{a}_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\begin{aligned}\pi_2^A &\equiv P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a], & \pi_2^R &\equiv L(y_2^* = S|t) + \beta \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)], \\ \pi_1^A &\equiv \mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}], & \pi_1^R &\equiv L(y_1^* = S|t') + \beta \mathbb{E}[v_1(x'', t'', \bar{a}', \epsilon'_1)].\end{aligned}\quad (93)$$

It is the case that:

$$\begin{aligned}P(d_2 = A|x, z, t, a, \bar{a}_{-1}) &= \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:} \\ \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a] &= \int_{\tilde{a}} \int_{t'} \int_{\tilde{x}'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_x(\tilde{x}') dF_t(\tilde{t}') dF_{\tilde{a}|a}(\tilde{a}|a), \text{ where:} \\ \mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}] &= P(a' = 1) \int_{\tilde{z}'} \sigma \log(e^{\pi_2^{A'}(1)/\sigma} + e^{\pi_2^{R'}(1)/\sigma}) dF_{z|x,t}(\tilde{z}'|x', t') \\ &\quad + [1 - P(a' = 1)] \int_{\tilde{z}'} \sigma \log(e^{\pi_2^{A'}(0)/\sigma} + e^{\pi_2^{R'}(0)/\sigma}) dF_{z|x,t}(\tilde{z}'|x', t'),\end{aligned}\quad (94)$$

where $\pi_2^{A'}(a')$ denotes next year's Round 2 acceptance payoff evaluated at $a' \in \{0, 1\}$.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only two places a appears in $P(d_2 = A|x, z, t, a, \bar{a}_{-1})$ are in the success probabilities, which are assumed to be equal for the $a = 1$ and $a = 0$ applicants, and in the distribution of $\bar{a}$ conditional

on a . I parameterize $\bar{a}|a \sim \mathcal{N}(a\beta_a, \sigma_a^2)$, where by assumption $\beta_a > 0$. β_a is positive because $a = 1$ applicants are more likely than $a = 0$ applicants to be in a year with a high percentage of accepted $a = 1$ applicants. As such, multiple years of data are necessary to identify β_a . Then:

$$F_{\bar{a}|a}(\bar{a}|1; \beta_a, \sigma_a^2) = \Phi\left(\frac{\bar{a} - \beta_a}{\sigma_a}\right) < \Phi\left(\frac{\bar{a}}{\sigma_a}\right) = F_{\bar{a}|a}(\bar{a}|0; \beta_a, \sigma_a^2). \quad (95)$$

Therefore, $\bar{a}|1$ first-order stochastically dominates $\bar{a}|0$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|a]$'s integrand consisting of all but the outermost integral is strictly increasing in $\bar{a}$. But the only place $\bar{a}$ appears there is in $\mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}]$, and the only place $\bar{a}$ appears in that expectation is in $P(y' = S|x', z', t', \bar{a}; \theta_S) = \frac{\exp(f_S(x', z', t', |a' - \bar{a}|)\theta_S)}{1 + \exp(f_S(x', z', t', |a' - \bar{a}|)\theta_S)}$.

By assumption, $f_S(\cdot)$ is strictly increasing and $\theta_{S, |a - \bar{a}_{-1}|} > 0$. $\theta_{S, |a - \bar{a}_{-1}|}$ is positive because a large value of $|a - \bar{a}_{-1}|$ means that applicant a is diverse relative to the previous year's accepted applicants and is therefore more likely to succeed. Recall $P(a' = 1) = p$. Taking the limit $p \rightarrow 0$:

$$\begin{aligned} & \mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}] \\ & \rightarrow \int_{\tilde{z}'} \sigma \log(e^{\{P(y'=S|x', z', t', \bar{a}) + \beta \mathbb{E}[v_1(x'', t'', \bar{a}, \epsilon'_1)|0]\}/\sigma} + e^{\{L(y_2'=S|t') + \beta \mathbb{E}[v_1(x'', t'', \bar{a}, \epsilon'_1)]\}/\sigma}) dF_{\tilde{z}|x, t}(\tilde{z}'|x', t'). \end{aligned} \quad (96)$$

$P(y' = S|x', z', t', \bar{a})$ is strictly increasing in $\bar{a}$, and by the continuity of $E[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}]$ in p , I have that it, and in turn $E[v_1(x', t', \bar{a}, \epsilon'_1)|a]$'s integrand, are also strictly increasing in $\bar{a}$ for sufficiently small p . Combining this result with $\bar{a}|1 \succ_{FOSD} \bar{a}|0$, I obtain the inequality:

$$\begin{aligned} \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|1] &= \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_{\tilde{x}}(\tilde{x}') dF_{\tilde{t}}(\tilde{t}') dF_{\bar{a}|a}(\tilde{a}|1) \\ &> \int_{\tilde{a}} \int_{\tilde{t}'} \int_{\tilde{x}'} \sigma \log(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}) dF_{\tilde{x}}(\tilde{x}') dF_{\tilde{t}}(\tilde{t}') dF_{\bar{a}|a}(\tilde{a}|0) \\ &= \mathbb{E}[v_1(x', t', \bar{a}, \epsilon'_1)|0]. \end{aligned} \quad (97)$$

Therefore, by Equation (94), Inequality (97), and $P(y = S|x^1, z^1, t, 1 - \bar{a}_{-1}) = P(y = S|x^0, z^0, t, \bar{a}_{-1})$, I have that $P(d_2 = A|x^1, z^1, t, 1, \bar{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \bar{a}_{-1})$. Intuitively, if the committee rejects either applicant, their a does not matter. But since their success probabilities are equal, accepting applicant $a = 1$ yields a higher expected payoff than accepting $a = 0$.

In this proof, I must assume $P(a' = 1)$ is sufficiently small, and not just $P(a' = 1) < \frac{1}{2}$. The reason is that if $P(a' = 1)$ is too large, then $\mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}]$ may not be strictly increasing in $\bar{a}$, considering $P(y' = S|x', z', t', 1 - \bar{a})$ is strictly decreasing in $\bar{a}$. This indeterminacy can be explored by calculating $\frac{\partial}{\partial \bar{a}} \mathbb{E}[v_2(x', z', t', a', \bar{a}, \epsilon'_2)|x', t', \bar{a}]$. As such, to be sure the committee is more likely to accept the $a = 1$ applicant, the committee must know that there will be sufficiently few $a' = 1$ applicants. This assumption is strong, but possible if $a' = 1$ is sufficiently rare for the committee to know this will happen. However, even without forward-looking dynamics, if $\bar{a}_{-1} < \frac{1}{2}$, then the committee will be more likely to accept an $a = 1$ applicant than an $a = 0$ applicant with identical x and z values. The reason is that $P(y = S|x, z, t, 1 - \bar{a}_{-1}) > P(y = S|x, z, t, \bar{a}_{-1})$.